Q1:- Find sum
$z_1 = 3+i \quad z_2 = -1+2i$

Solve
$z_1+z_2 = (3+i)+(-1+2i)$
$= 3-1+i+2i$
$= 2+3i$

Q2:- Find modulus of z
$z = 6+7i$

Solve
$|z| = \sqrt{6^2+7^2} = \sqrt{36+49} = \sqrt{85}$

Q3. $z = i-1$ Then find $f(z) = z^2+z+1$

Solv $f(z) = z^2+z+1$
$= (i-1)^2+(i-1)+1$
$= i^2+1-2i+i-1+1$
$= -1+1-2i+i-1+1$
$= -i$

Q4:- Solve $\frac{1+2i}{3-4i} + \frac{2-i}{5i}$

Solv $\frac{1+2i}{3-4i} \times \frac{3+4i}{3+4i} + \frac{2-i}{5i} \times \frac{i}{i}$

$\frac{-5+10i}{9+16} - \frac{1+2i}{5}$

$= \frac{-5+10i-5-10i}{25}$

$= \frac{-2}{5}$

Q5:- $z = re^{i\theta}$ Find $\frac{1}{z}$

Solve
$\frac{1}{z} = \frac{1}{re^{i\theta}} = \frac{1}{r}e^{-i\theta}$

Q6:- Find Product
$z_1 = -3+2i \quad z_2 = -1+11i$

Solve
$z_1 \cdot z_2 = (-3+2i)(-1+11i)$
$= -19-35i$

Q7: Divide $\frac{z_1}{z_2} = \frac{-8+i}{-3+8i}$

Solv $\frac{z_1}{z_2} = \frac{-8+i}{-3+8i} \times \frac{-3+8i}{-3-8i}$
$= \frac{24+64i-3i+8}{(-3)^2-(8i)^2}$
$= \frac{32+61i}{73}$

Q8: $z = -3+9i$
Find Re(z) and Im(z)

Solve
$Re(z) = -3 \quad Im(z) = 9$

Q9 Verify Commutative law w.r.t.M
$z_1 = 2+i \quad z_2 = -3+9i$

Solve $z_1 \cdot z_3 = z_2 \cdot z_1$
$(2+i)(-3+9i) = (-3+9i)(2+i)$
$-6+18i-3i+9i^2 = -6-3i+18i+9i^2$
$-15+15i = -15+15i$

Q10: Verify Associative law w.r.t M
$z_1 = 2+i \quad z_2 = -3+9i \quad z_3 = 8+i$

Solve
$z_1(z_2 \cdot z_3) = (z_1 \cdot z_2)z_3$
$(2+i)(-3+9i)(8+i) = (2+i)(-3+9i)(8+i)$
$(2+i)(-24-3i+72i+9i^2) = (-6+18i-3i+9i^2)(8+i)$
$(2+i)(-33+69i) = (-15+15i)(8+i)$
$-135+105i = -135+105i$

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Q11: Find Multiplative inverse. $z = 5+i$

Solve $z = 5+i$

$$z^{-1} = \frac{1}{z} = \frac{1}{5+i} \times \frac{5-i}{5-i} = \frac{5-i}{25+1} = \frac{5-i}{26}$$

Q12. Write distributive law × over +

Solve $z_1(z_2+z_3) = (z_1 \times z_2) + (z_1 \times z_3)$

Q13: Find distance of complex Number $z_1 = 2+i$ $z_2 = -3+4i$

Solve $z_1+z_2 = (2+i)+(-3+4i) = -1+8i$

$$|z_1+z_2| = \sqrt{(-1)^2+(8)^2} = \sqrt{65}$$

Q14: Verify $|z|^2 = z\bar{z}$ For $z = -3+4i$

Solu $|z|^2 = \left(\sqrt{(-3)^2+(4)^2}\right)^2 = 9+16 = 25$

$z\bar{z} = (-3+4i)(-3-4i) = 25$

Q15 $f(z) = 2z^3+2z$ Then find $f'(z)$

Solve $f(z) = 2z^3+2z$

$f'(z) = 6z^2+2$

Q16: $\lim_{(x,y)\to(2,4)} 3xy - y^2$

Sol $3(2)(4) - 4^2 = 24-16 = 8$

Q17 Express in the form of $a+ib$ $-\sqrt{3}+i$

Solu $-\sqrt{3}+i$

Q18 Binomial formula,

Sol $\binom{n}{k} = \frac{n!}{k!(n-k)!}$

Q19 Find Polar representation $z=(0,5)$

Sol $r = \sqrt{0+5^2} = 5$ $\quad \text{Arg } z = \pi/2$

Polar representation $= 5(\cos\pi/2 + i\sin\pi/2)$

Q20 calculate polar representation $z = -1-i$

Solu $r = \sqrt{(-1)^2+(-1)^2} = \sqrt{1+1} = \sqrt{2}$

$\tan^{-1}\frac{y}{x} = \tan^{-1}\frac{-1}{-1} = \tan^{-1}1$

$\theta = \pi/4$

$\text{Arg } z = \frac{\pi}{4} + \pi = -3\pi/4$

Polar representation $= \sqrt{2}\cos(-3\pi/4) + i\sin(-3\pi/4)$

Q21 Polar representation $-1-i$ is given by $\sqrt{2}\cos -3\pi/4 + i\sin -3\pi/4$ write in exponential form

solu $e^{i\theta} = \cos\theta + i\sin\theta$

$z = re^{i\theta} = \sqrt{2}\, e^{-3\pi/4\, i} = \sqrt{2}\exp\left[i\frac{3\pi}{4}\right]$

Q22: $e^{i\theta_1} \cdot e^{i\theta_2} = e^{i(\theta_1+\theta_2)}$

proof:

$$e^{i\theta_1} \cdot e^{i\theta_2} = (\cos\theta_1 + i\sin\theta_1)(\cos\theta_2 + i\sin\theta_2)$$

$$= \cos\theta_1\cos\theta_2 + i\cos\theta_1\sin\theta_2 + i\sin\theta_1\cos\theta_2 + i^2\sin\theta_1\sin\theta_2$$

$$= \cos\theta_1\cos\theta_2 - \sin\theta_1\sin\theta_2 + i(\sin\theta_1\cos\theta_2 + \cos\theta_1\sin\theta_2)$$

$$= \cos(\theta_1+\theta_2) + i\sin(\theta_1+\theta_2)$$

$$= e^{i\theta_1+\theta_2} \quad \because \cos(\alpha+\beta) = \cos\alpha\cos\beta - \sin\alpha\sin\beta$$

$$\sin(\alpha+\beta) = \sin\alpha\cos\beta - \cos\alpha\sin\beta$$

Q23: $z_1 = 2e^{i\pi/2}$ $\quad z_2 = 3e^{i\pi/2}$ Find $z_1 \cdot z_2$

Solve $z_1 \cdot z_2 = 2\times 3\, e^{i\pi/2 + i\pi/2} = 6e^{i2\pi/2} = 6e^{i\pi}$

Q24: $z_1 = 2e^{i\pi/3}$ $\quad z_2 = 4e^{i\pi/4}$ Solv. $\frac{z_1}{z_2}$

Solv $\frac{z_1}{z_2} = \frac{2e^{i\pi/3}}{4e^{i\pi/4}} = \frac{1}{2}e^{i(\frac{\pi}{3}-\pi/4)} = \frac{1}{2}e^{i\pi/12}$

Q25 $z = 7e^{i\pi/4}$ Find z^{100}

Sol $z^{100} = 7^{100}\, e^{i\pi/4(100)}$

$= 7^{100}\, e^{i\pi(25)}$ $\quad \therefore z^n = r^n e^{i\theta n}$

$= 7\, e^{}$

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Q26 $(\sqrt{3}+i)^3$ convert into exponential form

Solu $r=\sqrt{(\sqrt{3})^2+(1)^2} = \sqrt{3+1} = 2$

$\tan^{-1}\frac{1}{\sqrt{3}} = \pi/6$

$\sqrt{3}+i = re^{i\theta} = 2e^{i\pi/6}$

$(\sqrt{3}+i)^3 = r^3 e^{i3\theta} = 2^3 e^{i3\pi/6} = 8e^{i\pi/2}$

Q27 Demoivre formula

Sol $(\cos\theta+i\sin\theta)^n = \cos n\theta + i\sin n\theta$

Q28: Find Sqware root of $2i$

$r = \sqrt{0+2^2} = 2 \qquad \theta = \pi/2$

$z = 2(\cos\pi/2 + i\sin\pi/2)$

$z^{1/2} = \sqrt{2}(\cos\pi/4 + i\sin\pi/4)$

Q29 n^{th} root formula:

$z = \sqrt[n]{r}\ \exp\left(i\frac{\theta}{n} + \frac{2k\pi}{n}\right)$ $k=0,1,\dots,n-1$

30 Calculate the n^{th} root of $z=1$

Solu $r = \sqrt{1^2+0} = 1 \qquad \theta = 0$

$z = re^{i\theta} = e^{i(0)}$

$z^{1/n} = r^{1/n}\exp\left(i\left(\frac{\theta}{n}+\frac{2k\pi}{n}\right)\right) = \sqrt[n]{1}\ \exp\left[i\left(\frac{0}{n}+\frac{2k\pi}{n}\right)\right]$

$= \exp\left(i\frac{2k\pi}{n}\right)$

when $k=1$ $\quad z^{1/n} = e^{i2\pi/n} = \omega$

$k=2$ $\quad z^{1/n} = e^{i4\pi/n} = \omega^2$

Q31. Find Sqware root of unity.

Solu $z=1$

$z = e^{0i}$

$z^{1/2} = \sqrt{1}\ e^{i\left(\frac{0}{2}+\frac{2k\pi}{2}\right)} = e^{ik\pi}$

$k=0$ $\quad e^{i(0)\pi} = \cos 0 + i\sin 0 = 1$

$k=1$ $\quad e^{i\pi} = \cos\pi + i\sin\pi = -1$

32 Find Cube root of unity

Solu $z=1$

$z = e^{0i}$

$\sqrt[3]{z} = e^{i\left(\frac{0}{3}+\frac{2k\pi}{3}\right)} = e^{\frac{2}{3}k\pi i}$

$k=0 \quad c_0=1 \qquad k=2 \quad c_1 = e^{i2\pi/3}$

$k=3 \quad c_2 = e^{i4\pi/3}$

33 Find root of $z^2+(2+i)z+\frac{3}{4}=0$

Sol $a=1 \quad b=2+i \quad c=3/4$

$z = \frac{-(2+i) \pm \sqrt{(2+i)^2-4(1)(3/4)}}{2(1)}$

$= \frac{-(2+i) \pm \sqrt{4i}}{2}$

34. Open set:- A set is called open set if every points of S is interior point of S

35 closed set:- A set is said to be closed set if it contain all its boundary points

36 closed curve:- if intial and terminal points are eqvual the the curved is said to be closed curve.

37 Simple curve:- A curve is said to be simple curve if it does not cross itself.

38) Domain:- The connected open set is called domain.

39:- Express $f(z) = 4x^2+14y^2$ by involving the variable z and $\bar{z}$

Solu $f(z) = 4x^2+4y^2 i$

$x = \frac{z+\bar{z}}{2} \qquad y = \frac{z-\bar{z}}{2}$

$= 4\left(\frac{z+\bar{z}}{2}\right)^2 + 4\left(\frac{z-\bar{z}}{2}\right)^2 i$

$= z^2+\bar{z}^2+2z\bar{z}+(z^2+\bar{z}^2-2z\bar{z})i$

$= z^2+iz^2+\bar{z}^2+i\bar{z}^2+2z\bar{z}-2z\bar{z}i$

$= z^2(1+i)+\bar{z}^2(1+i2)+2z\bar{z}(1-i)$

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40) Triangle inequility
$|z_1+z_2| \leq |z_1|+|z_2|$

Proof:-
$|z_1+z_2|^2 \leq (z_1+z_2)(\overline{z_1+z_2})$ $\because |z|^2 = z\bar{z}$

$\leq (z_1+z_2)(\bar{z_1}+\bar{z_2})$

$\leq z_1\bar{z_1} + z_1\bar{z_2} + z_2\bar{z_1} + z_2\bar{z_2}$

$\leq |z_1|^2 + z_1\bar{z_2} + \overline{z_1}z_2 + |z_2|^2$

$\leq |z_1|^2 + 2Re(z_1\bar{z_2}) + |z_2|^2$

$\leq |z_1|^2 + 2|z_1||z_2| + |z_2|^2$

$\leq (|z_1|+|z_2|)^2$

Taking square root on b/s
$|z_1+z_2| \leq |z_1|+|z_2|$

41: verify $|z_1+z_2| \leq |z_1|+|z_2|$
$z_1 = -8+i \quad z_2 = -3+8i$

Sol $z_1+z_2 = (-8+i)+(-3+8i) = -11+9i$

$|z_1+z_2| = \sqrt{(-11)^2+9^2} = \sqrt{121+81} = \sqrt{202} = 14.2$

$|z_1| = \sqrt{64+1} = \sqrt{65}$

$|z_2| = \sqrt{9+64} = \sqrt{73}$

$|z_1|+|z_2| = \sqrt{65}+\sqrt{73} = 16.60$

$|z_1+z_2| \leq |z_1|+|z_2|$

42 $\overline{z_1+z_2} = \bar{z_1} + \bar{z_2}$ for
$z_1 = 2+i \quad z_2 = -3+9i$

Sol $z_1+z_2 = (2+i)+(-3+9i) = -1+10i$

$\overline{z_1+z_2} = \boxed{-1-10i}$

$\bar{z_1} = 2-i$

$\bar{z_2} = -3-9i$

$\bar{z_1}+\bar{z_2} = (2-i)+(-3-9i)$

$= \boxed{-1-10i}$

43 Euler formula.

$e^{i\theta} = \cos\theta + i\sin\theta$

Q44 $|z|^2 = z\bar{z}$

$z = x+iy$, $|z|^2 = (\sqrt{x^2+y^2})^2 = x^2+y^2$

$z\bar{z} = (x+iy)(x-iy) = x^2+y^2$

Q45 prove $\overline{z_1+z_2} = \bar{z_1}+\bar{z_2}$

Proof $z_1 = x_1+iy_1 \quad z_2 = x_2+iy_2$

$z_1+z_2 = (x_1+x_2)+i(y_1+y_2)$

$\overline{z_1+z_2} = (x_1+x_2)-i(y_1+y_2)$

$= x_1-iy_1+x_2-iy_2$

$= \bar{z_1}+\bar{z_2}$

Q46 $\overline{z_1-z_2} = \bar{z_1}-\bar{z_2}$

Proof $z_1 = x_1+iy_1 \quad z_2 = x_2+iy_2$

$z_1-z_2 = x_1+iy_1-x_2-iy_2$

$= (x_1-x_2)+i(y_1-y_2)$

$\overline{z_1-z_2} = (x_1-x_2)-i(y_1-y_2)$

$= x_1-iy_1-(x_2-iy_2)$

$= \bar{z_1}-\bar{z_2}$

Q47 $\overline{z_1z_2} = \bar{z_1}\bar{z_2}$

Proof $z_1 = x_1+iy_1 \quad z_2 = x_2+iy_2$

$z_1z_2 = (x_1+iy_1)(x_2+iy_2)$

$= x_1x_2 + ix_1y_2 + iy_1x_2 + i^2y_1y_2$

$= x_1x_2 + i^2y_1y_2 + i(x_1y_2+y_1x_2)$

$\overline{z_1z_2} = x_1x_2 + i^2y_1y_2 - i(x_1y_2+y_1x_2)$

$= (x_1-iy_1)(x_2-iy_2)$

$= \bar{z_1}\,\bar{z_2}$

Q48 $\overline{\frac{z_1}{z_2}} = \frac{\bar{z_1}}{\bar{z_2}}$

Proof $z_1 = x_1+iy_1 \quad z_2 = x_2+iy_2$

$\frac{z_1}{z_2} = \frac{x_1+iy_1}{x_2+iy_2} \times \frac{x_2-iy_2}{x_2-iy_2}$

$= \frac{x_1x_2-i^2y_1y_2}{(x_2)^2+(y_2)^2} + i\frac{(x_2y_1-x_1y_2)}{x_2^2+y_2^2}$

$\overline{\frac{z_1}{z_2}} = \frac{x_1x_2-i^2y_1y_2}{x_2^2+y_2^2} - i\frac{(x_2y_1-x_1y_2)}{x_2^2+y_2^2}$

$= \frac{(x_1-y_1i)(x_2+y_2i)}{(x_2+y_2i)(x_2-y_2i)}$

$= \frac{x_1-y_1i}{x_2-y_2i} = \frac{\bar{z_1}}{\bar{z_2}}$

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49 Complex number

A complex number can be defined as an order pair $z=(x,y)$ where x and y are real number.

So $|z_1+z_2| \geq |z_1|-|z_2|$

Proof

$|z_1| = |z_1+z_2-z_2|$

$|z_1| \leq |z_1+z_2|+|-z_2|$

$|z_1|-|z_2| \leq |z_1+z_2|$

Inside

Q51 $Re(z)$, $Im(z)$

$z = x+iy$ $Re(z)=x$ $Im(z)=y$

$\bar{z} = x-iy$

$z+\bar{z} = 2x \Rightarrow x = \frac{z+\bar{z}}{2} \Rightarrow Re(z) = \frac{z+\bar{z}}{2}$

$z-\bar{z} = 2iy \Rightarrow y = \frac{z-\bar{z}}{2i} \Rightarrow Im z = \frac{z-\bar{z}}{2i}$

Q52 (i) $Re(iz) = -Im(z)$

(ii) $Im\,iz = Re\,z$

Proof $z = x+iy$ $Re(z)=x$ $Im z=y$

$iz = ix+i^2y = -y+ix$

$\bar{iz} = -y-ix$

(i) $Re(iz) = \frac{iz+\bar{iz}}{2} = \frac{-y+ix-y-ix}{2}$

$= -\frac{2y}{2} = -y = -Im(z)$

(ii) $Im(iz) = \frac{iz-\bar{iz}}{2i} = \frac{-y+ix+y+ix}{2i}$

$= \frac{2ix}{2i} = x = Re(z)$

Q53 If complex no z satisfy $|z|<2$ then $|z^3+3z^2-2z+1|$ By using Δ equality

Sol $|z^3+3z^2-2z+1| \leq |z^3|+|3z^2|+|-2z|+|1|$

$\leq z^3+3|z|^2+2|z|+1$

$\leq 8+12+4+1$

≤ 25

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find $f'(z) = ?$

$$f(z) = \frac{(1+z^2)^4}{z^2}, \quad z \neq 0$$

$$f'(z) = \frac{4(1+z^2)^3 2z}{2z}$$

$$f'(z) = 4(1+z^2)^3 \quad \text{Ans.}$$

$$\lim_{z \to i} \frac{z-i}{z^2+1}$$

$$= \lim_{z \to i} \frac{z-i}{z^2+1}$$

$$= \lim_{z \to i} \frac{z-i}{z^2-(-1)}$$

$$= \lim_{z \to i} \frac{z-i}{z^2-(i)^2}$$

$$= \lim_{z \to i} \frac{z-i}{(z+i)(z-i)}$$

$$= \lim_{z \to i} \frac{1}{z+i}$$

$$= \frac{1}{i+i} = \frac{1}{2i} \quad \text{Ans.}$$

$f'(z) = ?$ at $z = i$

$$f(z) = \frac{z^3}{(z+i)^3}$$

$$f'(z) = \frac{3z^2}{3(z+i)^2(1)}$$

$$f'(z) = \frac{3z^2}{3(z+i)^2}$$

$$f'(i) = \frac{3(i)^2}{3(i+i)^2}$$

$$= \frac{3(-1)}{3(2i)^2}$$

$$= \frac{-3}{3 4i^2}$$

$$= \frac{-1}{4i^2}$$

$$= \frac{-1}{4(-1)}$$

$$= \frac{+1}{+4} = \frac{1}{4}$$

Find derivative at $z = 2i$ of

$$f(z) = (iz^3 + 3z^2)^3$$

$$f(z) = (iz^3 + 3z^2)^3$$

$$f'(z) = \frac{d}{dz}(iz^3 + 3z^2)^3$$

$$= 3(iz^3 + 3z^2)^2 (3z^2 i + 6z)$$

at $z = 2i$

$$= 3\left[i(2i)^3 + 3(2i)^2\right]^2 \left[3(2i)^2 i + 6(2i)\right]$$

$$= 3\left[i(8i^3) + 3(4i^2)\right]^2 (12i^3 + 12i)$$

$$= 3(8i^4 + 12i^2)^2 (12i^3 + 12i)$$

$= 3(8-12)^2(-12i+12i)$

$= 3(16)\quad(0)$

$= 0$

(5) Evaluate $\lim_{z\to 4i} \frac{z^2+16}{z-4i}$

$\lim_{z\to 4i} \frac{z^2-(-16)}{z-4i}$

$\lim_{z\to 4i} \frac{z^2-16i}{z-4i}$ $\quad a^2-b^2$ z(a+

$i^2=-1$

$\lim_{z\to 4i} \frac{(z)^2-(4i)^2}{z-4i}$

$= \lim_{z\to 4i} \frac{(z-4i)(z+4i)}{(z-4i)}$

$= \lim_{z\to 4i} (z+4i)$

$= 4i+4i = 8i$

(6) Evaluate $(-4-4i)^3$

$z = -4-4i$

$r = \sqrt{(4)^2+(4)^2}$

$= \sqrt{16+16}$

$= \sqrt{32}$

$= 4\sqrt{2}$

both are -ive

$\theta = \tan^{-1}(y/x)$

$= \tan^{-1}(1)$

$= \frac{\pi}{4}$

(4)

$\frac{\pi}{1} + \frac{\pi}{4} = \frac{4\pi + \pi}{4} = \frac{5\pi}{4}$

$\text{Arg}(z) = \theta = \pi + \theta$

$= \pi + \frac{\pi}{4} = \frac{5\pi}{4}$

$z = -4-4i = (4\sqrt{2}\, e^{i5\pi/4})$

$(-4-4i)^3 = (4\sqrt{2}\, e^{i5\pi/4})^3$

2. Prove $\lim_{z\to\infty} \frac{4z^2}{(z-1)^2} = 4$

$= \lim_{z\to\infty} \frac{4z^2}{z^2(1-\frac{1}{z})^2}$

$= \lim_{z\to\infty} \frac{4}{(1-\frac{1}{z})^2} = \frac{4}{(1-\frac{1}{\infty})^2} = \frac{4}{(1-0)}$

Simpfy by using De-Moivre's formula.

$$\frac{(\cos 2\theta + i\sin 2\theta)^5 (\cos 3\theta - i\sin 3\theta)^6}{(\cos 4\theta - i\sin 4\theta)^7 (\cos 5\theta + i\sin 5\theta)^8}$$

by using De-Moivre's formula.

$= \frac{(e^{i2\theta})^5 (e^{-i3\theta})^6}{(e^{-i4\theta})^7 (e^{i5\theta})^8} = \frac{(e^{i\theta})^{10-18}}{(e^{i\theta})^{-28+4}}$

$= (e^{i\theta})^{-8-12} \Rightarrow (e^{i\theta})^{-20}$

$= (\cos 20\theta - i\sin 20\theta)$ Ans.

(9) Find derivative $f(z) = (1+4z)^3$

$f'(z) = 3(1+4z)^2 (4)$

$= 12(1 + 16z^2 + 8z)$

$= 12 + 182z^2 + 96z$.

(10) $f(z) = z^4 - 9z^2 + iz - 2$

check continuity at $z = i$

$f(z) = z^4 - 9z^2 + iz - 2$

$= (i)^4 - 9(i)^2 + (i)(i) - 2$

$= (i^2)(i)^2 - 9(-1) + (i^2) - 2$

$= (-1)(-1) + 9 + (-1) - 2$

$= 1 + 9 - 1 - 2 \Rightarrow 9 - 2$

$= 7$

(11) $(\sqrt{3} - i)^{1/3}$ exponential form

Solve:-

in exponential form $r = \sqrt{3+1} = 2$

$\theta = \tan^{-1}\frac{1}{\sqrt{3}} = \frac{\pi}{6}$

$z = e^{i\theta}$

$\sqrt{3} - i = 2e^{i\frac{\pi}{6}}$

$(\sqrt{3} - i)^3 = 2^3 e^{i\frac{\pi}{6}(3)}$

$= 8e^{i\frac{\pi}{2}}$

(12) Verify That:-

$(\sqrt{2} - i) - i(1 - \sqrt{2}i) = -2i$

L.H.S $= (\sqrt{2} - i) - i(1 - \sqrt{2}i)$

$= \sqrt{2} - i - i + \sqrt{2}i^2$

$= \sqrt{2} - 2i + \sqrt{2}(-1)$

$= \sqrt{2} - 2i - \sqrt{2}$

$= -2i$ Ans.

(13) If $w = 1+7i$ and $z = 3+2i$
Then find $z \cdot w = ?$

Solve:-

$z \cdot w = (3+2i)(1+7i)$

$= 3 + 21i + 2i + 14i^2$

$= 3 + 23i + 14(-1)$

$= 3 + 23i - 14$

$= -11 + 23i$

(14) $z_1 = \frac{1}{e^{i\pi/3}}$, $z_2 = \frac{1}{e^{i\pi/2}}$

$\frac{z_1}{z_2} = ?$

$\frac{z_1}{z_2} = \frac{\frac{1}{e^{i\pi/3}}}{\frac{1}{e^{i\pi/2}}} = \frac{1}{e^{i\pi/3}} \times e^{i\pi/2}$

$= e^{i\pi/3} \cdot e^{i\pi/2}$

$= e^{i(\pi/3 \cdot \pi/2)}$

$= e^{i\pi/6}$.

(15) $f(z) = (z+3i+5)^7$ $z = -3i+5$

$f'(z) = ?$

$f'(z) = 7(z+3i+5)^6$

$z = -3i+5$

$f'(z) = 7(-3i+5+3i+5)^6$

$= 7(10)^6$

(16) magnitude of $\exp(-2+3i)$

let $z = -2+3i$

$|z| = \sqrt{(-2)+(3)^2}$

$= \sqrt{4+9}$

$= \sqrt{13}$

(17) Verify

$(3,1)(3,-1)\left(\frac{1}{5}, \frac{1}{10}\right) = (2,1)$

$L.H.S = (3,1)(3,-1)\left(\frac{1}{5}, \frac{1}{10}\right)$

$= (3+i)(3-i)\left(\frac{1}{5} + \frac{1}{10}i\right)$

$= 9 - 3i + 3i - i^2\left(\frac{1}{5} + \frac{1}{10}i\right)$

$= 9 - (-1)\left(\frac{1}{5} + \frac{1}{10}i\right)$

$= 10\left(\frac{1}{5} + \frac{1}{10}i\right)$

$= 2 + i$

$= (2,1)$ R.H.S

① $(\sqrt{3}+i)^3$

$\sqrt{3}+i = 2\left[\cos\frac{\pi}{6} + i\sin\frac{\pi}{6}\right]$

$(\sqrt{3}+i)^3 = 2^3\left[\cos\frac{\pi}{6} + i\sin\frac{\pi}{6}\right]^3$

By De moviex

$= 8\left[\cos 3\frac{\pi}{6} + i\sin\frac{3\pi}{6}\right]$

$= 8\left[\cos\frac{\pi}{2} + i\sin\frac{\pi}{2}\right]$

$(\sqrt{3}+i)^3 = 8[0 + i]$

$= 8i$

②

$$\frac{(\cos 2\theta + i\sin 2\theta)^5 (\cos 3\theta - i\sin 3\theta)^6}{(\cos 4\theta - i\sin 4\theta)^7 (\cos 5\theta + i\sin 5\theta)^8}$$

$$= \frac{\left[(\cos\theta + i\sin\theta)^2\right]^5 \left[(\cos\theta - i\sin\theta)^3\right]^6}{\left[(\cos\theta - i\sin\theta)^4\right]^7 \left[(\cos\theta + i\sin\theta)^5\right]^8}$$

$$= \frac{(\cos\theta + i\sin\theta)^{10} (\cos\theta + i\sin\theta)^{18}}{(\cos\theta + i\sin\theta)^{28} (\cos\theta + i\sin\theta)^{40}}$$

$$= (\cos\theta + i\sin\theta)^{10+18-28-40}$$

$$= (\cos\theta + i\sin\theta)^{-20}$$

$$= \cos(-20\theta) + i\sin(-20\theta)$$

$$= \cos 20\theta - i\sin 20\theta$$

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(12) $$\lim_{z\to\infty} \frac{4z^2}{(z-1)^2}$$

$$= \lim_{z\to\infty} \frac{4z^2}{z^2+1-2z}$$

$$= \lim_{z\to\infty} \frac{4z^2}{z^2\left(1+\frac{1}{z^2}-\frac{2z}{z^2}\right)}$$

$$= \lim_{z\to\infty} \frac{4}{1+\frac{1}{z^2}-\frac{2z}{z^2}}$$

$$= \lim_{z\to\infty} \frac{4}{1+\frac{1}{\infty}-\frac{2\infty}{\infty^2}}$$

$$= \lim_{z\to\infty} \frac{4}{1+0-0} = 4 \text{ Ans}$$

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⑤ $(3,1)(3,-1)\left(\frac{1}{5},\frac{1}{10}\right) = (2,1)$

Sol

$= (3+i)(3-i)\left(\frac{1}{5}+\frac{1}{10}i\right)$

$= (9-3i+3i-i^2)\left(\frac{1}{5}+\frac{1}{10i}\right)$

$= (9-i^2)\left(\frac{1}{5}+\frac{1}{10i}\right)$

$= (10)\left(\frac{1}{5}+\frac{1}{10i}\right)$

$= (2+i)$

$(2,1)$ ✓

---

⑥ $f(z) = \frac{1}{\gamma^4}[\cos 4\theta + i\sin 4\theta]$

find $f'(z)$ = ?

Sol

$= \frac{1}{\gamma^4}[\cos 4\theta + i\sin 4\theta]$

$= \frac{1}{\gamma^4}[\cos\theta + i\sin\theta]^4$

$f(z) = \frac{1}{\gamma^4} e^{i4\theta}$

$f'(z) =$ No idea

---

⑦ $w = 2+4i \quad z = 4+9i$

find $w-z$ = ?

$w-z = (2+4i)-(4+9i)$

$= 2+4i-4-9i$

$= -2-5i$ ✓

---

⑧ $z = 3-5i \quad \bar{z}$ = ?

$\bar{z}$

$\bar{z} = \overline{3-5i}$

$= 3+5i$ ✓

---

⑨ $\lim_{n\to\infty} \frac{1}{n}\left[\cos n\frac{\pi}{2} + i\sin n\frac{\pi}{2}\right]$

Sol

$= \lim_{n\to\infty} \frac{1}{n}\left[\cos\frac{\pi}{2} + i\sin\frac{\pi}{2}\right]^n$

$= \frac{1}{\infty}\left[\cos\frac{\pi}{2} + i\sin\frac{\pi}{2}\right]^{\infty}$

$= 0$ ✓

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(9) Evaluate $\lim_{z \to 1+i} z^2 + 2$

Sol

(10) $\lim_{n \to \infty} z_n$ $\quad z_n = \frac{1}{n}\left(\cos\left(\frac{n\pi}{3}\right) + i\sin\right)$

Sol

$= \lim_{n \to \infty} \frac{1}{n}\left(\cos\frac{n\pi}{3} + i\sin\frac{n\pi}{3}\right)$

$= \lim_{n \to \infty} \frac{1}{n}\left(\cos\frac{\pi}{3} + i\sin\frac{\pi}{3}\right)^n$

$= \lim_{n \to \infty} \frac{1}{\infty}\left(\cos\frac{\pi}{3} + i\sin\frac{\pi}{3}\right)^{\infty}$

$= 0$ A

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Date:

$(-4-4i)^3$ $\quad a=-4$

$\quad\quad\quad

1) $(-3i)^4$ a+ib ki form me likhna

2) $\lim_{z \to \infty} \frac{z^2+2}{z^2+z-i}$

3) $\overline{z_1} - \overline{z_2}$ ≠ $z_1 = 5+i$, $z_2 = -8+3i$

4) $z = \sqrt{2}\, e^{\frac{\pi i}{4}}$ find z^{-1}

5) Evaluate $(-4-4i)^3$

6) $(-\sqrt{3}+i)^2$ in exponential form.

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Date: _____

(1) $(-3i)^4$ $\quad (0, -3)$

let $z = -3i$

$= 0 - 3i$

$r = |z| = \sqrt{(0)^2 + (-3)^2}$

$= \sqrt{9}$

$= 3$

θ lies in 4th Quadrant So

$= -\theta$

$= -\frac{\pi}{2}$

$z = r(\cos\theta + i\sin\theta)$

$-3i = 3\left(\cos\left(-\frac{\pi}{2}\right) + i\sin\left(-\frac{\pi}{2}\right)\right)$

$(-3i)^4 = 3^4\left(\cos\left(-\frac{\pi}{2}\right) + i\sin\left(-\frac{\pi}{2}\right)\right)^4$

$= 81(0 - 1)^4$

$(-3i)^4 = +81$ A

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$z_1 = 5 + i$ $\quad z_2 = -8 + 3i$

$\bar{z}_1 - \bar{z}_2$

$\bar{z}_1 = \overline{5 + i}$ $\quad \bar{z}_2 = \overline{-8 + 3i}$

$\bar{z}_1 = 5 - i$ $\quad \bar{z}_2 = -8 - 3i$

Put in

$= 5 - i - 8 + 3i$

$= -3 + 2i$ A

$$z = \sqrt{2}\, e^{\frac{\pi}{4} i}$$

$$z^{-1} = \frac{1}{z} = \frac{1}{\sqrt{2}\, e^{\frac{\pi}{4} i}}$$

$$= \frac{1}{\sqrt{2}\left(\cos\frac{\pi}{4} + i \sin\frac{\pi}{4}\right)}$$

$$z^{-1} = \frac{1}{\sqrt{2}}\left(\cos\frac{\pi}{4} - i \sin\frac{\pi}{4}\right)$$

$$= \frac{1}{\sqrt{2}}\left(\frac{\sqrt{2}}{2} - i\,\frac{\sqrt{2}}{2}\right)$$

$$z^{-1} = \frac{1}{2}(1 - i)$$

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$(-4-4i)^3$

$(-4-4i)(-4-4i)^2$

$(-4-4i)(16+16i^2-2(4)(-4i))$

$(-4-4i)(16-16-32i)$

$(-4-4i)(-32i)$

$128i + 128i^2$

$128i - 128$ A

ISS KO KARNY KA METHOD ALAG HAI
BUT SOMETHING IS BETTER THAN NOTHING

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Solve $(-\sqrt{3}+i)^2$ =

(1)

$= (-\sqrt{3})^2 + i^2 - 2(-\sqrt{3})(i)$

$= 3 + i^2 + i2\sqrt{3}$

$= 2 + 2\sqrt{3}i$

$= 2(1 + \sqrt{3}i)$

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Date: 29-Mar-2020

MTH632

Ch#01

Ex #2

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Q#01 Reduce each of these quantities to a real number:

a) $\frac{1+2i}{3-4i} + \frac{2-i}{5i}$

Solution:

$$= \frac{1+2i}{3-4i} \times \frac{3+4i}{3+4i} + \frac{2-i}{5i} \times \frac{-5i}{-5i}$$

$$= \frac{3+4i+6i+(8i^2)}{3^2-(4i)^2} + \frac{-10i+5i^2}{-25i^2}$$

$$= \frac{3+4i+6i-8}{9+16} + \frac{-10i-5}{25}$$

$$= \frac{-5+10i}{25} + \frac{-10i-5}{25}$$

$$= \frac{-5+10i-10i-5}{25}$$

$$= -\frac{10}{25}$$

$$= \boxed{-\frac{2}{5}}$$

b) $\frac{5i}{(1-i)(2-i)(3-i)}$

Solution:

$$= \frac{5i}{(2-i-2i+i^2)(3-i)}$$

$$= \frac{5i}{(2-3i-1)(3-i)}$$

$$= \frac{5i}{(1-3i)(3-i)}$$

$$= \frac{5i}{3-i-9i+3i^2}$$

$$= \frac{5i}{3-10i-3}$$

$$= \frac{5i}{-10i}$$

$$= \boxed{-1/2}$$

jathol

c) $(1-i)^4$

Solution:

$= ((1-i)^2)^2$

$= ((1-i)(1-i))^2$

$= (1-i-i+i^2)^2$

$= (1-2i-1)^2$

$= (-2i)^2$

$= 4i^2$

$= -4$

Q#2 Show That

$$\frac{1}{1/z} = z \quad (z \neq 0)$$

Solution:

$= \frac{1}{z^{-1}}$ $\quad \because \frac{1}{z} = z^{-1}$

$$= \frac{1}{z^{-1}} \cdot \frac{z}{z}$$

$$\therefore z^{-1} = \frac{1}{z}$$

$$= \frac{z}{z^{-1} \cdot z}$$

$$= \frac{z}{1}$$

$$= z \qquad (z \neq 0)$$

Q#3 Use The associative and commutative laws for multiplication to show That

$$(z_1 z_2)(z_3 z_4) = (z_1 z_3)(z_2 z_4)$$

Solution:

$$(z_1 z_2)(z_3 z_4) = z_1[z_2(z_3 z_4)]$$

$$= z_1[(z_2 z_3) z_4]$$
$$= z_1[(z_3 z_2) z_4]$$
$$= z_1[z_3 (z_2 z_4)]$$

$$(z_1 z_2)(z_3 z_4) = (z_1 z_3)(z_2 z_4)$$

Proved

Q#6 Drive the identity

$$\left(\frac{Z_1}{Z_3}\right)\left(\frac{Z_2}{Z_4}\right) = \frac{Z_1 Z_2}{Z_3 Z_4} \quad (Z_3, Z_4 \neq 0)$$

Solution:

$$\frac{Z_1 Z_2}{Z_3 Z_4} = Z_1 Z_2 \left(\frac{1}{Z_3}\right)\left(\frac{1}{Z_4}\right)$$

$$= Z_1 \left(\frac{1}{Z_3}\right)\left(\frac{1}{Z_4}\right) Z_2$$

$$\frac{Z_1 Z_2}{Z_3 Z_4} = \left(\frac{Z_1}{Z_3}\right)\left(\frac{Z_2}{Z_4}\right)$$

Proved

Q#7 Drive by cancellation law

$$\frac{Z_1 Z}{Z_2 Z} = \frac{Z_1}{Z_2} \quad (Z_2 \neq 0, Z \neq 0)$$

Solution:

$$\frac{Z_1 Z}{Z_2 Z} = \left(\frac{Z_1}{Z_2}\right)\left(\frac{Z}{Z}\right)$$

$$= \left(\frac{Z_1}{Z_2}\right) Z_1 \left(\frac{1}{Z}\right)$$

$$= \frac{Z_1}{Z_2} Z Z^{-1}$$

$$= \frac{Z_1}{Z_2} \cdot 1 \qquad \because Z Z^{-1} = 1$$

$$\frac{Z_1 Z}{Z_2 Z} = \frac{Z_1}{Z_2}$$

Proved

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29-03-2020

⇒ Mth632 (Practice Question)

Question NO:01

Find the sum of two complex numbers, $z = 2+i$, $w = -3+9i$

$$z+w = 2+i-3+9i$$
$$= 10i - 1$$

Question NO:02.

Diff $z_1 = -5+i$, $z_2 = -3+9i$

$$z_1 - z_2 = -5+i+3-9i$$
$$= -2-8i$$

Question NO:03

Product ; $z_1 = -3+2i$, $z_2 = -1+11i$

$$z_1 z_2 = (-3+2i)(-1+11i)$$
$$= +3-33i-2i+22i^2$$
$$= 3-22-35i$$
$$= -19-35i$$

Question No:04

Divide :- $z_1 = -8+i$, $z_2 = -3+8i$

$$\frac{z_1}{z_2} = \frac{-8+i}{-3+8i} \times \frac{-3-8i}{-3-8i}$$
$$= \frac{(-8+i)(-3-8i)}{9+64}$$

$$\frac{z_1}{z_2} = \frac{24 + 64i - 3i - 8i^2}{73}$$

$$= \frac{24 + 8 + 61i}{73}$$

$$= \frac{32 + 61i}{73}$$

Question NO:05

Simplify:

$$(\sqrt{3} + i3) - i(5 - \sqrt{3}i)$$

$$= \sqrt{3} + i3 - 5i + \sqrt{3}i^2$$

$$= \cancel{\sqrt{3}} + 3i - 5i - \cancel{\sqrt{3}}$$

$$= -2i$$

Question NO:06

$Re(z) = ?$ $Im(z) = ?$

$$z = -3 + 9i$$

$Re(z) = -3$, $Im(z) = 9$.

Question NO:07

$z_1 = 2 + i$, $z_2 = -3 + 9i$

Commutative law for multiplication

$z_1 z_2 = z_2 z_1$

L.H.S → $z_1 z_2 = (2+i)(-3+9i)$

$= -6 + 18i - 3i + 9i^2$

$= -6 + 15i - 9$

$= \boxed{-15 + 15i}$

R.H.S → $z_2 z_1$

$(-3 + 9i)(2 + i) = z_2 z_1$

$\boxed{z_2 z_1 = -15 + 15i}$

L.H.S = R.H.S.

Q#8 $z^{-1} = ?$

$z = -5 + i$

$z^{-} = \frac{1}{-5+i} \times \frac{-5-i}{-5-i}$

$= \frac{-5-i}{25-i^2} \Rightarrow \frac{-5-i}{25+1} \Rightarrow \frac{-5-i}{26}$ Ans--

Q#9 Write down distributive law of multiplication over addition.

$z_1(z_2 + z_3) = z_1 z_2 + z_1 z_3$

Q#10 $|z| = ?$ $z = -7 + 5i$

$|z| = \sqrt{49 + 25} \Rightarrow$

$|z| = \sqrt{74}$ Ans.

Date: __/__/____

Mon Tue Wed Thu Fri Sat

Q#11 $z_1 = 2+i$, $z_2 = -3+9i$

$$= \sqrt{(-3-2)^2 + (9-1)^2}$$

$$= \sqrt{25+64} \Rightarrow \sqrt{89}$$

$= \sqrt{(-3-2)^2 + (9-1)^2}$

$= \sqrt{25+64} \Rightarrow \sqrt{89}$

File→

Practice Question #03.

Q#1 Verify that $|z|^2 = z\bar{z}$

$z = 3+4i$

L.H.S $= |z|^2$

$|z| = \sqrt{3^2+4^2} = \sqrt{9+16} = \sqrt{25} = 5$

$|z|^2 = (5)^2 = 25$

R.H.S $z\bar{z}$

$\bar{z} = 3-4i$

$z\bar{z} = (3+4i)(3-4i)$

$= 9 - 12i + 12i - 16i^2$

$= 9 - 16(-1)$ $\quad \because i^2 = -1$

$= 9 + 16$

$z\bar{z} = 25$

L.H.S = R.H.S.

Q#2. $\overline{z_1 + z_2} = \bar{z}_1 + \bar{z}_2$

$z_1 = 2+i$, $z_2 = 3+9i$

L.H.S = $\overline{z_1 + z_2}$

$z_1 + z_2 = 2+i+3+9i$

$= 5 + 10i$

$\overline{z_1 + z_2} = 5 - 10i$

R.H.S $\Rightarrow$ $\bar{z}_1 + \bar{z}_2$

$\bar{z}_1 = 2 - i$, $\bar{z}_2 = 3 - 9i$

$\bar{z}_1 + \bar{z}_2 = 2 - i + 3 - 9i$

$\bar{z}_1 + \bar{z}_2 = 5 - 10i$

L.H.S = R.H.S

Q#3 Show that:

$Re(z) = \frac{z + \bar{z}}{2}$, $Im(z) = \frac{z - \bar{z}}{2i}$

$z = x + iy$ - (1) , $\bar{z} = x - iy$ - (2)

Now add (1) and (2) and we get Re(z)

$z = x + iy$

$\bar{z} = x - iy$

$z + \bar{z} = 2x$ - (A)

and Subtract (1) and (2)

$z = x + iy$

$-\bar{z} = -x + iy$

$z - \bar{z} = +2iy$ - (B)

From (A) and (B)

(A) → $\bar{z} + z = 2x$, (B) → $z - \bar{z} = 2iy$

NOW;

$$\boxed{x = \frac{z + \bar{z}}{2}}, \quad \boxed{y = \frac{z - \bar{z}}{2i}}$$

Proved.

Q#4:- $Re(iz) = -Im(z)$

and $Im(iz) = Re(z)$

L.H.S $Re(iz) = -Im(z)$

$Re(i(x+iy)) \Rightarrow Re(ix + i^2 y)$

$= Re(ix - y)$

$\boxed{= -y}$

NOW;

R.H.S $-Im(z) = -Im(x+iy)$

$= -(+y)$

$\boxed{= -y}$

Hence L.H.S = R.H.S

NOW $Im(iz) = Re(z)$.

$Im(iz) = Im(i(x+iy))$

$= Im(ix + i^2 y)$

$= Im(ix - y)$

Im(iz

Now;

Re(

Re(

L.H

Q#5 |z

$z_1 + z_2$

$|z_1 + z$

$|z_1 +$

NOW

$|z_1|$

$|z_1|$

$|z_1|$

$|z_1|$

Hen

Pro

$Im(iz) = x$

Now;

$Re(z) = Re(a+iy)$

$Re(z) = a$

L.H.S = R.H.S

Q#5 $|z_1+z_2| \leq |z_1| + |z_2|$

$z_1 = -8+i$, $z_2 = -3+8i$

$z_1+z_2 = -8+i-3+8i$

$= -11+9i$

$|z_1+z_2| = \sqrt{(-11)^2+9^2} = \sqrt{121+81}$

$|z_1+z_2| = \sqrt{202} = 14.212$

Now; $R.H.S = |z_1|+|z_2|$

$|z_1| = \sqrt{8^2+1^2} \Rightarrow \sqrt{64+1} = \sqrt{67}$

$|z_2| = \sqrt{9+64} = \sqrt{73}$

$|z_1|+|z_2| = \sqrt{67}+\sqrt{73}$

$|z_1|+|z_2| = 16.729$

Hence

$14.212 \leq 16.729$

Proved

Q#6 $z = -7+5i$ in polar coordinates:

$r = |z| = \sqrt{(-7)^2+(5)^2}$

$= \sqrt{49+25}$

$r = \sqrt{74}$

$\tan\theta = \frac{y}{x} \Rightarrow \theta = \tan^{-1}\left(\frac{y}{x}\right)$

$\theta = \tan^{-1}\left(\frac{5}{-7}\right)$

$\theta = \tan$ $\boxed{\theta = -35.5}$

in polar coordinates;

$z = r(\cos\theta + i\sin\theta)$

$\boxed{z = \sqrt{74}\left(\cos(-35.5)+i\sin(-35.5)\right)}$

Q#7: $z = (0, 11) = 11i$ in polar rep.

when $x = 0$ then following rule:- $\theta = \frac{\pi}{2}$ if $Im(z) > 0$

Here $Im(z) = 11 > 0$

So; $\theta = \pi/2$.

$r = \sqrt{0^2 + 121^2} = \sqrt{121}$

$r = 11$

in polar form,

$$z = 11\left[\cos\frac{\pi}{2} + i\sin\frac{\pi}{2}\right]$$

$$z = 11\left[0 + i\right]$$

$$z = 11i$$

PRACTICE QUESTIONS
(SOLUTION)
MTH632
Lecture No. 1

Q1: Find the sum of two complex numbers, $z = 2 + i$, $w = -3 + 9i$.

Solution: $z + w = (2 + i) + (-3 + 9i)$

$= 2 + (-3) + i + 9i$

$= -1 + 10i$

Q2: Find the difference of two complex numbers, $z_1 = -5 + i$, $z_2 = -3 + 9i$.

Solution: $z_1 - z_2 = (-5 + i) - (-3 + 9i)$

$= -5 - (-3) + i - 9i$

$= -5 + 3 + i - 9i$

$= -2 - 8i$

Q3: Find the product of two complex numbers, $z_1 = -3 + 2i$, $z_2 = -1 + 11i$.

Solution: $z_1 z_2 = (-3 + 2i)(-1 + 11i)$

$= 3 - 33i - 2i + 22i^2$

$= 3 - 33i - 2i + 22(-1)$

$= -19 - 35i$

Q4: Divide of two complex numbers, $z_1 = -8 + i$, $z_2 = -3 + 8i$.

Solution:

z_1 戔 8戔+ 戔 z_2 戔8 + i

$$\frac{z_1}{z_2} = \frac{8+i}{3+8i}$$

$$= \frac{8+i}{3+8i} \times \frac{3-8i}{3-8i}$$

$$= \frac{24+64i-3i-8i^2}{(3)^2-(8i)^2}$$

$$= \frac{24+64i-3i-8(-1)}{9-64(-1)}$$

$$= \frac{24+64i-3i+8}{9-64(-1)}$$

$$= \frac{32+61i}{73}$$

Q5: Simplify $(\sqrt{3}+i\sqrt{3}) - i(5-\sqrt{3}i)$

Solution:

$$
\begin{aligned}
& (\sqrt{3} + i\ 3) - i(5 - \sqrt{3}i) \\
= \ & \sqrt{3} + 3i - 5i + \sqrt{3}i^2 \\
= \ & \sqrt{3} - 2i + \sqrt{3}(-1) \\
= \ & \sqrt{3} - 2i - \sqrt{3} \\
= \ & \sqrt{3} - \sqrt{3} - 2i \\
= \ & 0 - 2i \\
= \ & -2i
\end{aligned}
$$

Q6: Find *Re(z)* and *Im(z)* complex numbers, $z = -3 + 9i$.

Solution:

$Re(z) = -3$ and $Im(z) = 9$

PRACTICE QUESTIONS

MTH632

Lecture No. 2

Modules (5 – 7)

Q1: Verify commutative law with respect to multiplication for complex numbers $z_1 = 2 + i$, $z_2 = -3 + 9i$.

Solution: The commutative law is $z_1 + z_2 = z_2 + z_1$

Consider $z_1 + z_2 = (2 + i) + (-3 + 9i)$

$= 2 - 3 + i + 9i$

$= -1 + 10i$

Now $z_2 + z_1 = (-3 + 9i) + (2 + i)$

$= -3 + 2 + 9i + i$

$= -1 + 10i$

Hence $z_1 + z_2 = z_2 + z_1$

Q2: Verify the associative law with respect to multiplication for complex numbers $z_1 = 2 + i$, $z_2 = -3 + 9i$ and $z_3 = -8 + i$.

Solution: The associative law is $(z_1 z_2) z_3 = z_1 (z_2 z_3)$

Consider $z_1 z_2 = (2 + i)(-3 + 9i)$

$= 2(-3) + 2(9i) + i(-3) + (i)(9i)$

$= -6 + 18i - 3i + 9i^2$

$= -6 + 18i - 3i + 9(-1)$

$= -6 - 9 + 18i - 3i$

$= -12 + 15i$

Now $(z_1 z_2) z_3 = (-12 + 15i)(-8 + i)$

$= -12(-8) + (-12(i)) + 15i(-8) + (15i)(i)$

$= 96 - 12i - 120i + 15i^2$

$= 96 - 12i - 120i + 15(-1)$

$= 81 - 132i$

Similarly, you can find $z_1 (z_2 z_3)$.

Q3: Find the multiplicative inverse z^{-1} of the complex numbers $z = -5 + i$.

Solution:

$z^{-1} =$

$$\frac{1}{z} = \frac{1}{-5 + i} = \frac{-5}{(-5)^2 + i^2} - \frac{1}{(-5)^2 + i^2}$$

by formula

$$= \frac{-5}{25\ -1} - i\frac{1}{25-1}$$

$$= \frac{-5}{24} - i\frac{1}{24}$$

Q4: Write down the Distributive law of multiplication over addition.

Solution: The Distributive law is $z_1 (z_2 + z_3) = z_1 z_2 + z_1 z_3$

Q5: Let $z_1 = -8 + i$, $z_2 = -3 + 8i$, then calculate $\frac{z_1}{z_2}$.

Solution:

$$\frac{z_1}{z_2} = (x_1 + iy_1)\left(\frac{x_2}{x_2^2+y_2^2} - i\frac{y_2}{x_2^2+y_2^2}\right)$$

$$= \left(\frac{x_1x_2+y_1y_2}{x_2^2+y_2^2}, \frac{y_1x_2-x_1y_2}{x_2^2+y_2^2}\right)$$

$$= \left(\frac{(-8)(-3)+(1)(8)}{(-3)^2+(8)^2}, \frac{(1)(-3)-(-8)(8)}{(-3)^2+(8)^2}\right)$$

$$= \left(\frac{32}{(-3)^2+(8)^2}, \frac{61}{(-3)^2+(8)^2}\right)$$

Q6: Find the modulus $| z |$ of the complex numbers $z = -7 + 5i$.

Solution: $| z | = | -7 + 5i | = \sqrt{7^2+5^2}$

$= \sqrt{49+25} = \sqrt{74}$

Q7: Find the distance of the complex number $z_1 = 2 + i$ from $z_2 = -3 + 9i$.

Solution: Distance

$= \sqrt{(2-(-3))^2+(1-9)^2} = \sqrt{(5)^2+(-8)^2}$

$= \sqrt{25+64} = \sqrt{89}$

Q8: Show that $|z_1 + z_2| \geq ||z_1| - |z_2||$

Proof

$|z_1| = |z_1 + z_2 - z_2| \leq |z_1 + z_2| + |-z_2|$ $\quad \because |z| = |-z|$

$\Rightarrow |z_1| - |z_2| \leq |z_1 + z_2|$ — I

Interchanging z_1 & z_2

$|z_2| - |z_1| \leq |z_1 + z_2|$

$-\alpha < x < \alpha$

$|x| < \alpha$

$\Rightarrow -(|z_1| - |z_2|) \leq |z_1 + z_2|$

$|z_1| - |z_2| \geq -|z_1 + z_2|$ — II

$-|z_1 + z_2| \leq |z_1| - |z_2| \leq |z_1 + z_2|$ — III

PRACTICE QUESTIONS

MTH632

Lecture No. 3

Modules (8 – 9)

Q1: Verify that $|z|^2 = z\bar{z}$ for $z = 3+4i$.

Solution:

$z = -3 + 4i$, so $\bar{z} = -3 + 4i = -3 - 4i$

$z\bar{z} = (-3+4i)(-3-4i)$

$= (-3)(-3) + (-3)(-4i) + (4i)(-3) + (4i)(-4i)$

$= 9 + 12i - 12i - 16\, i^2$

$= 9 - 16\,(-1)$

$z\bar{z} = 25$

Now $|z| = \sqrt{(-3)^2+(4)^2} = \sqrt{25} = 5$

$|z|^2 = 5^2 = 25$

Hence $|z|^2 = z\bar{z}$

Q2: Verify that

$\overline{z_1 + z_2} = \overline{z_1} + \overline{z_2}$ for $z_1 = 2+i$, $z_2 = -3+9i$.

Solution:

$\overline{z_1 + z_2} = (2+i) + (-3+9i)$

$= (2-3) + (i+9i)$

$= -1 + 10i$

$= -1 - 10i$

$\overline{z_1} + \overline{z_2} = (2+i) + (-3+9i)$

$= 2 - i + (-3-9i)$

$= -1 - 10i$

Hence $\overline{z_1 + z_2} = \overline{z_1} + \overline{z_2}$

Q3: Show that $Re(z) = \frac{z+\overline{z}}{2}$ and $Im(z) = \frac{z-\overline{z}}{2i}$.

Solution:

$z = x + yi$, and $\overline{z} = x - yi$

$Re(z) = x$, $Im(z) = y$

Now $z + \overline{z} = (x + yi) + (x-yi)$

$= (x + x) + (yi-yi)$

$= 2x + 0i$

$$z + \bar{z} = 2x$$

$$\frac{z + \bar{z}}{2} = x$$

$$\mathrm{Re}(z) = \frac{z + \bar{z}}{2}$$

Now $z - \bar{z} = (x + yi) - (x - yi)$

$= (x - x) + (yi + yi)$

$= 0 + 2yi$

$$z - \bar{z} = 2yi$$

$$\frac{z - \bar{z}}{2i} = y$$

$$\mathrm{Im}(z) = \frac{z - \bar{z}}{2i}$$

Q4: Show that $\mathrm{Re}(iz) = -\mathrm{Im}(z)$ and $\mathrm{Im}(iz) = \mathrm{Re}(z)$.

Solution:

Since $\mathrm{Re}(z) = \frac{z + \bar{z}}{2}$

So $\mathrm{Re}(iz) = \frac{iz + \overline{iz}}{2}$

$$= \frac{iz - i\bar{z}}{2}$$

$$= i\left(\frac{z - \bar{z}}{2}\right)$$

$$= i.i\left(\frac{z - \bar{z}}{2i}\right)$$

$Re(iz) = -Im(z)$

Since $Im(z) = \frac{z - \bar{z}}{2i}$

So $Im(iz) = \frac{iz - \overline{iz}}{2i}$

$$= \frac{iz + i\bar{z}}{2i}$$

$$= i\left(\frac{z + \bar{z}}{2i}\right)$$

$$= \left(\frac{z + \bar{z}}{2}\right)$$

$Im(iz) = Re(z)$

Q5: Verify that the triangle inequality $|z_1 + z_2| \leq |z_1| + |z_2|$ for $z_1 = -8 + i$, $z_2 = -3 + 8i$.

Solution:

$z_1 = -8 + i, \quad |z_1| = \sqrt{(-8)^2+1^2} = \sqrt{65} = 8.06$

$z_2 = -3 + 8i, \quad |z_2| = \sqrt{(-3)^2+8^2} = \sqrt{73} = 8.5$

$z_1 + z_2 = (-8 + i) + (-3 + 8i) = -11 + 9i$

$|z_1+z_2| = \sqrt{(-11)^2+9^2} = \sqrt{202} = 14.12$

Clearly $|z_1+z_2| < |z_1| + |z_2|$

14.12 < 8.06+8.5

14.12 < 16.56

Q6: Find the Polar representation of the complex numbers $z = -7 + 5i$.

Solution:

We represent $z = (x, y)$ in (r, θ)in polar form.

For $z = x + iy$, $r = |z| = \sqrt{(-7)^2+5^2} = \sqrt{74}$

$\theta = \tan^{-1}\left(\frac{y}{x}\right) = \tan^{-1}\left(\frac{5}{-7}\right) = -\tan^{-1}\left(\frac{5}{7}\right)$

$z = r(\cos\theta + i\sin\theta)$

$$z = \sqrt{74}\left(\cos\left(-\tan^{-1}\left(\frac{5}{7}\right)\right)+i\sin\left(-\tan^{-1}\left(\frac{5}{7}\right)\right)\right)$$

Q7: Find the Polar representation of the complex numbers $z = (0, 11) = 11i$.

Solution:

We represent $z = (x, y)$ in (r, θ)in polar form.

For $z = x + iy$, $r = |z| = \sqrt{(0)^2+11^2} = \sqrt{121} = 11$

$\theta = \tan^{-1}\left(\frac{y}{x}\right) = \tan^{-1}\left(\frac{11}{0}\right) = \infty$, so

$\theta = \frac{\pi}{4}$

$z = r(\cos\theta + i\sin\theta)$

$z = 11\left(\cos\left(\frac{\pi}{4}\right)+i\sin\left(\frac{\pi}{4}\right)\right)$

PRACTICE QUESTIONS

MTH632

Lecture No. 4

Modules (10 – 12)

Q1: Express the complex number $-1+i$ in the exponential form.

Solution:

We represent z = (x, y) in (r, θ)in polar form which is further converted to exponential form $z = re^{-i\theta}$.

For z = x + i y, z = -1 + i, so x = -1, y = 1

$$r = |z| = \sqrt{(-1)^2+1^2} = \sqrt{2}$$

Since x is −ve and y is +ve, so θ is in 2nd quadrant.

$$\theta = \tan^{-1}\left(\frac{y}{x}\right) = \tan^{-1}\left(\frac{-1}{1}\right) = -\tan^{-1}(1) = -\frac{\pi}{4}$$

$$= \pi - \frac{\pi}{4} = \frac{3\pi}{4}$$

$$z = re^{-i\theta}$$

$$z = \sqrt{2}e^{-i\frac{3\pi}{4}}$$

Q2: Let $z=8\,e^{\frac{\pi i}{2}}$, then find z^{-1} .

Solution:

$$z^{-1} = \frac{1}{re^{i\theta}} = \frac{1}{8e^{i\frac{\pi}{2}}} = \frac{1}{8}\,e^{-i\frac{\pi}{2}}$$

Q3: Let $z=7\,e^{\frac{i\pi}{4}}$, then find z^{50}.

Solution:

$$z^n = r^n e^{in\theta}$$

Here $n = 50,\ r = 7,\ \theta = \frac{\pi}{4}$

$$z^{50} = 7^{50}e^{i50\left(\frac{\pi}{4}\right)} = 7^{50}e^{i\left(\frac{25\pi}{2}\right)}$$

Q4: Use De Moivres' formula to simplify $\left[\cos\left(\frac{\pi}{3}\right)+i\sin\left(\frac{\pi}{3}\right)\right]^{15}$.

Solution:

By De Moivres' formula, $(\cos\theta + i\sin\theta)^n = (\cos n\theta + i\sin n\theta)$

So, $\left(\cos\frac{\pi}{4} + i\sin\frac{\pi}{4}\right)^{15} = \left(\cos 15\left(\frac{\pi}{4}\right) + i\sin 15\left(\frac{\pi}{4}\right)\right)$

Q5: Let $z_1 = \sqrt{3}\, e^{\frac{5\pi i}{3}}$ *and* $z_2 = 2\, e^{\frac{\pi i}{3}}$, then find Arg ($z_1 + z_2$).

Solution:

Arg ($z_1 + z_2$) =?

$\text{Arg}(z_1) = \theta_1$, $\text{Arg}(z_2) = \theta_2$

$\text{Arg}(z_1+z_2) = \theta_1 + \theta_2$

Here,

$$\text{Arg}(z_1) = \text{Arg}\left(\sqrt{3}\, e^{i\frac{5\pi}{3}}\right) = \frac{5\pi}{3}$$

$$\text{Arg}(z_2) = \text{Arg}\left(2\, e^{i\frac{\pi}{3}}\right) = \frac{\pi}{3}$$

So

$$\text{Arg}(z_1+z_2) = \theta_1 + \theta_2 = \frac{5\pi}{3} + \frac{\pi}{3} =$$

$$\frac{6\pi}{3} = 2\pi$$

PRACTICE QUESTIONS

MTH632

Lecture No. 5

Modules (13 – 14)

Q1: Find square roots of $2i$.

Solution:

Please follow the solved example at **page 12** of **Module 14.**

Q2: Write down the formula of nth roots of complex number z.

Solution:

Please follow the solved example at **page 13** of **Module 13.**

Q3: Find 6th roots of $2i$.

Solution:

Please follow the solved example at **page 12** of **Module 14.**

Q4: Find cube roots of 1.

Solution:

Please follow the solved example at

page 3 of **Module 14.**

Q5: Find all values of $(3i)^{1/3}$.

Solution:

Please follow the solved example at **page 12** of **Module 14.**

PRACTICE QUESTIONS

MTH632

Lecture No. 6

Modules 14-15

Q1: Find 7th roots of $z_0 = 1$.

Solution:

Please follow the solved example at **page 10** of **Module 14.**

Q2: Find all values of $(5i)^{1/3}$.

Solution:

Please follow the solved example at **page 12** of **Module 14.**

Q3: Find roots of the following equation

$$z^2 - (3 + i)z + \frac{1}{4} = 0$$

Solution:

Please follow the solved example at **page 14** of **Module 14.**

PRACTICE QUESTIONS

MTH632

Lecture No. 8

Modules 18-19

Q1: Explain domain and range of a complex valued function with the help of an example.

Solution:

Please follow the solved example at **pages 6, 11 and 12** of **Module 18.**

Q2: Express the function $f(z) = \bar{z}\,\mathrm{Im}(z) + z\,\mathrm{Re}(z)$ in the form $f(z) = u(x, y) + i\,v(x, y)$.

Solution:

Please follow the solved example at **pages 5-8** of **Module 19.**

Q3: Express the function $f(z) = 8(x + y) + i\,4y^2$ in the formula, involving z and $\bar{z}$.

Solution:

Please follow the solved example at **pages 9-12** of **Module 19.**

Q4: Define polynomial and give reason why

$f(z) = 1 + z + z^2 + z^3 + z^4 + \cdots$ is NOT a polynomial.

Solution:

Please follow the solved example at **pages 28 and 31** of **Module 19.**

PRACTICE QUESTIONS

MTH632

Lecture No. 9

Modules 20

Q1: Show that the function $w = f(z) = iz - 3$ is one-to-one.

Solution:

Please follow the solved example at **pages 18 and 19** of **Module 20.**

Q2: Show whether the function $w = f(z) = z^2$ is one-to-one or not.

Solution:

Please follow the solved example at **pages 21 and 22** of **Module 20.**

Q3: Show that the function $f(z) = -z$ maps the line $y = -x + 1$ to the line $v = -u - 1$.

Solution:

Please follow the solved example at **pages 30-36** of **Module 20.**

SOLVED EXERCISE MTH-632

Complex Analysis & Differential Geometry

Instructor

Dr. Sohail Iqbal

Midterm's Syllabus Solved Exercises

Book Name: Brown-Churchill-Complex Variables and Application 8th edition

Chapter 1 Section 2,3,4,8,10,

Chapter 2 Section 12,18,20,**23**,25,26

Section 2 ***Basic Algebraic Operations***

Question No.1: Verify that

a): $\left(\sqrt{2}-i\right)-i\left(1-\sqrt{2}\right)=-2i$

proof:

$$\begin{aligned} L.H.S &= \left(\sqrt{2}-i\right)-i\left(1-\sqrt{2}i\right) \\ &= \sqrt{2}-i-i+\sqrt{2}i^2 \\ &= \sqrt{2}-2i+\sqrt{2}\left(-1\right) \\ &= \sqrt{2}-2i-\sqrt{2} \\ &= -2i = R.HS \end{aligned}$$

b):

$$
\begin{aligned}
L.H.S &= (2,-3)(-2,1) \\
&= (2-3i)(-2+i) \\
&= 2(-2+i)-3i(-2+i) \\
&= -4+2i+6i-3i^2 \\
&= -4+8i-3(-1) \\
&= -4+8i+3 \\
&= -1+8i = R.HS
\end{aligned}
$$

c):

$$
\begin{aligned}
L.H.S &= (3,1)(3,-1)\left(\frac{1}{5},\frac{1}{10}\right) \\
&= (3+i)(3-i)\left(\frac{1}{5}+\frac{1}{10}i\right) \\
&= \left(3(3-i)+i(3-i)\right)\left(\frac{1}{5}+\frac{1}{10}i\right) \\
&= \left(9-3i+3i-i^2\right)\left(\frac{1}{5}+\frac{1}{10}i\right) \\
&= \left(9-(-1)\right)\left(\frac{1}{5}+\frac{1}{10}i\right) \\
&= 10\left(\frac{1}{5}+\frac{1}{10}i\right) \\
&= 2+i \\
&= (2,1) = R.HS
\end{aligned}
$$

Question No.2:

a): Show that *Re(iz)=-ImgZ;*

Proof:

$$
\begin{aligned}
L.H.S &= \mathrm{Re}(iz) \\
&= \mathrm{Re}\left(i\left(x+iy\right)\right) \\
&= \mathrm{Re}\left(ix+i^2y\right) \\
&= \mathrm{Re}(ix-y) \\
&= -y \\
&= -\mathrm{Im}(z) = R.H.S
\end{aligned}
$$

b): Show that Img(iz)=Re z;

Proof:

$$\begin{aligned} L.H.S &= \mathrm{Im}(iz) \\ &= \mathrm{Im}\left(i\left(x+iy\right)\right) \\ &= \mathrm{Im}\left(ix+i^2y\right) \\ &= \mathrm{Im}(ix-y) \\ &= \mathrm{Re}(z) = R.H.S \end{aligned}$$

Question No.3: Show that $(1+z)^2 = 1+2z+z^2$

Ans:

Question No.4: verify that each of the two numbers $z = 1 \pm i$ satisfies the equation $z^2 - 2z + 2 = 0$

Proof:

a): when $z = 1+i$ then

$$\begin{aligned} L.H.S &= z^2 - 2z + 2 \\ &= (1+i)^2 - 2(1+i) + 2 \\ &= (1)^2 + (i)^2 + 2(1)(i) - 2 - 2i + 2 \\ &= 1 + (-1) + 2i - 2 - 2i + 2 \\ &= 0 = R.H.S \end{aligned}$$

b): when $z = 1-i$ then

$$\begin{aligned} L.H.S &= z^2 - 2z + 2 \\ &= (1-i)^2 - 2(1-i) + 2 \\ &= (1)^2 + (i)^2 - 2(1)(i) - 2 + 2i + 2 \\ &= 1 + (-1) - 2i - 2 + 2i + 2 \\ &= 0 = R.H.S \end{aligned}$$

Question No.5: Prove that multiplication of complex numbers is commutative.

Proof:

Let z_1 and z_2 be two complex numbers such that $z_1 = x_1 + iy_1$ *and* $z_2 = x_2 + iy_2$ where $x_1, x_2, y_1, y_2 \in \mathbb{R}$

$$\begin{aligned} z_1 z_2 &= (x_1 + iy_1)(x_2 + iy_2) \\ &= (x_1)(x_2 + iy_2) + (iy_1)(x_2 + iy_2) \\ &= (x_1)(x_2) + (x_1)(iy_2) + (iy_1)(x_2) + (iy_1)(iy_2) \\ &= (x_1)(x_2) + i(x_1)(y_2) + i(y_1)(x_2) + (i)(i)(y_1)(y_2) \\ &= (x_2)(x_1) + i(y_2)(x_1) + i(x_2)(y_1) + (i)(i)(y_2)(y_1) \end{aligned}$$

As real numbers are commutative

$$\begin{aligned} &= (x_2)(x_1) + i(x_2)(y_1) + i(y_2)(x_1) + (i)(i)(y_2)(y_1) \\ &= (x_2)(x_1 + iy_1) + (iy_2)(x_1 + iy_1) \\ &= (x_2 + iy_2).(x_1 + iy_1) \\ &= z_2 z_1 \end{aligned}$$

Hence complex numbers are commutative.

Question No.6: Verify the associative law for addition of complex numbers

Proof:

Let z_1, z_2 and z_3 be three complex numbers such that

$z_1 = x_1 + iy_1$, $z_2 = x_2 + iy_2$ *and* $z_3 = x_3 + iy_3$ where $x_1, x_2, x_3, y_1, y_2, y_3 \in \mathbb{R}$

$$\begin{aligned} (z_1 + z_2) + z_3 &= \big((x_1 + iy_1) + (x_2 + iy_2)\big) + (x_3 + iy_3) \\ &= \big((x_1 + x_2) + i(y_1 + y_2)\big) + (x_3 + iy_3) \\ &= \big((x_2 + x_1) + i(y_2 + y_1)\big) + (x_3 + iy_3) \end{aligned}$$

As real numbers are commutative

$$= (x_1 + x_2) + x_3 + i(y_1 + y_2) + iy_3$$
$$= x_1 + (x_2 + x_3) + i((y_1 + y_2) + y_3)$$
$$= x_1 + (x_2 + x_3) + i(y_1 + (y_2 + y_3))$$
$$= x_1 + (x_2 + x_3) + iy_1 + i(y_2 + y_3)$$
$$= x_1 + iy_1 + (x_2 + x_3) + i(y_2 + y_3)$$
$$= z_1 + (z_2 + z_3)$$

Question No.7:

Ans:

Question No.8:

a): Write $(x, y) + (u, v) = (x, y)$ and point out how it follows that the complex number 0=(0,0) is unique as an additive identity.

Ans:

$$(x, y) + (u, v) = (x, y)$$
$$(x + iy) + (u + iv) = (x + iy)$$
$$(x + u) + i(y + v) = (x + iy)$$

Comparing the real and imaginary numbers on both sides,

$$x + u = x \ \ and \ \ y + v = y$$

$$\Rightarrow \quad x + u = x \ \ and \ \ y + v = y$$
$$u = x - x \ \ and \ \ v = y - y$$
$$u = 0 \quad and \ \ v = 0$$

Hence $(u, v) = (0, 0) = 0$

Suppose there is another additive $(c + id)$ identity then

$$(x, y) + (c, d) = (x, y)$$
$$(x + iy) + (c + i) = (x + iy)$$
$$(x + c) + i(y + d) = (x + iy)$$

$$\Rightarrow \quad x + c = x \ \ and \ \ y + d = y$$
$$c = x - x \ \ and \ \ d = y - y$$
$$c = 0 \quad and \ \ d = 0$$

Hence $(c,d)=(0,0)=0=(u,v)$ proved that additive identity is a unique number.

Question No.9: use -1=(-1,0) and z=(x,y) to show that (-1)z=-z

Ans:

Given that $-1=(-1,0)=-1+i0$ and $z=(x,y)=x+iy$

$$\begin{aligned} L.H.S &= (-1)z \\ &= (-1,0)(x,y) \\ &= (-1+i0)(x+iy) \\ &= (-1)(x+iy)+i0(x+iy) \\ &= -x-iy+i0-0 \\ &= -(x+iy) \\ &= -z = R.H.S \end{aligned}$$

Question No.10: use $i=(0,1)$ *and* $y=(y,0)$ to verify that –(iy)=(-i)y. Thus show that additive inverse of a complex number z=x+iy can be written –z=-x-iy without ambiguity.

Proof:

$$i=(0,1) \text{ and } y=(y,0)$$

$$\begin{aligned} L.H.S &= -(iy) \\ &= -\big((0,1)(y,0)\big) \\ &= -\big((0+i)(y+i0)\big) \\ &= -\big((0)(y+i0)+(i)(y+i0)\big) \\ &= -\big(0+i0+iy-0\big) \\ &= -\big((0-0)+(i)(y+0)\big) \\ &= (-i)y = R.H.S \end{aligned}$$

Additive inverse gives the zero number in complex numbers

If $z=x+iy$ then let $w=u+iv$ be the additive inverse of the z , so

$$\begin{aligned} z+w &= 0+i0 \\ (x+iy)+(u+iv) &= 0+i0 \\ (x+u)+i(y+v) &= 0+i0 \end{aligned}$$

Comparing real and imaginary parts

$$x+u=0 \quad and \quad y+v=0$$
$$u=-x \quad and \quad v=-y$$

As

$$w=u+iv$$
$$=-x-iy$$
$$=-(x+iy)$$
$$=-z$$

Hence additive inverse of a complex number z=x+iy can be written –z=-x-iy

Question No.11:solve the equation $z^2+z+1=0$ *for* $z=(x,y)$ by writing

$$(x,y)(x,y)+(x,y)+(1,0)=(0,0)$$

Ans:

$$z^2+z+1=(x,y)(x,y)+(x,y)+(1,0)$$
$$=(x^2-y^2,yx+xy)+(x,y)+(1,0)$$
$$=(x^2-y^2,yx+xy)+(x,y)+(1,0)$$
$$=(x^2-y^2+x+1,yx+xy+y+0)$$

Given $z^2+z+1=0$

Comparing real and imaginary parts

$$(x^2-y^2+x+1,yx+xy+y+0)=(0,0)$$

The real part is $x^2-y^2+x+1=0$ …..……..1 and

The imaginary part is $yx+xy+y+0=0$ ……..…2

Solving equation no. 2

$$2xy+y=0$$
$$y(2x+1)=0$$
$$y=0 \quad or \quad 2x+1=0$$
$$x=\frac{-1}{2}$$

When y=0 then this equation does not satisfied.

Put this value of x in equation 1

$$\left(\frac{-1}{2}\right)^2 - y^2 + \left(\frac{-1}{2}\right) + 1 = 0$$

$$\frac{1}{4} - y^2 - \frac{1}{2} + 1 = 0$$

$$\frac{1-2+4}{4} - y^2 = 0$$

$$\frac{3}{4} - y^2 = 0$$

$$\sqrt{y^2} = \sqrt{\frac{3}{4}}$$

$$y = \pm\frac{\sqrt{3}}{4}$$

Hence $x = \frac{-1}{2}$ and $y = \pm\frac{\sqrt{3}}{4}$

$$z = (x, y) = \left(-\frac{1}{2}, \pm\frac{\sqrt{3}}{4}\right)$$

When y=0 then this equation does not satisfied.

Complex Numbers: Trig Identities: 1

De Moivre's Theorem states that for whole number n,

$$(\cos\theta + i\sin\theta)^n = \cos n\theta + i\sin n\theta.$$

We can use this fact to derive certain trig identites: an example of a use of complex numbers to do real calculations that would oth we express $\cos 5 \theta$ in terms of powers of $\cos \theta. $

$$\begin{aligned}\cos 5\theta + i\sin 5\theta &= (\cos\theta + i\sin\theta)^5 \\ &= \cos^5\theta + 5i\cos^4\theta\sin\theta + 10i^2\cos^3\theta\sin^2\theta + \\ &\quad 10i^3\cos^2\theta\sin^3\theta + 5i^4\cos\theta\sin^4\theta + i^5\sin^5\theta \\ &= [\cos^5\theta - 10\cos^3\theta\sin^2\theta + 5\cos\theta\sin^4\theta] + \\ &\quad i[5\cos^4\theta\sin\theta - 10\cos^2\theta\sin^3\theta + \sin^5\theta].\end{aligned}$$

Comparing real parts,

$$\cos 5\theta = \cos^5\theta - 10\cos^3\theta\sin^2\theta + 5\cos\theta\sin^4\theta.$$

Using the substitution $\sin^2\theta = 1 - \cos^2\theta$,

$$\begin{aligned}\cos 5\theta &= \cos^5\theta - 10\cos^3\theta(1 - \cos^2\theta) + 5\cos\theta(1 - \cos^2\theta)^2 \\ &= 16\cos^5\theta - 20\cos^3\theta + 5\cos\theta.\end{aligned}$$

Section No. 3

Solved Exercise

1. Reduce each of these quantities to a real number.

$$\frac{1+2i}{3-4i}+\frac{2-i}{5i}=\left(\frac{1+2i}{3-4i}\right)\cdot\left(\frac{3+4i}{3+4i}\right)+\frac{(2-i)i}{(5i)i}$$

$$=\frac{3+4i+6i-8}{(3)^2-(4i)^2}+\frac{2i-i^2}{5i^2}$$

$$=\frac{-5+10i}{9-16(-1)}+\frac{2i-(-1)}{5(-1)}$$

$$=\frac{-5+10i}{25}-\frac{1+2i}{5}$$

$$=\frac{(-5+10i)-5(1+2i)}{25}$$

$$=\frac{-5+10i-5-10i}{25}=\frac{-10}{25}=-\frac{2}{5}$$

b):

$$\frac{5i}{(1-i)(2-i)(3-i)}=\frac{5i(1+i)(2+i)}{\left((1)^2-(i)^2\right)\left((2)^2-(i)^2\right)(3-i)}$$

$$=\frac{5i(2+i+2i+i^2)}{\left(1-(-1)\right)\left(4-(-1)\right)(3-i)}$$

$$=\frac{5i(2+3i-1)}{\left(1+1\right)\left(4+1\right)(3-i)}$$

$$=\frac{5i(1+3i)}{10(3-i)}$$

$$=\frac{i+3i^2}{2(3-i)}$$

$$=\frac{i-3}{2(3-i)}$$

$$=\frac{-(3-i)}{2(3-i)}=-\frac{1}{2}$$

c): $(1-i)^4$ =?

$$(1-i)^2 = (1)^2 + (i)^2 - 2(1)(i)$$
$$= 1\text{-}1\text{-}2i = 2i$$
$$(1-i)^4 = (1-i)^2(1-i)^2$$
$$= (2i)(2i)$$
$$= 4i^2 = 4(-1) = -4$$

2. Show that

$$L.H.S = \frac{1}{1/z} = \frac{1}{z^{-1}} \qquad \because \frac{1}{z} = z^{-1}$$
$$= \frac{1}{z^{-1}} \cdot \frac{z}{z} = \frac{1.z}{z^{-1}.z}$$
$$= \frac{z}{1} = z = R.H.S$$

3. use the associative and commutative laws for multiplication to show that

$$L.H.S = (z_1 z_2)(z_3 z_4)$$
$$= (z_2 z_1)(z_3 z_4) \qquad \because (z_1 z_2) = (z_2 z_1) \quad \textit{commutative law}$$
$$= (z_2)(z_1 z_3)(z_4) \qquad \because (z_1)(z_3 z_4) = (z_1 z_3)(z_4) \quad \textit{associative law}$$
$$= (z_1 z_3)(z_2)(z_4) \qquad \because (z_2)(z_1 z_3) = (z_1 z_3)(z_2) \quad \textit{commutative law}$$
$$= (z_1 z_3)(z_2 z_4) \qquad \because (z_2)(z_4) = (z_2 z_4)$$
$$= R.H.S$$

5. Drive expression

$$\frac{z_1}{z_2} = \frac{x_1 x_2 - y_1 y_2}{x_2^2 + y_2^2} + i\frac{y_1 x_2 - x_1 y_2}{x_2^2 + y_2^2} \quad (z_2 \neq 0)$$

Let $z_1 = x_1 + iy_1$ and $z_2 = x_2 + iy_2$ where $(z_2 \neq 0)$

$$L.H.S = \frac{z_1}{z_2}$$

$$= \frac{x_1 + iy_1}{x_2 + iy_2} \cdot \frac{x_2 - iy_2}{x_2 - iy_2}$$

$$= \frac{x_1x_2 + i^2y_1y_2 + iy_1x_2 - ix_1y_2}{(x_2 + iy_2)(x_2 - iy_2)}$$

$$= \frac{x_1x_2 + (-1)y_1y_2 + i(y_1x_2 - x_1y_2)}{(x_2)^2 - (iy_2)^2}$$

$$= \frac{x_1x_2 - y_1y_2 + i(y_1x_2 - x_1y_2)}{x_2^{\,2} - (-1)\,y_2^{\,2}}$$

$$= \frac{(x_1x_2 - y_1y_2) + i(y_1x_2 - x_1y_2)}{x_2^{\,2} + y_2^{\,2}}$$

$$= \frac{x_1x_2 - y_1y_2}{x_2^2 + y_2^2} + i\frac{y_1x_2 - x_1y_2}{x_2^2 + y_2^2} = R.H.S$$

6. Drive the identity

$$\left(\frac{z_1}{z_3}\right)\left(\frac{z_2}{z_4}\right) = \frac{z_1z_2}{z_3z_4}$$

$$L.H.S = \left(\frac{z_1}{z_3}\right)\left(\frac{z_2}{z_4}\right)$$

$$= (z_1)\left(\frac{1}{z_3}\right)(z_2)\left(\frac{1}{z_4}\right) \qquad \because \left(\frac{z_1}{z_2}\right) = (z_1)\left(\frac{1}{z_2}\right)$$

$$= (z_1)\left(z_3^{-1}\right)(z_2)\left(z_4^{-1}\right) \qquad \because \left(z_3^{-1}\right)(z_2) = (z_2)\left(z_3^{-1}\right) \text{ commutative law}$$

$$= (z_1)(z_2)\left(z_3^{-1}\right)\left(z_4^{-1}\right)$$

$$= (z_1z_2)(z_3)^{-1}(z_4)^{-1}$$

$$= (z_1z_2)(z_3)^{-1}(z_4)^{-1}$$

$$= (z_1z_2)(z_3z_4)^{-1}$$

$$= \frac{z_1z_2}{z_3z_4} = R.H.S$$

7. Use the identity to drive the cancellation law,

$$\frac{z_1 z}{z_2 z} = \frac{z_1}{z_2}$$

$$L.H.S. = \frac{z_1 z}{z_2 z}$$

$$= \frac{z_1 z}{z_2 z} \cdot \frac{z^{-1}}{z^{-1}}$$ multiplying numerator and denomerator by z^{-1}

$$= \frac{z_1}{z_2} \frac{z z^{-1}}{z z^{-1}}$$ associative law

$$= \frac{z_1}{z_2} \frac{1}{1}$$ $\because z z^{-1} = 1$

$$= \frac{z_1}{z_2} = R.H.S$$

Section 8

Exponential form

1. Find the principal argument Arg z when

a): $z = \dfrac{i}{-2-2i}$

Answer:

Definition: (principle value of arg of z):-

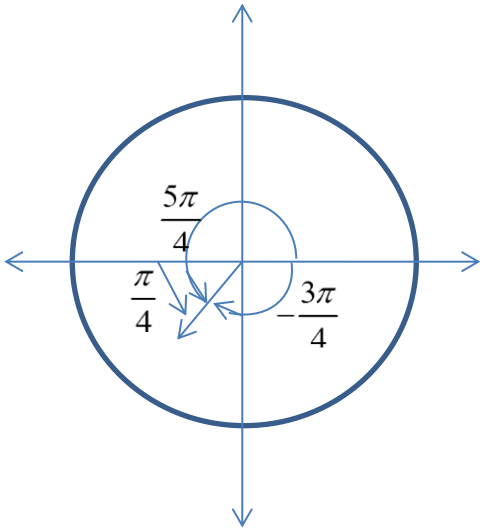
Let $z \neq 0$ be a complex number. Then principle value of argument of z, denoted by $Arg\ z$ is a unique value of θ such that

$$z = r(\cos\theta + i\sin\theta) \qquad -\pi < \theta \leq \pi$$

Principle value of argument of z is also referred as **"The argument of z"**

$$\arg z = \text{Arg}\, z + 2n\pi \qquad n = 0, \pm 1, \pm 2, \pm 3, -----$$

$$z = \frac{i}{-2-2i}$$
$$= \frac{i}{-2(1+i)} \cdot \frac{(1-i)}{(1-i)}$$
$$= \frac{i-i^2}{-2\left((1)^2-(i)^2\right)}$$
$$= \frac{i-(-1)}{-2\left(1-(-1)\right)}$$
$$= \frac{1+i}{-2\left(1+1\right)}$$
$$= \frac{1+i}{-4} = -\frac{1}{4} - i\frac{1}{4}$$



$$z = x + iy = -\frac{1}{4} - i\frac{1}{4}$$
$$x = -\frac{1}{4} \quad , \quad y = -\frac{1}{4}$$
$$\theta = \tan^{-1}\left(\frac{y}{x}\right) = \tan^{-1}\left(\frac{-1/4}{-1/4}\right) = \tan^{-1}(1) = \frac{\pi}{4}$$

As x and y are negative so angle lies in the 3rd quadrant, so we add π to $\frac{\pi}{4}$.

$$Arg\ z = \pi + \frac{\pi}{4} = \frac{5\pi}{4}$$

this angle in anti-clockwise direction.

For clockwise direction from positive axis the angle is

$$Arg\ z = \frac{5\pi}{4} - 2\pi = -\frac{3\pi}{4}$$

See example *z=-1-i* on page 23 in handouts of MTH 632 written by Fiddling writer.

Second method.

$$z = \frac{i}{-2-2i}$$

$Arg\ z = Arg\ (i) - Arg(-2 - 2i)$

For $Arg(i) = (0+i)$

If $z \neq 0$ *and* $x = 0$, then we use the following rule:

$$Arg\ z = \frac{\pi}{2} \qquad If \quad \text{Im}\ z > 0$$

$$Arg\ z = -\frac{\pi}{2} \qquad If \quad \text{Im}\, z < 0$$

So, for $Arg(i) = (0+i) = \frac{\pi}{2}$

For Arg(-2-2i)=$\tan^{-1}\left(\frac{-2}{-2}\right) = \tan^{-1}(1) = \frac{\pi}{4}$

As x and y are negative so angle lies in the 3rd quadrant, so we add π to $\frac{\pi}{4}$.

$$Arg(-2 - 2i) = \pi + \frac{\pi}{4} = \frac{5\pi}{4}$$

$$Arg\ z = Arg\ (i) - Arg(-2 - 2i)$$

$$Arg\ z = \frac{\pi}{2} - \frac{5\pi}{4}$$

$$Arg\ z = \frac{2\pi - 5\pi}{4} = -\frac{3\pi}{4}$$

b): $z = (\sqrt{3} - i)^6$

$$r = |z| = \sqrt{x^2 + y^2}$$

$$= \sqrt{\left(\sqrt{3}\right)^2 + (-1)^2} = \sqrt{3+1} = 2$$

θ=$\tan^{-1}\left(\frac{1}{\sqrt{3}}\right) = \frac{\pi}{6}$

To write in rectangular form.

$$(\sqrt{3}-i)=re^{i\theta}=2e^{i\frac{\pi}{6}}$$

$$z=(\sqrt{3}-i)^6=\left(2e^{i\frac{\pi}{6}}\right)^6$$

$$=2^6e^{i\left(\frac{\pi}{6}\right)6}=64e^{i\pi}$$

$$\theta=\pi$$

Check.(For principal angle we subtract angle $\frac{\pi}{6}$ from π and here we just find the angle so we do not subtract.)

2. show that $\left|e^{i\theta}\right|=1$

Solution:

$$L.H.S=\left|e^{i\theta}\right|=\left|\cos\theta+i\sin\theta\right|$$
$$=\sqrt{(\cos\theta)^2+(\sin\theta)^2}$$
$$=\sqrt{\cos^2\theta+\sin^2\theta}=\sqrt{1}$$
$$=1=R.H.S$$

show that $\overline{e^{i\theta}}=e^{-i\theta}$

Solution:

$$L.H.S=\overline{e^{i\theta}}=\overline{\cos\theta+i\sin\theta}$$
$$=\cos\theta-i\sin\theta=\cos\theta+(-i)\sin\theta$$
$$=e^{-i\theta}=R.H.S$$

4. show that $\left|e^{i\theta}-1\right|=2$

Solution:

$$L.H.S=\left|e^{i\theta}-1\right|$$
$$=\left|\cos\theta+i\sin\theta-1\right| \qquad 0\le\theta\le 2\pi$$
$$when \quad \theta=\pi$$
$$=\left|\cos\pi+i\sin\pi-1\right|$$
$$=\left|-1+0-1\right|$$
$$=\left|-2\right|=2=R.H.S$$

5. By writing the individual factors on the left in exponential form, performing the needed operations and changing back to rectangular coordinates, show that

a): $i(1-\sqrt{3})(\sqrt{3}+i)=2(1+\sqrt{3}i)$

Solution:

Exponential form are

$$i=e^{i\frac{\pi}{2}}=\cos\frac{\pi}{2}+i\sin\frac{\pi}{2}=i$$
$$(1-\sqrt{3}i)=2\cos\frac{\pi}{3}-2i\sin\frac{\pi}{3}=2e^{-i\frac{\pi}{3}}$$
$$(\sqrt{3}+i)=2\cos\frac{\pi}{6}+2i\sin\frac{\pi}{6}=2e^{i\frac{\pi}{6}}$$

$$L.H.S=i(1-\sqrt{3}i)(\sqrt{3}+i)$$
$$=e^{i\frac{\pi}{2}}\left(2e^{-i\frac{\pi}{3}}\right)\left(2e^{i\frac{\pi}{6}}\right)$$
$$=4e^{i\frac{\pi}{2}-i\frac{\pi}{3}+i\frac{\pi}{6}}$$
$$=4e^{i\frac{3\pi-2\pi+\pi}{6}}$$
$$=4e^{i\frac{\pi}{3}}$$
$$=2(2e^{i\frac{\pi}{3}})$$
$$=2(1+\sqrt{3}i)$$
$$=R.H.S$$

b): $$L.H.S = \frac{5i}{2+i} = \frac{5e^{i\frac{\pi}{2}}}{\sqrt{3}e^{i}}$$

8. Prove: $$\exp\left(i\frac{\theta_1+\theta_2}{2}\right)\exp\left(i\frac{\theta_1-\theta_2}{2}\right) = \exp(i\theta_1)$$

Solution:

$$L.H.S = \exp\left(i\frac{\theta_1+\theta_2}{2}\right)\exp\left(i\frac{\theta_1-\theta_2}{2}\right)$$
$$= \left(\cos\left(\frac{\theta_1+\theta_2}{2}\right)+i\sin\left(\frac{\theta_1+\theta_2}{2}\right)\right)\left(\cos\left(\frac{\theta_1-\theta_2}{2}\right)+i\sin\left(\frac{\theta_1-\theta_2}{2}\right)\right)$$
$$= \left(\cos\left(\frac{\theta_1+\theta_2}{2}\right)\cos\left(\frac{\theta_1-\theta_2}{2}\right)-\sin\left(\frac{\theta_1+\theta_2}{2}\right)\sin\left(\frac{\theta_1-\theta_2}{2}\right)\right)$$
$$+\mathrm{i}\left(\sin\left(\frac{\theta_1+\theta_2}{2}\right)\cos\left(\frac{\theta_1-\theta_2}{2}\right)+\cos\left(\frac{\theta_1+\theta_2}{2}\right)\sin\left(\frac{\theta_1-\theta_2}{2}\right)\right)$$
$$= \cos\left(\frac{\theta_1+\theta_2}{2}+\frac{\theta_1-\theta_2}{2}\right)+i\sin\left(\frac{\theta_1+\theta_2}{2}+\frac{\theta_1-\theta_2}{2}\right)$$
$$= \cos(\theta_1)+i\sin(\theta_1)$$
$$= e^{i\theta_1} = \exp(i\theta_1) = R.H.S$$

b): $$L.H.S = \exp\left(i\frac{\theta_1+\theta_2}{2}\right)\overline{\exp\left(i\frac{\theta_1-\theta_2}{2}\right)}$$

$$= \left(\cos\left(\frac{\theta_1+\theta_2}{2}\right)+i\sin\left(\frac{\theta_1+\theta_2}{2}\right)\right)\overline{\left(\cos\left(\frac{\theta_1-\theta_2}{2}\right)+i\sin\left(\frac{\theta_1-\theta_2}{2}\right)\right)}$$
$$= \left(\cos\left(\frac{\theta_1+\theta_2}{2}\right)+i\sin\left(\frac{\theta_1+\theta_2}{2}\right)\right)\left(\cos\left(\frac{\theta_1-\theta_2}{2}\right)-i\sin\left(\frac{\theta_1-\theta_2}{2}\right)\right)$$
$$= \left(\cos\left(\frac{\theta_1+\theta_2}{2}\right)\cos\left(\frac{\theta_1-\theta_2}{2}\right)+\sin\left(\frac{\theta_1+\theta_2}{2}\right)\sin\left(\frac{\theta_1-\theta_2}{2}\right)\right)$$
$$+\mathrm{i}\left(\sin\left(\frac{\theta_1+\theta_2}{2}\right)\cos\left(\frac{\theta_1-\theta_2}{2}\right)-\cos\left(\frac{\theta_1+\theta_2}{2}\right)\sin\left(\frac{\theta_1-\theta_2}{2}\right)\right)$$
$$= \cos\left(\frac{\theta_1+\theta_2}{2}-\frac{\theta_1-\theta_2}{2}\right)+i\sin\left(\frac{\theta_1+\theta_2}{2}-\frac{\theta_1-\theta_2}{2}\right)$$
$$= \cos(\theta_2)+i\sin(\theta_2)$$
$$= e^{i\theta_2} = \exp(i\theta_2) = R.H.S$$

10. use the de Moivre's Formula to drive

a): $\cos 3\theta = \cos^3\theta - 3\cos\theta\sin^2\theta$

Proof:

$$\begin{aligned}
\sin 3\theta &= 3\sin\theta - 4\sin^3\theta \\
\cos 3\theta &= 4\cos^3\theta - 3\cos\theta \\
\tan 3\theta &= \frac{3\tan\theta - \tan^3\theta}{1 - 3\tan^2\theta}
\end{aligned}$$

$$L.H.S = \cos^3\theta - 3\cos\theta\sin^2\theta = \cos^3\theta - 3\cos\theta(1-\cos^2\theta)$$
$$= \cos^3\theta - 3\cos\theta + 3\cos^3\theta$$
$$= 4\cos^3\theta - 3\cos\theta = \cos 3\theta = R.H.S$$

$$(\cos\theta + i\sin\theta)^3 = (cos\theta + isin\theta)(cos\theta + isin\theta)(cos\theta + isin\theta)$$
$$= \left(cos^2\theta - sin^2\theta + i\sin\theta cos\theta + i\cos\theta sin\theta\right)(cos\theta + isin\theta)$$
$$= cos^3\theta + icos^2\theta\sin\theta - sin^2\theta cos\theta - i\sin^3\theta + icos^2\theta\sin\theta - sin^2\theta\cos\theta + i\cos^2\theta\sin\theta - \sin^2\theta\cos\theta$$
$$= cos^3\theta - 3sin^2\theta cos\theta + 3icos^2\theta\sin\theta - i\sin^3\theta$$
$$= cos^3\theta - 3sin^2\theta cos\theta + i(3cos^2\theta\sin\theta - \sin^3\theta)$$

using de Moivre's law

$$\cos 3\theta + i\sin 3\theta = cos^3\theta - 3cos\theta sin^2\theta + i(3cos^2\theta\sin\theta - \sin^3)$$

Hence

$$\cos 3\theta + i\sin 3\theta = cos^3\theta - 3cos\theta sin^2\theta$$

And

$$\sin 3\theta = 3cos^2\theta\sin\theta - \sin^3\theta$$

Section 10

Examples:

1. Find the square root of 2i, Topic 12 time 15 mint

Answer:

We can write square root of 2i as $(2i)^{\frac{1}{2}}$

$$2i = 2\left(\cos\frac{\pi}{2} + i\sin\frac{\pi}{2}\right) = 2\exp\left[i\left(\frac{\pi}{2} + 2k\pi\right)\right] \qquad \text{k=0,1}$$

$$=2e^{i\left(\frac{\pi}{2}+2k\pi\right)}$$

Taking square root

$$(2i)^{\frac{1}{2}} = \left(2e^{i\left(\frac{\pi}{2}+2k\pi\right)}\right)^{\frac{1}{2}} = \sqrt{2}e^{i\left(\frac{\pi}{4}+k\pi\right)}$$

When k=0

$$\sqrt{2}e^{i\left(\frac{\pi}{4}+k\pi\right)} = \sqrt{2}e^{i\left(\frac{\pi}{4}\right)} = \sqrt{2}\left(\cos\frac{\pi}{4} + i\sin\frac{\pi}{4}\right) = 1+i$$

When k=1

$$\sqrt{2}e^{i\left(\frac{\pi}{4}+(1)\pi\right)} = \sqrt{2}e^{i\left(\frac{5\pi}{4}\right)} = \sqrt{2}\left(\cos\frac{5\pi}{4} + i\sin\frac{5\pi}{4}\right) = -1-i = -(1+i)$$

So the roots of 2i are $\pm(1+i)$

1. Find the square root of $1-\sqrt{3}i$, Topic 12 time 15 mint

Solution:

$$1-\sqrt{3}i = \cos\left(-\frac{\pi}{6}\right) + i\sin\left(-\frac{\pi}{6}\right) = 2e^{i\left(-\frac{\pi}{6}+2k\pi\right)}$$

Taking square root

$$\left(1-\sqrt{3}i\right)^{\frac{1}{2}} = \left(2e^{i\left(-\frac{\pi}{6}+2k\pi\right)}\right)^{\frac{1}{2}} = \sqrt{2}e^{i\left(-\frac{\pi}{12}+k\pi\right)} \qquad \text{k=0,1}$$

When k=0

$$\left(1-\sqrt{3}i\right)^{\frac{1}{2}} = \sqrt{2}e^{i\left(-\frac{\pi}{12}+k\pi\right)}$$

$$= \sqrt{2}e^{i\left(-\frac{\pi}{12}\right)} = \sqrt{2}\left(\cos\left(-\frac{\pi}{12}\right)+i\sin\left(-\frac{\pi}{12}\right)\right)$$

$$= \sqrt{2}\left(\cos\left(\frac{\pi}{12}\right)-i\sin\left(\frac{\pi}{12}\right)\right) = -\sqrt{2}\left(\frac{\sqrt{3}-i}{2}\right) = -\frac{\sqrt{3}-i}{\sqrt{2}}$$

When k=1

$$\left(1-\sqrt{3}i\right)^{\frac{1}{2}} = \sqrt{2}e^{i\left(-\frac{\pi}{12}+(1)\pi\right)}$$

$$= \sqrt{2}e^{i\left(\frac{11\pi}{12}\right)} = \sqrt{2}\left(\cos\left(\frac{11\pi}{12}\right)+i\sin\left(\frac{11\pi}{12}\right)\right)$$

$$= \sqrt{2}\left(\frac{\sqrt{3}-i}{2}\right) = +\frac{\sqrt{3}-i}{\sqrt{2}}$$

So the roots of $1-\sqrt{3}i$ are $\pm\frac{\sqrt{3}-i}{\sqrt{2}}$

2. $(-16)^{\frac{1}{4}} = (-1\times16)^{\frac{1}{4}} = \left(i^2\times2^4\right)^{\frac{1}{4}} = 2\left(i\right)^{\frac{1}{2}} = 2\sqrt{i}$

We find the square root of i.

Topic 25 examples:

Complex Limits

Nonexistence of limits

Example: Show that the limit

$$\lim_{z\to 0} \frac{z}{\bar{z}}$$

Does not exist.

Solution: First we calculate the limit when z approaches to 0 along the real axis

$$\lim_{z\to 0} \frac{z}{\bar{z}} = \lim_{x\to 0} \frac{x+i0}{x-i0} = \lim_{x\to 0} 1 = 1.$$

If z approaches 0 along the imaginary axis.

$$\lim_{z\to 0} \frac{z}{\bar{z}} = \lim_{y\to 0} \frac{0+iy}{0-iy} = \lim_{x\to 0} -1 = -1.$$

As 1 is not equal to -1 so, limit does not exist.

Complex Limits

Example: Show that

$$\lim_{z\to z_0} \bar{z} = \bar{z}_0$$

Solution:

$\varepsilon > 0$

$|f(z) - \bar{z}_0| < \varepsilon$

$\Leftrightarrow |\bar{z} - \bar{z}_0| < \varepsilon$

$\Leftrightarrow |\overline{z - z_0}| < \varepsilon$

$\Leftrightarrow |z - z_0| < \varepsilon$

$\delta = \varepsilon$

$|z - z_0| < \delta = \varepsilon$

$\Rightarrow |f(z) - \bar{z}_0| < \varepsilon$

$|f(z) - \bar{z}_0| < \varepsilon$ whenever $|z - z_0| < \delta$

Example: Show that

$$\lim_{z \to z_o} \overline{z} = \overline{z_o}$$

Solution:

Let ε be any real non-negative number. $\varepsilon > 0$.

$$\left| f(z) - \overline{z_o} \right| < \varepsilon$$

$$\Leftrightarrow \quad \left| \overline{z} - \overline{z_o} \right| < \varepsilon$$

$$\Leftrightarrow \quad \left| \overline{z - z_o} \right| < \varepsilon$$

$$\Leftrightarrow \quad \left| z - z_o \right| < \varepsilon$$

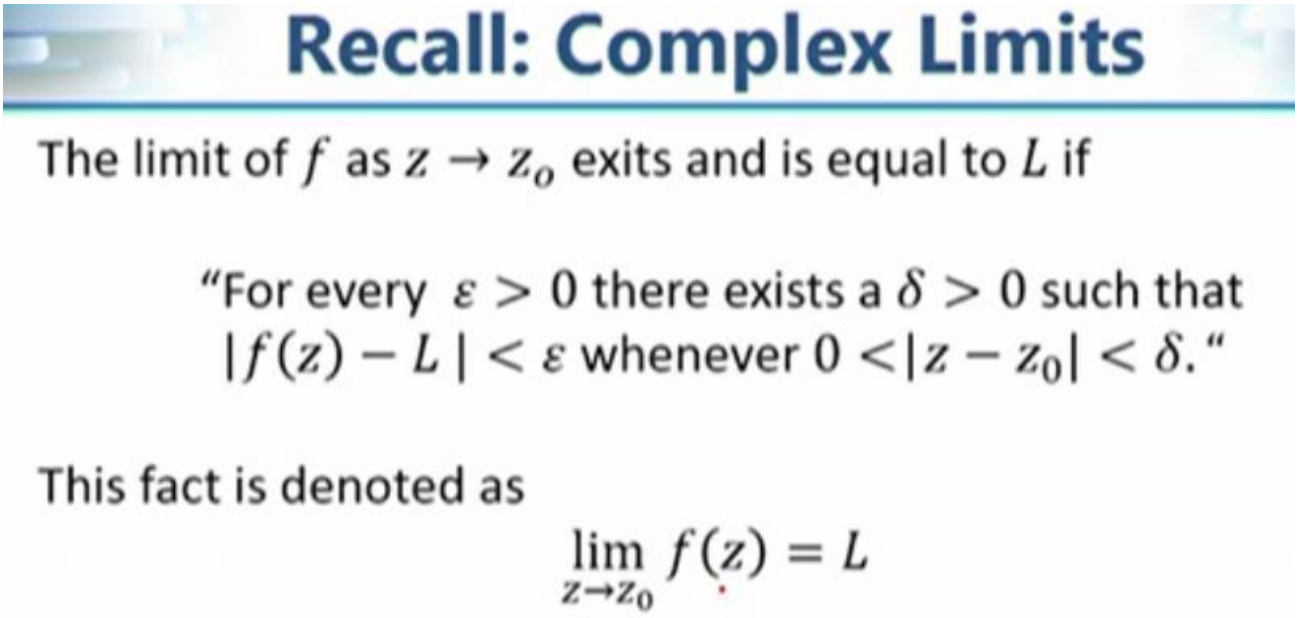
If $\delta = \varepsilon$

$$\left| z - z_o \right| < \delta = \varepsilon$$

$$\Rightarrow \quad \left| f(z) - \overline{z_o} \right| < \varepsilon$$

$\left| f(z) - z_o \right| < \varepsilon$ whenever $\left| z - z_o \right| < \delta s$

Topic 26



Complex Limits

A complex function is of the form

$$f(z) = f(x+iy) = u(x,y) + iv(x,y)$$

We find a relation between the complex limit of $f(z)$ and the real limits of the multivariable real function $u(x,y)$ and $v(x,y)$.

Relation between Real and Complex Limits

Main motivation

Theorem

Let $f(z) = u(x,y) + iv(x,y)$ be a complex function that is defined in some neighborhood of z_0, except perhaps at $z_0 = x_0 + iy_0$. Then

$$\lim_{z \to z_0} f(z) = u_0 + iv_0$$

iff

$$\lim_{(x,y)\to(x_0,y_0)} u(x,y) = u_0 \text{ and } \lim_{(x,y)\to(x_0,y_0)} v(x,y) = v_0 .$$

Example: if $u(x,y) = \dfrac{2x^3}{(x^2+y^2)}$ then show that.

$$\lim_{(x,y)\to(0,0)} u(x,y) = 0$$

Solution:

$x = r\cos\theta,$ y=rsinθ

$$u(r\cos\theta,\ r\sin\theta) = \frac{2r^3\cos^3\theta}{r^2\cos^2\theta + r^2\sin^2\theta} = 2r\cos^3\theta$$

In polar form:

$$|u(x,y)-0|$$
$$\left|2r\cos^3\theta\right| = 2r\left|\cos^3\theta\right| \le 2r$$
$$2r < \varepsilon \quad \Rightarrow \quad r < \frac{\varepsilon}{2}$$

To find $\delta > 0$ such that $|u(x,y)-0| < \varepsilon$

$$|u(x,y)-0| \le 2r < \varepsilon \text{ when ever } \; 0 < \sqrt{(x-0)^2+(y-0)^2} < \delta$$

Exampe: Show that the function

$$u(x,y) = \frac{xy}{x^2+y^2}$$

Does not have a limit as (x,y) approaches (0,0).

Solution:

For y=-x

$$\lim_{(x,y)\to(0,0)} \frac{xy}{x^2+y^2} = \lim_{x\to 0} \frac{-x^2}{x^2+y^2}$$
$$= \lim_{x\to 0} \frac{-x^2}{x^2+(-x)^2} = \lim_{x\to 0} \frac{-x^2}{2x^2}$$
$$= \lim_{x\to 0} \frac{-1}{2} = -\frac{1}{2}$$

Topic 29

Real Limits of Multivariable functions

Properties of real limits of functions of two variables

Theorem: If $\lim_{(x,y)\to(x_0,y_0)} F(x,y) = A$

and

$$\lim_{(x,y)\to(x_0,y_0)} G(x,y) = B$$

then

1. $\lim_{(x,y)\to(x_0,y_0)} cF(x,y) = cA$, where c is a real constant
2. $\lim_{(x,y)\to(x_0,y_0)} F(x,y) \pm G(x,y) = A \pm B$
3. $\lim_{(x,y)\to(x_0,y_0)} F(x,y).G(x,y) = A.B$
4. $\lim_{(x,y)\to(x_0,y_0)} \frac{F(x,y)}{G(x,y)} = \frac{A}{B}, B \neq 0$

Example:

Calculate the limit

$$\lim_{(x,y)\to(1,2)} 3xy^2 - y$$

Solution:

$$\lim_{(x,y)\to(1,2)} 3xy^2 - y$$

$$= \lim_{(x,y)\to(1,2)} 3xy^2 - \lim_{(x,y)\to(1,2)} y$$

$$= \left(\lim_{(x,y)\to(1,2)} 3x\right)\left(\lim_{(x,y)\to(1,2)} y^2\right) - \left(\lim_{(x,y)\to(1,2)} y\right)$$

$$= 3(1)(2)^2 - 2 = 12 - 2 = 10$$

Example: ***Compute the limit***

$$\lim_{z\to i} z^2 + z + 1$$

Solution:

$$f(z) = z^2 + z + 1$$

$$f(x+iy) = (x+iy)^2 + (x+iy) + 1$$

$$= (x^2 - y^2) - 2ixy + (x+iy) + 1$$

$$= (x^2 - y^2 + x + 1) - i(-2xy + y)$$

$$\lim_{(x,y)\to(0,1)} x^2 - y + x + 1 = -1 + 1 = 0$$

$$\lim_{(x,y)\to(0,1)} -2xy + y = 1$$

$$\Rightarrow \quad \lim_{z\to i} f(z) = i$$

Properties of Complex Limits

Theorem: Suppose that

$$\lim_{z\to z_0} f(z) = A$$

and

$$\lim_{z\to z_0} g(z) = B$$

then

1. $\lim_{z\to z_0} c\,f(z) = cA$, c a complex constant.
2. $\lim_{z\to z_0} (f(z) \pm g(z)) = A \pm B.$
3. $\lim_{z\to z_0} (f(z)\,.\,g(z)) = A.B$
4. $\lim_{z\to z_0} \frac{f(z)}{g(z)} = \frac{A}{B}$ provided $B \neq 0.$

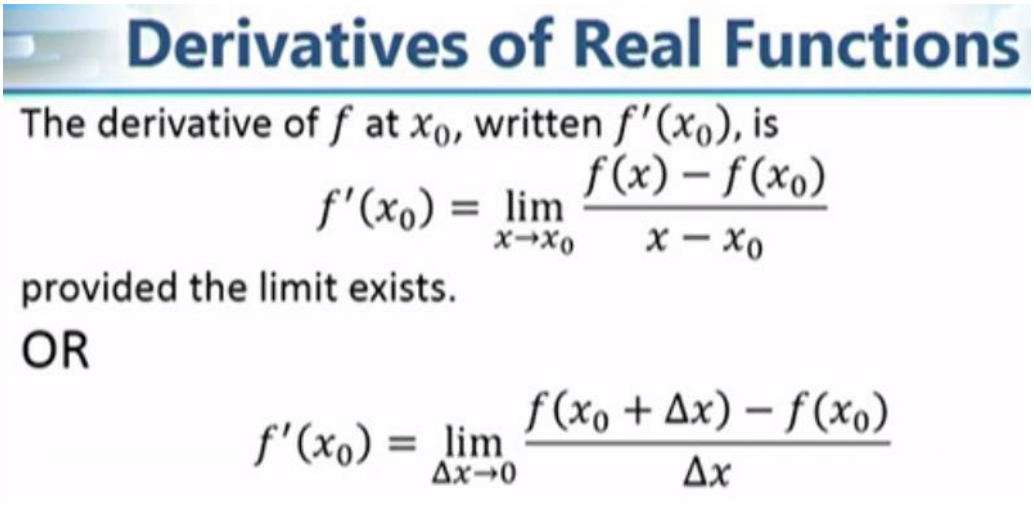
Example:

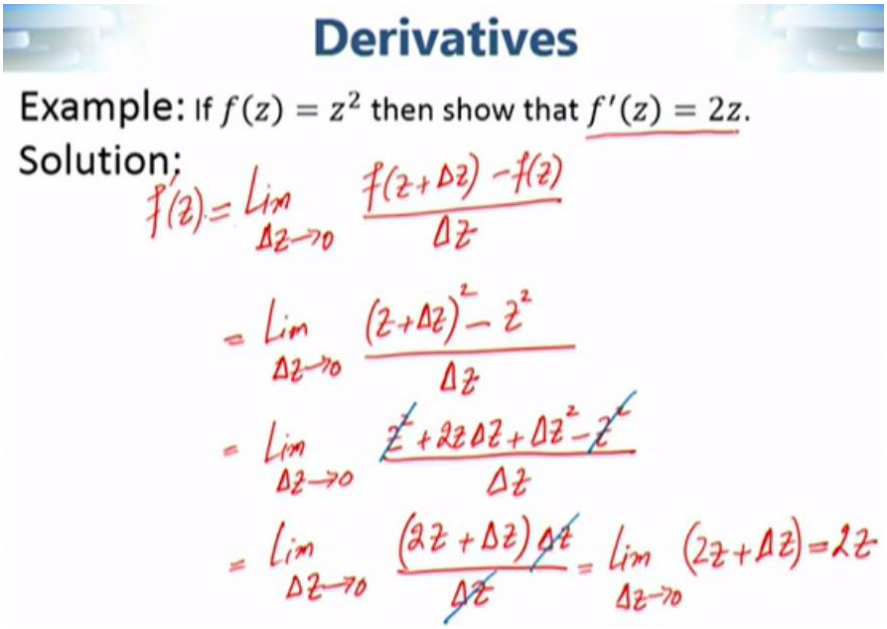
Compute the limit

$$\lim_{z\to i} z^2+z+1$$
$$=\left(\lim_{z\to i} z\right)\left(\lim_{z\to i} z\right)+\left(\lim_{z\to i} z\right)+\left(\lim_{z\to i} 1\right)$$
$$=(i)(i)+i+1$$
$$=-1+i+1=i$$

Topic 34

The mean value theorem for complex numbers does not hold.





Derivatives

Example: Show that the function $f(z) = \mathrm{Re}(z)$ is differentiable nowhere.

Solution: First we approach $z_0 = x_0 + iy_0$ along a line parallel to x −axis. In this case $z = x + iy_0$.

$$f'(z_0) = \lim_{z \to z_0} \frac{f(z) - f(z_0)}{\Delta z}$$

$$= \lim_{(x+iy_0) \to (x_0+iy_0)} \frac{\mathrm{Re}(x + iy) - \mathrm{Re}(x_0 + iy_0)}{(x + iy_0) - (x_0 + iy_0)}$$

$$= \lim_{(x+iy_0) \to (x_0+iy_0)} \frac{x - x_0}{x - x_0}$$

$$= 1$$

Derivatives

Solution (cont.): Next we approach z_0 along the line parallel to y −axis by forcing z to be of the form $z = x_0 + iy$.

$$f'(z_0) = \lim_{z \to z_0} \frac{f(z) - f(z_0)}{\Delta z}$$

$$= \lim_{(x_0+iy) \to (x_0+iy_0)} \frac{Re(x_0 + iy) - Re(x_0 + iy_0)}{(x_0 + iy) - (x_0 + iy_0)}$$

$$= \lim_{(x+iy_0) \to (x_0+iy_0)} \frac{x_0 - x_0}{i(y - y_0)}$$

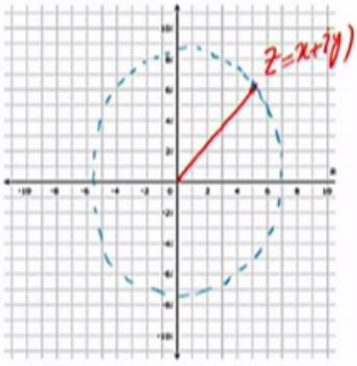
$$= 0$$

Hence the function f(z)=Re(z) is differentiable nowhere.

Derivatives

Example: Consider the real-valued function $f(z) = |z|^2$.
Find points where $f(z)$ is differentiable.

Solution: If $z = x+iy$, $f(z) = f(x+iy) = |x+iy|^2 = x^2+y^2$
Here $f(z)$ is real-valued function.



$f(z) = |z|^2 = z\bar{z}$

Derivatives

Example: Consider the real-valued function $f(z) = |z|^2 = z\bar{z}$
Find points where $f(z)$ is differentiable.

Solution: We know that $f'(z) = \lim_{\Delta z \to 0} \frac{f(z+\Delta z) - f(z)}{\Delta z}$

$$f'(z) = \lim_{\Delta z \to 0} \frac{(z+\Delta z)(\overline{z+\Delta z}) - z\bar{z}}{\Delta z}$$

$$= \lim_{\Delta z \to 0} \frac{(z+\Delta z)(\bar{z}+\overline{\Delta z}) - z\bar{z}}{\Delta z}$$

$$= \lim_{\Delta z \to 0} \frac{z\bar{z} + z\overline{\Delta z} + \bar{z}\Delta z + \Delta z\,\overline{\Delta z} - z\bar{z}}{\Delta z}$$

$$= \lim_{\Delta z \to 0} \left(z\frac{\overline{\Delta z}}{\Delta z} + \bar{z} + \overline{\Delta z} \right)$$

Path 1

Derivatives

Example: Consider the real-valued function $f(z) = |z|^2$.
Find points where $f(z)$ is differentiable.

Solution: Path 1: z moves parallel to x-axis

$$\lim_{\Delta z \to 0} \bar{z} + \overline{\Delta z} + z \frac{\overline{\Delta z}}{\Delta z}$$

In this case $\overline{\Delta z} = \Delta z$

$$= \lim_{\Delta z \to 0} \bar{z} + \Delta z + z \frac{\Delta z}{\Delta z}$$

$$= \lim_{\Delta z \to 0} (\bar{z} + \Delta z + z) = \bar{z} + z$$

Derivatives

Example: Consider the real-valued function $f(z) = |z|^2$.
Find points where $f(z)$ is differentiable.

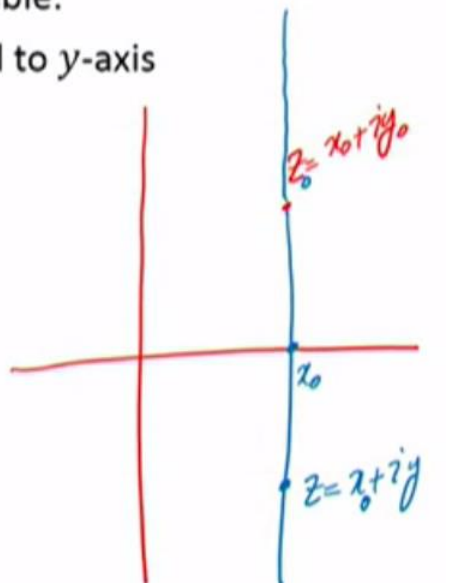
Solution: Path 2: z moves parallel to y-axis

$$\Delta z = z - z_0$$

$$= (x_0 + iy) - (x_0 + iy_0)$$

$$= i(y - y_0)$$

$$\Rightarrow \overline{\Delta z} = -\Delta z$$



Derivatives

Example: Consider the real-valued function $f(z) = |z|^2$.
Find points where $f(z)$ is differentiable.

Solution: Path 2: z moves parallel to y-axis

$$\lim_{\Delta z \to 0} \bar{z} + \overline{\Delta z} + z\frac{\overline{\Delta z}}{\Delta z}$$

In this case $\overline{\Delta z} = -\Delta z$

$$= \lim_{\Delta z \to 0} \bar{z} - \Delta z + z\frac{(-\Delta z)}{\Delta z}$$

$$= \lim_{\Delta z \to 0} \left(\bar{z} - \Delta z - z\right) = \bar{z} - z.$$

Derivatives

Example: Consider the real-valued function $f(z) = |z|^2$.
Find points where $f(z)$ is differentiable.

Solution: Along path 1: parallel to x-axis

$$\lim_{\Delta z \to 0} \frac{f(z+\Delta z) - f(z)}{\Delta z} = z + \bar{z}$$

along path 2: parallel to y-axis

$$\lim_{\Delta z \to 0} \frac{f(z+\Delta z) - f(z)}{\Delta z} = \bar{z} - z$$

$$\Rightarrow \quad z + \bar{z} = \bar{z} - z \Rightarrow z = 0.$$

Derivatives

Observations

Example: Consider the real-valued function $f(z) = |z|^2$. Find points where $f(z)$ is differentiable.

① $f(z) = f(x+iy) = (x^2+y^2)+i(0)$

$u(x,y) = x^2+y^2$, $u: \mathbb{R}^2 \to \mathbb{R}$

$v(x,y) = 0$, $v: \mathbb{R}^2 \to \mathbb{R}$

② $f(z)$ is cont but not diff..

Differentiation Formulas

The proofs are very similar to differentiation formulas for real-valued functions

$$\frac{d}{dz}C = 0$$

where C is a constant.

$\frac{d}{dz}(10) = 0$,

$\frac{d}{dz}(i) = 0$

Differentiation Formulas

$$\frac{d}{dz} z^n = n\, z^{n-1}$$

where n is a positive integer.

Example: $\frac{d}{dz}(z^{15})$, $n = 15$

$= 15\, z^{15-1}$

$= 15\, z^{14}$

Differentiation Formulas

$$\frac{d}{dz}[Cf(z)] = C\frac{d}{dz} f(z)$$

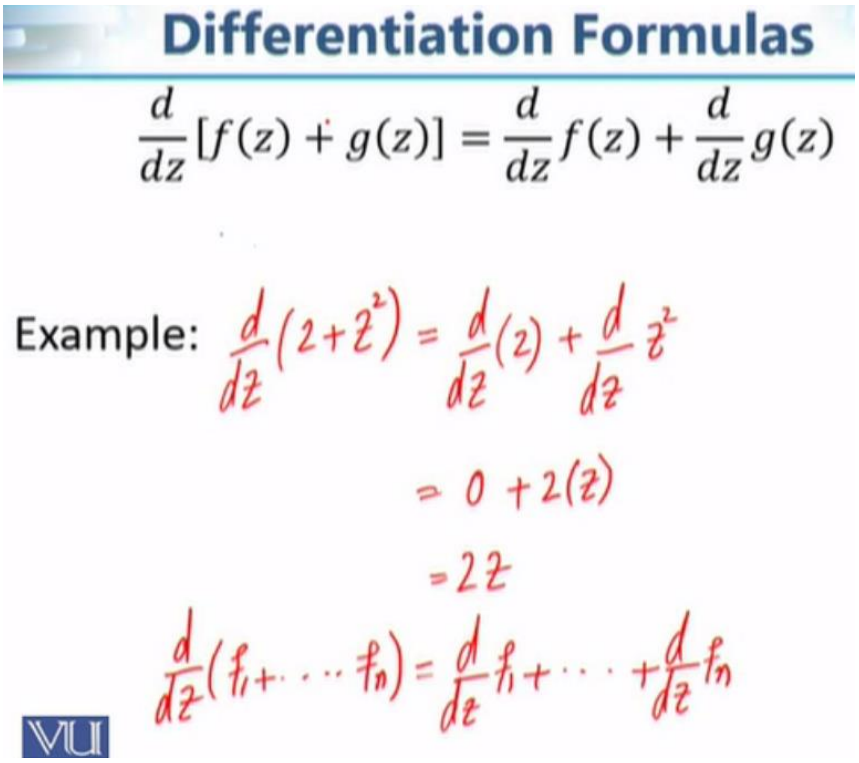
Example:

$\frac{d}{dz}(10 f(z)) = 10 \frac{d}{dz}(f(z))$

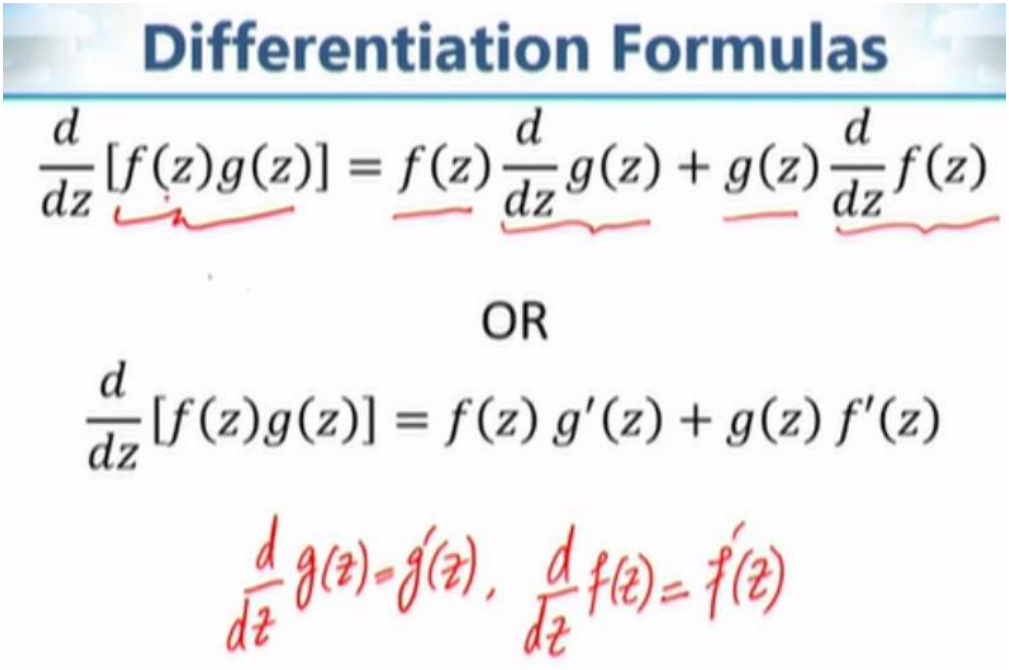
$\frac{d}{dz}(10 z^3) = 10 \frac{d}{dz} z^3 = 10(3z^2)$

$= 30 z^2$.

The Sum Rule



The product Rule



Differentiation Formulas

$$\frac{d}{dz}[f(z)g(z)] = f(z)\,g'(z) + g(z)\,f'(z)$$

Proof: Let $w = f(z)g(z)$

$$w'(z_0) = \lim_{\Delta z \to 0} \frac{w(z+\Delta z) - w(z)}{\Delta z}$$

Now

$$w(z+\Delta z) = f(z+\Delta z)g(z+\Delta z)$$

$$w(z+\Delta z) - w(z) = f(z+\Delta z)g(z+\Delta z) - f(z)\,g(z)$$

Differentiation Formulas

$$w(z+\Delta z) = f(z+\Delta z)g(z+\Delta z)$$

$$w(z+\Delta z) - w(z) = f(z+\Delta z)g(z+\Delta z) - f(z)\,g(z)$$

Adding and subtracting $f(z)g(z+\Delta z)$, we get

$$w(z+\Delta z) - w(z) = f(z+\Delta z)g(z+\Delta z) - f(z)g(z+\Delta z) + f(z)g(z+\Delta z) - f(z)\,g(z)$$

$$w(z+\Delta z) - w(z) = g(z+\Delta z)\,[f(z+\Delta z) - f(z)] + f(z)[g(z+\Delta z) - g(z)]$$

Differentiation Formulas

$$w(z+\Delta z)-w(z)=g(z+\Delta z)\,[f(z+\Delta z)-f(z)] + f(z)[g(z+\Delta z)-g(z)]$$

Dividing by Δz

$$\frac{w(z+\Delta z)-w(z)}{\Delta z}=\frac{g(z+\Delta z)\,[f(z+\Delta z)-f(z)]}{\Delta z} + \frac{f(z)[g(z+\Delta z)-g(z)]}{\Delta z}$$

Taking limit $\Delta z \to 0$

$$\lim_{\Delta z\to 0}\frac{w(z+\Delta z)-w(z)}{\Delta z}=\lim_{\Delta z\to 0} g(z+\Delta z)\,\frac{[f(z+\Delta z)-f(z)]}{\Delta z} + \lim_{\Delta z\to 0} f(z)\,\frac{[g(z+\Delta z)-g(z)]}{\Delta z}$$

Differentiation Formulas

$$\lim_{\Delta z\to 0}\frac{w(z+\Delta z)-w(z)}{\Delta z}=\lim_{\Delta z\to 0} g(z+\Delta z)\,\frac{[f(z+\Delta z)-f(z)]}{\Delta z} + \lim_{\Delta z\to 0} f(z)\,\frac{[g(z+\Delta z)-g(z)]}{\Delta z}$$

$$= \lim_{\Delta z\to 0} g(z+\Delta z)\ \lim_{\Delta z\to 0}\frac{f(z+\Delta z)-f(z)}{\Delta z} + \lim_{\Delta z\to 0} f(z)\ \lim_{\Delta z\to 0}\frac{g(z+\Delta z)-g(z)}{\Delta z}$$

Hence proved the following

$$w'(z)=g(z)f'(z)+f(z)g'(z)$$

Differentiation Formulas

Example: Given $f(z) = a_0 + a_1 z + a_2 z^2 + \cdots + a_n z^n$ then find

$$\frac{d}{dz} f(z)$$

Solution:

$$\frac{d}{dz}(a_0 + a_1 z + \cdots + a_n z^n) = \frac{d}{dz} a_0 + \frac{d}{dz} a_1 z + \cdots + \frac{d}{dz} a_n z^n$$

$$= a_1 \frac{d}{dz} z + a_2 \frac{d}{dz} z^2 + \cdots + a_n \frac{d}{dz} z^n$$

$$= a_1 + 2a_2 z + 3a_3 z^2 + \cdots + n a_n z^{n-1}$$

Differentiation Formulas

Chain Rule

Chain rule is useful when we want to differentiate composition of two functions.

Given two functions $f(z)$ and $g(z)$ composition of f and g is given by

$$f \circ g(z) = f\big(g(z)\big)$$

Differentiation Formulas

Chain Rule

$$\frac{d}{dz}[f(g(z))] = (f \circ g)'(z) = f'(g(z))\, g'(z)$$

Derivative of outside function

Derivative of inside function

Using Chain Rule

Differentiation Formulas

Example: Given $f(z) = z^3 + i2z^2 + 4$ then find

$$\frac{d}{dz}[f(z)]^3$$

Solution:

$h(z) = z^3$, $h'(z) = 3z^2$

$(f(z))^3 = h \circ f(z) = h(f(z)) = (f(z))^3$

$$\frac{d}{dz}(f(z))^3 = 3(f(z))^2 f'(z)$$

$$= 3(f(z))^2 (3z^2 + 2i(z) + 0)$$

The Cauchy-Riemann Equations

Recall

Theorem: Suppose that

$$f(z) = f(x + iy) = u(x, y) + iv(x, y)$$

If $f(z)$ differentiable at a point $z_0 = x_0 + iy_0$ then u and v satisfy **Cauchy-Riemann equations**

$$u_x(x_0, y_0) = v_y(x_0, y_0) \text{ and } u_y(x_0, y_0) = -v_x(x_0, y_0).$$

The Cauchy-Riemann Equations

Example: We know $f(z) = z^2$ is differentiable and $f'(z) = 2z$. Check whether Cauchy-Riemann equations are satisfied by $f(z)$.

Solution: If $z = x+iy$, $f(z) = z^2 = (x+iy)^2 = x^2 - y^2 + i(2xy)$

Here $u(x,y) = x^2 - y^2$, $v(x,y) = 2xy$

Check Cauchy-Riemann equations $u_x = v_y$, $u_y = -v_x$

$u_x = 2x$, $u_y = -2y$, $v_x = 2y$, $v_y = 2x$

$u_x = 2x = v_y$ ✓ $u_y = -2y = -v_x$ ✓

Hence Cauchy Riemann Equations are satisfied.

The Cauchy-Riemann Equations

Example: We Show that $f(z) = \bar{z}$ is nowhere differentiable.

Solution: If $z = x+iy$, $f(z) = f(x+iy) = \overline{(x+iy)} = x+i(-y)$

$\Rightarrow u(x,y) = x$, $v(x,y) = -y$

check Cauchy-Riemann equations $\boxed{u_x = v_y}$, $u_y = -v_x$

$u_x = 1$, $u_y = 0$, $v_x = 0$, $v_y = -1$

$u_x = 1$, $v_y = -1$. If $u_x = v_y$ is satisfied then $1 = -1$

$\Rightarrow$ Cauchy-Riemann are nowhere satisfied.

C-R equations are not satisfied.

The Cauchy-Riemann Equations

Solution: We can rewrite $f(z)$ in the following form

$$f(z) = \begin{cases} \dfrac{(\bar{z})^2}{z} = \dfrac{x^3 - 3xy^2}{x^2+y^2} + i\dfrac{y^3 - 3x^2y}{x^2+y^2} & \text{when } z \neq 0 \\ 0 & \text{when } z = 0 \end{cases}$$

$u(x,y)$

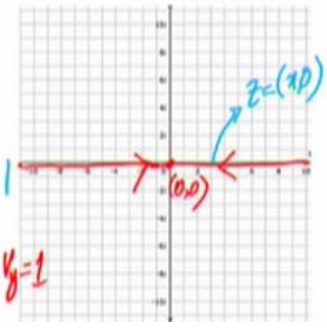
Step1: Checking CR equations

u_x, u_y, v_x, v_y

$u_x(0,0) = \lim_{(x,0)\to(0,0)} \dfrac{u(x,0) - u(0,0)}{x-0}$

$= \lim_{x\to 0} \dfrac{\frac{x^3}{x^2}}{x} = \lim_{x\to 0} \dfrac{x^3}{x^3} = \lim_{x\to 0} 1 = 1$

$\Rightarrow u_x = 1$. Similarly $u_y(0,0) = 0$, $v_x(0,0) = 0$, $v_y = 1$



Uy is equal to vy so it satisfied the equation.

The Cauchy-Riemann Equations

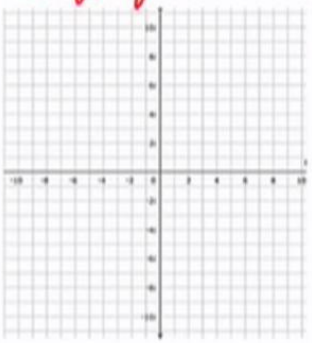
Solution: We can rewrite $f(z)$ in the following form

$$f(z) = \begin{cases} \dfrac{(\bar{z})^2}{z} = \dfrac{x^3 - 3xy^2}{x^2 + y^2} + i\dfrac{y^3 - 3x^2y}{x^2 + y^2} & when\ z \neq 0 \\ 0 & when\ z = 0 \end{cases}$$

Step1: Checking CR equations

$u_x(0,0) = 1,\ u_y(0,0) = 0,\ v_x(0,0) = 0,\ v_y(0,0) = 1$ $\quad u_x = v_y,\ u_y = -v_x$.

Hence Cauchy-Riemann equations are satisfied by $f(z)$



VU

The Cauchy-Riemann Equations

Solution: We can rewrite $f(z)$ in the following form

$$f(z) = \begin{cases} \dfrac{(\bar{z})^2}{z} = \dfrac{x^3 - 3xy^2}{x^2 + y^2} + i\dfrac{y^3 - 3x^2y}{x^2 + y^2} & when\ z \neq 0 \\ 0 & when\ z = 0 \end{cases}$$

Path1: Along x-axis. $z = x + i0$.

$$\lim_{(x,0)\to(0,0)} \frac{f(x+i0) - f(0)}{x - 0}, \quad f(x+i0) = \frac{x^3}{x^2}$$

$$= \lim_{(x,0)\to(0,0)} \frac{\frac{x^3}{x^2}}{x} = \lim_{(x,0)\to(0,0)} \frac{x^3}{x^3} = \lim_{(x,0)\to(0,0)} 1 = 1.$$

The function is not differentiable at the origin.

The Cauchy-Riemann Equations

Cauchy-Riemann Conditions for differentiability

Theorem: Let $f(z) = f(x+iy) = u(x,y) + iv(x,y)$ be defined in some neighborhood of the point $z_0 = x_0 + iy_0$ and suppose that

1. If all the partial derivatives u_x, v_y, u_y and v_x exist everywhere in the neighbourhood.
2. those partial derivatives are continuous at (x_0, y_0) and satisfy Cauchy-Riemann equations

$$u_x = v_y, \qquad u_y = -v_x$$

Then f is differentiable at z_0, and the derivative $f'(z_0)$ can be computed with either of the following equations

$$f'(z_0) = u_x(x_0, y_0) + iv_x(x_0, y_0).$$

and also

$$f'(z_0) = v_y(x_0, y_0) - iu_y(x_0, y_0).$$

VU

CR Equations in Polar Coordinates

Cauchy-Riemann equations

$$u_x = v_y, \qquad u_y = -v_x$$

Let us write

$$u_x, u_y, v_x, v_y$$

in terms of

$$u_r, u_\theta, v_r, v_\theta.$$

CR Equations in Polar Coordinates

Chain Rule

Suppose that $u(x, y)$ is a differentiable function of x and y, where $x = x(r, \theta)$ and $y = y(r, \theta)$ are differentiable functions of r and θ. Then

$$\frac{\partial u}{\partial r} = \frac{\partial u}{\partial x}\frac{\partial x}{\partial r} + \frac{\partial u}{\partial y}\frac{\partial y}{\partial r}$$

and

$$\frac{\partial u}{\partial \theta} = \frac{\partial u}{\partial x}\frac{\partial x}{\partial \theta} + \frac{\partial u}{\partial y}\frac{\partial y}{\partial \theta}$$

CR Equations in Polar Coordinates

Given $u(x, y)$ and $v(x, y)$ be two component functions of complex valued function

$$f(x + iy) = u(x, y) + iv(x, y)$$

Then

$$\frac{\partial u}{\partial r} = \frac{\partial u}{\partial x}\frac{\partial x}{\partial r} + \frac{\partial u}{\partial y}\frac{\partial y}{\partial r}$$

where $x = r\cos\theta, y = r\sin\theta$

$\frac{\partial x}{\partial r} = \cos\theta, \frac{\partial y}{\partial r} = \sin\theta$

$u_r = u_x \cos\theta + u_y \sin\theta$

CR Equations in Polar Coordinates

So we have $u_r = u_x \cos\theta + u_y \sin\theta$; $v_r = v_x \cos\theta + v_y \sin\theta$

$u_\theta = -u_x r\sin\theta + u_y r\cos\theta$; $v_\theta = -v_x r\sin\theta + v_y r\cos\theta$

$u_\theta = -u_x r\sin\theta + u_y r\cos\theta = -r(u_x \sin\theta - u_y \cos\theta)$

$= -r(v_y \sin\theta + v_x \cos\theta)$

$= -rv_r$

$\Rightarrow \boxed{u_\theta = -rv_r,\ ru_r = v_\theta} \Longleftrightarrow u_x = v_y,\ u_y = -v_x$

CR Equations in Polar Coordinates

Theorem: Let the function

$$f(z) = u(r,\theta) + iv(r,\theta)$$

be defined throughout some neighbourhood of a non-zero point $z_0 = r_0\, e^{i\theta_0}$ and suppose that

1. the first order partial derivates of the functions u and v with respect to r and θ exist everywhere in the neighbourhood
2. those partial derivatives are continuous at (r_0, θ_0) and satisfy polar form of Cauchy-Riemann equations

$$ru_r = v_\theta, \qquad u_\theta = -rv_r \quad .$$

Then $f'(z_0)$ exists and its value is

$$f'(z_0) = e^{-i\theta}(u_r + iv_r)$$

where the right side is to be evaluated at (r_0, θ_0)

CR Equations in Polar Coordinates

Example: Show that if f is the principal cube root function given by

$$f(re^{i\theta}) = f(z) = z^{\frac{1}{3}} = r^{\frac{1}{3}}\cos\frac{\theta}{3} + i\, r^{\frac{1}{3}}\sin\frac{\theta}{3},$$

where the domain is restricted to be

$$\{re^{i\theta} : r > 0 \text{ and } -\pi < \theta < \pi\}$$

then the derivative is given by

$$f'(z) = \frac{1}{3z^{\frac{2}{3}}} = \frac{1}{3}r^{-\frac{2}{3}}\cos\frac{\theta}{3} + i\frac{1}{3}r^{-\frac{2}{3}}\sin\frac{\theta}{3}$$

for every point in the domain.

CR Equations in Polar Coordinates

Solution:

$$f(re^{i\theta}) = r^{\frac{1}{3}}\cos\frac{\theta}{3} + i\, r^{\frac{1}{3}}\sin\frac{\theta}{3},$$

be defined throughout some neighbourhood of a non-zero point $z_0 = r_0\, e^{i\theta_0}$ and suppose that

CR Equations in Polar Coordinates

Solution:

$$f(re^{i\theta}) = r^{\frac{1}{3}} \cos\frac{\theta}{3} + i\, r^{\frac{1}{3}} \sin\frac{\theta}{3},$$

the first order partial derivatives of the functions u and v with respect to r and θ exist everywhere in the neighbourhood

$u(r,\theta) = r^{1/3}\cos\frac{\theta}{3}$, $v(r,\theta) = r^{1/3}\sin\frac{\theta}{3}$

$u_r = \frac{1}{3} r^{-2/3}\cos\frac{\theta}{3}$, $v_r = \frac{1}{3} r^{-2/3}\sin\frac{\theta}{3}$

$u_\theta = \frac{-1}{3} r^{1/3}\sin\frac{\theta}{3}$, $v_\theta = \frac{1}{3} r^{1/3}\cos\frac{\theta}{3}$

CR Equations in Polar Coordinates

Solution:

$$f(re^{i\theta}) = r^{\frac{1}{3}} \cos\frac{\theta}{3} + i\, r^{\frac{1}{3}} \sin\frac{\theta}{3},$$

those partial derivatives are continuous at (r_0, θ_0) and satisfy polar form of Cauchy-Riemann equations

$ru_r = v_\theta$ ✓ $u_\theta = -rv_r$

$u_r = \frac{1}{3} r^{-2/3}\cos\frac{\theta}{3}$, $v_\theta = \frac{1}{3} r^{1/3}\cos\frac{\theta}{3} = r\left(\frac{1}{3} r^{-2/3}\cos\frac{\theta}{3}\right)$

$v_\theta = \frac{1}{3} r^{1/3}\cos\frac{\theta}{3}$ $= \frac{1}{3} r^{1/3}\cos\frac{\theta}{3}$

Similarly $u_\theta = -r v_r$ is satisfied

First C-R equation is satisfied

CR Equations in Polar Coordinates

Solution:

$$f(re^{i\theta}) = r^{\frac{1}{3}}\cos\frac{\theta}{3} + i\, r^{\frac{1}{3}}\sin\frac{\theta}{3},$$

Then $f'(z_0)$ exists and its value is

$$f'(z_0) = e^{-i\theta}(u_r + iv_r)$$

where the right side is to be evaluated at (r_0, θ_0)

$u_r = \frac{1}{3} r^{-2/3}\cos\frac{\theta}{3}$, $v_r = \frac{1}{3} r^{-2/3}\sin\frac{\theta}{3}$

$$f'(z_0) = e^{-i\theta_0}\left(\frac{1}{3} r_0^{-2/3}\cos\frac{\theta_0}{3} + i\frac{1}{3} r_0^{-2/3}\sin\frac{\theta_0}{3}\right)$$

CR Equations in Polar Coordinates

Solution:

$$f(re^{i\theta}) = r^{\frac{1}{3}}\cos\frac{\theta}{3} + i\, r^{\frac{1}{3}}\sin\frac{\theta}{3},$$

$$\begin{aligned} f'(z) &= e^{-i\theta}\left[\tfrac{1}{3}r^{-\frac{2}{3}}\cos\tfrac{\theta}{3} + i\tfrac{1}{3}r^{-\frac{2}{3}}\sin\tfrac{\theta}{3}\right] \\ &= e^{-i\theta}\left[\tfrac{1}{3}r^{-\frac{2}{3}}e^{\frac{i\theta}{3}}\right] \\ &= \tfrac{1}{3}r^{-\frac{2}{3}}e^{-i\frac{2}{3}} \end{aligned}$$

$$f'(z) = \frac{1}{3z^{\frac{2}{3}}}$$

$z^{2/3} = r^{2/3} e^{i\frac{2}{3}\theta}$

$f(z) = z^{1/3}$

The Cauchy-Riemann Equations

Cauchy-Riemann Conditions for differentiability

Theorem: Let $f(z) = f(x + iy) = u(x, y) + iv(x, y)$ be defined in some neighborhood of the point $z_0 = x_0 + iy_0$ and suppose that

1. If all the partial derivatives u_x, v_y, u_y and v_x exist everywhere in the neighbourhood.
2. those partial derivatives are continuous at (x_0, y_0) and satisfy Cauchy-Riemann equations

$$u_x = v_y, \qquad u_y = -v_x$$

Then f is differentiable at z_0, and the derivative $f'(z_0)$ can be computed with either of the following equations

$$f'(z_0) = u_x(x_0, y_0) + iv_x(x_0, y_0).$$

and also

$$f'(z_0) = v_y(x_0, y_0) - iu_y(x_0, y_0).$$

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Exmaples are

The Cauchy-Riemann Equations

Example: Show that the function

$$f(z) = e^{-y}\cos x + i\, e^{-y}\sin x$$

is differentiable for all z and find its derivative.

Solution: Let (x_0, y_0) be an arbitrary point.

defined in some neighborhood of the point $z_0 = x_0 + iy_0$ ✓
and suppose that

$f(z) = f(x+iy) = u(x,y) + iv(x,y)$

$\Rightarrow u(x,y) = e^{-y}\cos x, \quad v(x,y) = e^{-y}\sin x$

$\Rightarrow u(x,y)$ & $v(x,y)$ are defined for all (x,y).

The Cauchy-Riemann Equations

Example: Show that the function

$$f(z) = e^{-y}\cos x + i\, e^{-y}\sin x$$

is differentiable for all z and find its derivative.

Solution:

If all the partial derivatives u_x, v_y, u_y and v_x exist everywhere in the neighbourhood.

$u(x,y) = e^{-y}\cos x$ | $v(x,y) = e^{-y}\sin x$

$u_x(x,y) = -e^{-y}\sin x$ | $v_x(x,y) = e^{-y}\cos x$

$u_y(x,y) = -e^{-y}\cos x$ | $v_y(x,y) = -e^{-y}\sin x$

The Cauchy-Riemann Equations

Example: Show that the function

$$f(z) = e^{-y}\cos x + i\, e^{-y}\sin x$$

is differentiable for all z and find its derivative.

Solution:

those partial derivatives are continuous at (x_0, y_0) and satisfy Cauchy-Riemann equations

$$u_x = v_y, \qquad u_y = -v_x$$

$u_x = -e^{-y}\sin x$, $v_x = e^{-y}\cos x$

$u_y = -e^{-y}\cos x$, $v_y = -e^{-y}\sin x$

$u_x = -e^{-y}\sin x = v_y$ ✓, $u_y = -e^{-y}\cos x = -v_x$ ✓

The Cauchy-Riemann Equations

Example: Show that the function

$$f(z) = e^{-y}\cos x + i\, e^{-y}\sin x$$

is differentiable for all z and find its derivative.

Solution:

Then f is differentiable at z_0, and the derivative $f'(z_0)$ can be computed with either of the following equations

$$f'(z_0) = u_x(x_0, y_0) + iv_x(x_0, y_0).$$ ✓

and also

$$f'(z_0) = v_y(x_0, y_0) - iu_y(x_0, y_0).$$ ✓

$u_x = -e^{-y}\sin x$, $v_x = e^{-y}\cos x$

$f'(z_0) = -e^{-y_0}\sin x_0 + i(e^{-y_0}\cos x_0)$

(4) $\lim_{z \to \infty} \frac{4z^2}{(z-1)^2} = 4$

(5) Ak question C.R

Q4) SOL:-

Date:

$$\lim_{z \to \infty} \frac{4z^2}{(z-1)^2} = 4$$

$$\frac{4z^2}{z^2+2z+1}$$

$$\frac{z^2(4)}{z^2\left(1-\frac{2}{z}+\frac{1}{z^2}\right)}$$

$$\frac{z^2(4)}{z^2\left(1-\frac{2}{z}+\frac{1}{z}\right)}$$

$$\lim_{z \to \infty} \frac{4}{1-\frac{2}{\infty}+\frac{1}{\infty}}$$

$$\frac{4}{1-0+0} = 4 \quad \text{R.H.S}$$

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Date:

Q) $\lim_{2+i} z^2 + 2z + 9 - 8 + 5i$

Sol $(2+i)^2 + 2(2+i) - 8 + 5i$

$4 + i^2 + 2i + 4 + 2i - 8 + 5i$

$4 - 1 + 4i + 4 + 2i - 8 + 5i$

$4 - 1 + 4 - 8 + 4i + 2i + 5i$

$-1 + 11i$

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Q) $f(z) = x^2 - y^2 - 2y + i(2x - 2xy)$

$x = \frac{z+\bar{z}}{2}$ $\quad y = \frac{z-\bar{z}}{2i}$

$$\left(\frac{z+\bar{z}}{2}\right)^2 - \left(\frac{z-\bar{z}}{2i}\right)^2 - 2\left(\frac{z-\bar{z}}{2i}\right) + i\left(2\left(\frac{z+\bar{z}}{2}\right) - 2\left(\frac{z+\bar{z}}{2}\right)\left(\frac{z-\bar{z}}{2i}\right)\right)$$

$$= \frac{z^2+\bar{z}^2+2z\bar{z}}{4} - \left(\frac{z^2+\bar{z}^2-2z\bar{z}}{4}\right) - \frac{2(z-\bar{z})}{2i}$$

$$\frac{4z\bar{z}}{4} - \frac{2(z-\bar{z})}{2i} + i\left(\frac{z+\bar{z}-(z^2-\bar{z}^2)}{i}\right)$$

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$$|z|^2 = z\bar{z}$$

$z = x + iy$

$|z|^2 = \left(\sqrt{x^2 + y^2}\right)^2$

$|z|^2 = x^2 + y^2$

$z = x + iy$

$\bar{z} = x - iy$

$z\bar{z} = x^2 - i^2 y^2$

$z\bar{z} = x^2 + y^2$

$z\bar{z} = x^2 + y^2$

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$$(1+\sqrt{3}i)^4$$

$1+\sqrt{3}i =$

$$r = \sqrt{1+3} = 2$$

$$\theta = \tan^{-1}(\sqrt{3}/1)$$

$$\pi/3$$

$$= 2e^{i(\pi/3)}$$

$$(1+\sqrt{3}i)^4 = \left(2(\cos\pi/3 + i\sin\pi/3)\right)^4$$

$$2^4\left(\cos\frac{4\pi}{3} + i\sin\frac{4\pi}{3}\right)$$

$$2^4 e^{i4\pi/3}$$

$$16e^{i4\pi/3}$$

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Real Limits of Multivariable functions

Example: Show that the function $u(x,y) = \frac{xy}{x^2+y^2}$ does not have a limit as (x,y) approaches $(0,0)$.

Solution: Along $y = x$

$$\lim_{(x,y)\to(0,0)} \frac{xy}{x^2+y^2} = \lim_{(x,y)\to(0,0)} \frac{x^2}{x^2+x^2}$$

along $y = x$

$$= \lim_{x\to 0} \frac{x^2}{2x^2}$$

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Tools View Help

Real Limits of Multivariable functions

Example: If $u(x,y) = \frac{2x^3}{(x^2+y^2)}$ then show that

$$\lim_{(x,y)\to(0,0)} u(x,y) = 0 .$$

Solution:

Find $\delta > 0$ such that

$|u(x,y)-0| \le 2h < \varepsilon$ whenever $0 < \sqrt{(x-0)^2+(y-0)^2} < \delta$

h

or $|u(x,y)-0| \le 2h < \varepsilon$ $\quad h < \delta$

choose $\boxed{\delta = \frac{\varepsilon}{2}} \Rightarrow h < \frac{\varepsilon}{2} \Rightarrow \boxed{2h < \varepsilon} \Rightarrow |u-0| < \varepsilon$

VU

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Continuity of $u(x, y)$

nple: Check whether the following function is nuous at (0,0)

$$u(x,y) = \begin{cases} \dfrac{x^3}{x^2+y^2} & (x,y) \neq (0,0) \\ 0 & (x,y) = (0,0) \end{cases}$$

tion:

$$\lim_{(x,y)\to(0,0)} \frac{x^3}{x^2+y^2} = 0$$

$u(0,0) = 0$

$u(x,y)$ is cont. at $(0,0)$

15:02

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inuity of Complex Functions

: Show that the function $f(z) = \bar{z}$ is continuous.

:

$\lim_{z \to z_0} \bar{z} = \bar{z}_0$ (from previous discussion)

$f(z_0) = \bar{z}_0$

$\Rightarrow$ $f(z)$ is Cont. at z_0

$\Rightarrow$ $f(z)$ is Cont. at each Complex number:

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8:07 4G 81%

You
Yesterday, 6:44 pm

(ix) $\lim_{z \to (i+1)} (z^2 - 2z + 1)$

(x) magnitude of $\exp(-2+3i)$

(xi) $\frac{1}{re^{i\theta}}$ The $U_r = ?$

U_r tha
U_y tha
U_x tha kuch clear nai tha.

(xii) close disk ki mathematical form puchi hui thi.

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$$|z^3 + 3z^2 - 2z + 1|$$

$$\leq |z^3| + |3z^2| + |-2z| + 1$$

$$\because \quad |z^3| = |z|^3 \qquad |z_1 z_2| = |z_1||z_2|$$

$$|z|^3 + 3|z|^2 + 2|z| + 1$$

$$< z^3 + 3z^2 + 2z + 1$$ upper bound

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Date

$$f(z) = (1+z^2)^4$$

$$\cdot z^2$$

$$= 4(1+z^2)^{4-1} \cdot 2z$$

$$2z$$

$$= 4(1+z^2)^3$$

OR Chain Rule

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$+i)^3$

$f(z) = x^2 - y^2 - 2y + i(2x - 2xy)$

$x = \frac{z+\bar{z}}{2} \qquad y = \frac{z-\bar{z}}{2i}$

$\left(\frac{z+\bar{z}}{2}\right)^2 - \left(\frac{z-\bar{z}}{2i}\right)^2 - 2\left(\frac{z-\bar{z}}{2i}\right)$

$+ i\left(2\left(\frac{z+\bar{z}}{2}\right) - 2\left(\frac{z+\bar{z}}{2}\right)\left(\frac{z-\bar{z}}{2i}\right)\right.$

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Q1 Re z and Im z

$z = x + iy$ $\quad \bar{z} = x - iy$

$z + \bar{z} = x + iy + x - iy$

$z + \bar{z} = 2x$

$2x = z + \bar{z}$

$x = \frac{z + \bar{z}}{2}$

Im z $\quad z - \bar{z} = x + iy - x$

$z - \bar{z} = +2iy$

$y = \frac{z - \bar{z}}{2i}$

$Re z = \frac{z + \bar{z}}{2}$ $\quad Im z =$

Q NO 2

Quadratic Formula

mple: Find roots of the following equation

$$z^2 + (2+i)z + \frac{3}{4} = 0$$

$a=1,\ b=2+i,\ c=3/4$

$$\frac{-(2+i) \pm [(2+i)^2 - 4(1)(3/4)]^{1/2}}{2(1)}$$

$$\frac{-(2+i) \pm [4+i^2+2(2i) - 3]^{1/2}}{2}$$

$$\frac{-(2+i) \pm [4-1+4i-3]^{1/2}}{2}$$

$$\frac{-(2+i) + [4i]^{1/2}}{2}$$

Here $[4i]^{1/2} = \sqrt{2} + i\sqrt{2}$, $-\sqrt{2} - i\sqrt{2}$

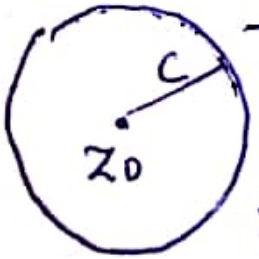
Roots of Complex Numbers (Part II)

So, $\frac{-(2+i) + (\sqrt{2}+i\sqrt{2})}{2}$, $\frac{(-2+i) + (-\sqrt{2}-\sqrt{2}i)}{2}$

END

- Roots of Unity
- Relation between roots of complex numbers and roots of unity
- Quadratic formula

A circle in plane is a set of points whose distance from a fixed point, say z_0 is constant



The point z_0 is called centre, the constant is called radius

* Equation of a circle of radius ε cent z_0 is $|z - z_0| = \varepsilon$

A closed disc with centre z_0 and ra $|z - z_0| \leq \varepsilon$

in the Complex Pla

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given by

$$z = \sqrt[n]{r_0}\, exp\left[i\left(\frac{\theta_0}{n} + \frac{2k\pi}{n}\right)\right], k = 0, \pm 1, \pm 2, \ldots$$

Example: Find square roots of $2i$

$2i = 2(\cos \pi/2 + i \sin \pi/2)$

$z_0 = 2e^{i\pi/2}$

$Z = \sqrt{2}\, exp\left[i\left(\frac{\pi/2}{n} + \frac{2k\pi}{2}\right)\right] k \in \mathbb{Z}$

$k = 0$

$z_1 = 1 + i$, $z_2 = -1 - i$

$\theta_0 + 2k\pi, k \in \mathbb{Z}$

$\frac{\theta_0}{n} + \frac{2k\pi}{n}, k \in \mathbb{Z}$

$exp\left[i\frac{\theta_0}{n} + \frac{2k\pi}{n}\right], k \in \mathbb{Z}$

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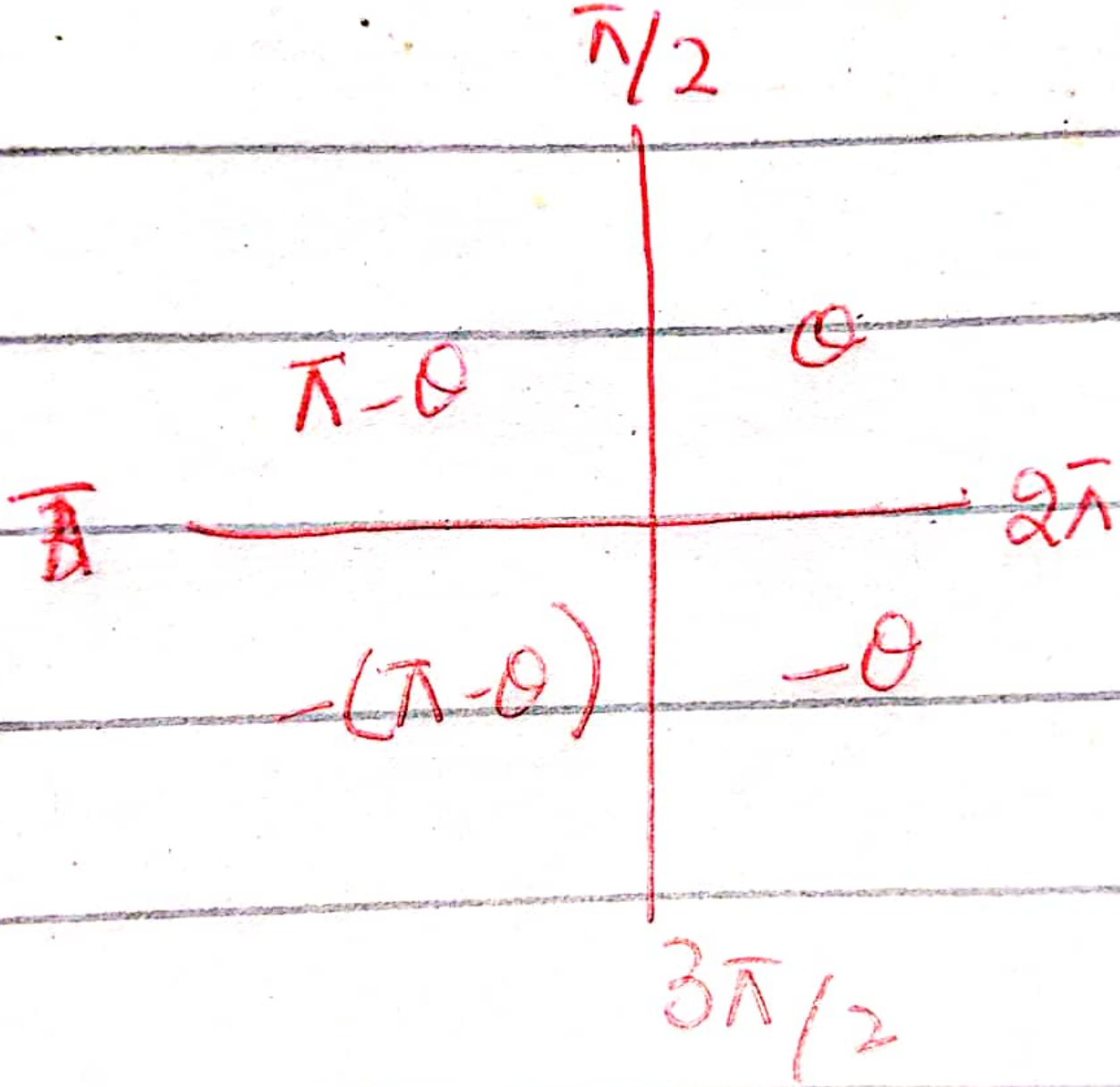
Polynomial Functions

Example

$$f(z) = \frac{1}{z} = z^{-1}$$

is NOT a polynomial.

Power of function in polynomial is non-negative and finite



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Relation between Real & Complex Limit

Example: Compute the limit

$$\lim_{z \to i} z^2 + z + 1 .$$

Solution:

$f(z) = z^2 + z + 1$

$f(x+iy) = (x+iy)^2 + (x+iy) + 1$

$= (x^2 - y^2) - 2ixy + (x+iy) + 1$

$= \underbrace{(x^2 - y^2 + x + 1)}_{u(x,y)} + i\underbrace{(-2xy + y)}_{v(x,y)}$

$\lim_{(x,y) \to (0,1)} x^2 - y^2 + x + 1 = -1 + 1 = 0$

$\lim_{(x,y) \to (0,1)} -2xy + y = 1$

$\Rightarrow \lim_{z \to i} f(z) = i$

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$(2i)^{1/2}$ $\quad 2i$

$2i =$

$r = 2$

$\theta = \pi/2$

Now

$2i = 2e^{i\pi/2}$

$(2i)^{1/2} = (2e^{i\pi/2})^{1/2}$

$= \sqrt{2}\, e^{i(\pi/4) + \frac{2k\pi}{2}}$

$k = 0$

$\sqrt{2}\, e^{i\pi/4}$

$\sqrt{2}(\cos\pi/4 + i\sin\pi/4)$

$\sqrt{2}\left(\frac{1}{\sqrt{2}} + i\frac{1}{\sqrt{2}}\right)$

$z_1 = 1 + i$

$-\sqrt{2}(\cos\pi/4 - i\sin\pi/4)$

$z_2 = -1 - i$

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an angle)

Q1 Find Square roots of $2i$

$2i = 2(\cos\frac{\pi}{2} + i\sin\pi/2)$

$z_0 = 2e^{i\pi/2}$

$z = \exp[i(\frac{\theta_0}{n} + \frac{2k\pi}{n})]$

$z = \sqrt{2}[$

Let $z = \sqrt{2i}$

by using exponential form $z = re^{i\theta}$

$\Rightarrow r^2 = e^{i(2\theta)} = 2i = 2e^{i(\pi/2 + 2n\pi)}$

$r^2 = 2 \Rightarrow r = \sqrt{2}$

$2\theta = \frac{\pi}{2} + 2\pi\pi \Rightarrow \theta = \frac{\pi}{4} + n\pi$

So, $z = \sqrt{2}e^{i(\pi/4 + 2n\pi)}$

$z = \sqrt{2}\left[\cos(\frac{\pi}{4} + 2n\pi) + i\sin(\frac{\pi}{4} + 2n\pi)\right]$

$z = \sqrt{2}\left[\pm\frac{1}{\sqrt{2}} \pm \frac{1}{\sqrt{2}}i\right]$

$z = \pm 1 \pm i$

$z = 1+i$, $z = -1-i$

Powers in Exponen

onential form

er $z = re^{i\theta}$

$e^{2\pi i}$ for $n = 0, \pm 1, \pm 2$

tical induction)

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Derivatives

Example: If $f(z) = z^2$ then show that $f'(z) = 2z$.

Solution;

$$f'(z) = \lim_{\Delta z \to 0} \frac{f(z+\Delta z) - f(z)}{\Delta z}$$

$$= \lim_{\Delta z \to 0} \frac{(z+\Delta z)^2 - z^2}{\Delta z}$$

$$= \lim_{\Delta z \to 0} \frac{z^2 + 2z\Delta z + \Delta z^2 - z^2}{\Delta z}$$

$$= \lim_{\Delta z \to 0} \frac{(2z + \Delta z)\Delta z}{\Delta z} = \lim_{\Delta z \to 0} (2z + \Delta z) = 2z$$

VU

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Properties of Complex Limits

Theorem: Suppose that $\lim_{z \to z_0} f(z) = A$ and $\lim_{z \to z_0} g(z) = B$

Then

1. $\lim_{z \to z_0} c\,f(z) = cA$, c a complex constant.

$f(z) = u(x,y) + i\,v(x,y)$, $A = u_0 + iv_0$

$\Rightarrow \lim_{(x,y)\to(x_0,y_0)} u(x,y) = u_0$ & $\lim_{(x,y)\to(x_0,y_0)} v(x,y) = v_0$

$cA = (c_0 + ic_1)(u_0 + iv_0) = (c_0u_0 - c_1v_0) + i(c_1u_0 + v_0c_0)$

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Consequences of Continuity

Theorem: If a function f is continuous throughout a region R that is both closed and bounded, there exists a nonnegative real number M such that

$$|f(z)| \leq M \text{ for all points } z \text{ in } R$$

where equality holds for at least one such z.

Proof: Given f is continuous. We know that $g(z) = |z|$ is continuous $g \circ f(z) = g(f(z)) = |f(z)|$

Since composition of continuous functions is continuous so $|f(z)|$ is continuous. If $f(z) = u(x,y) + iv(x,y)$ then $\sqrt{(u(x,y))^2 + (v(x,y))^2}$ is continuous.

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$$z = \pm 1 \pm i$$

$z = 1+i$, $z = -1-i$

Q2: formula of n^{th} root of Complex number

$$z_0^{1/n} \; z = \sqrt[n]{r_0}\, \exp\left[i\left(\frac{\theta_0}{n} + \frac{2k\pi}{n}\right)\right]$$

$k = 0, \pm 1, \pm 2, \cdots$

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View Help

Complex Limits

Example: Show that

$$\lim_{z \to z_0} \bar{z} = \bar{z}_0$$

Solution:

$\varepsilon > 0$ ✓

$|f(z) - \bar{z}_0| < \varepsilon$

$\Leftrightarrow |\bar{z} - \bar{z}_0| < \varepsilon$

$\Leftrightarrow |\overline{z - z_0}| < \varepsilon$

$\Leftrightarrow |z - z_0| < \varepsilon$ ✓

$\delta = \varepsilon$

$|z - z_0| < \delta = \varepsilon$

$\Rightarrow |f(z) - \bar{z}_0| < \varepsilon$

$|f(z) - \bar{z}_0| < \varepsilon$ whenever $|z - z_0| < \delta$

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oots of Compl...

Relation between roots of a complex number and roots of unity

ample: Find all values of $(2i)^{1/3}$.

step 1: $z_0 = 2i = 2\cos(\pi/2) + i\sin\pi/2$

$z_0 = 2e^{i(\pi/2)}$

$z_0 = 2e^{i(\pi/2 + 2k\pi)} \quad k \in \mathbb{Z}$

2: $z_0^{1/3} = \sqrt[3]{2}\, e^{i\left(\frac{\pi}{6} + \frac{2k\pi}{3}\right)} \quad k \in \mathbb{Z}$

3: $k = 0, 1, 2$

$c_0 = \sqrt[3]{2}\, e^{i(\pi/6)} = \sqrt[3]{2}\left(\frac{\sqrt{3}}{2} + i/2\right)$

$c_1 = c_0 w_3, \quad c_2 = c_0 w_3^2$

Quadratic

Consider equation

Then the quadratic form... the above equation

ax^2

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nuity of Complex Functions

Show that the function $f(z) = \bar{z}$ is continuous.

$\lim_{z \to z_0} \bar{z} = \bar{z_0}$ (From previous discussion)

$f(z_0) = \bar{z_0}$

$\Rightarrow$ $f(z)$ is Cont. at z_0

$\Rightarrow$ $f(z)$ is Cont. at each Complex number.

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$$\lim_{z \to 4i} \frac{z^2 + 16}{z - 4i}$$

$$\frac{z^2 - (-16)}{z^2 - 4i^2}$$

$$\frac{z^2 - 4i^2}{z - 4i}$$

$$\frac{(z - 4i)(z + 4i)}{(z - 4i)}$$

$$8i$$

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Q1 Re z and Im z

$z = x + iy \qquad \bar{z} = x - iy$

$z + \bar{z} = x + iy + x - iy$

$z + \bar{z} = 2x$

$2x = z + \bar{z}$

$x = \frac{z + \bar{z}}{2}$

Im z $\quad z - \bar{z} = x + iy - x + iy$

$z - \bar{z} = +2iy$

$y = \frac{z - \bar{z}}{2i}$

$Re\, z = \frac{z + \bar{z}}{2} \qquad Im\, z = \frac{z - \bar{z}}{2i}$

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$$\lim_{z \to \infty} \frac{z^2}{z^2 + z - 1}$$

$$= \lim_{z \to \infty} \frac{z^2\left(1 + \frac{2}{z}\right)}{z^2\left(1 + \frac{1}{z} - \frac{1}{z^2}\right)}$$

$$= \frac{\left(1 + \frac{2}{\infty}\right)}{1 + \left(\frac{1}{\infty}\right) - \left(\frac{1}{\infty^2}\right)}$$

$$= 1$$

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c) $(1-i)^4 = ?$

$$(1-i)^2 = (1)^2 + (i)^2 - 2(1)(i)$$
$$= 1 - 1 - 2i = 2i$$
$$(1-i)^4 = (1-i)^2(1-i)^2$$
$$= (2i)(2i)$$
$$= 4i^2 = 4(-1) = -4$$

2. Show that

$$L.H.S = \frac{1}{1/z} = \frac{1}{z^{-1}} \qquad \because \frac{1}{z} = z^{-1}$$
$$= \frac{1}{z^{-1}} \cdot \frac{z}{z} = \frac{1.z}{z^{-1}.z}$$
$$= \frac{z}{1} = z = R.H.S$$

3. use the associative and commutative laws for multiplication to

$$L.H.S = (z_1 z_2)(z_3 z_4)$$
$$= (z_2 z_1)(z_3 z_4) \qquad \because (z_1 z_2) = (z_2 z_1) \quad \textit{commutative la}$$
$$= (z_2)(z_1 z_3)(z_4) \qquad \because (z_1)(z_3 z_4) = (z_1 z_3)(z_4) \quad \textit{associati}$$
$$= (z_1 z_3)(z_2)(z_4) \qquad \because (z_2)(z_1 z_3) = (z_1 z_3)(z_2) \quad \textit{commut}$$
$$= (z_1 z_3)(z_2 z_4) \qquad \because (z_2)(z_4) = (z_2 z_4)$$
$$= R.H.S$$

5. Drive expression

$$\frac{z_1}{z_2} = \frac{x_1 x_2 - y_1 y_2}{x_2^2 + y_2^2} + i\frac{y_1 x_2 - x_1 y_2}{x_2^2 + y_2^2} \quad (z_2 \neq 0)$$

Let $z_1 = x_1 + iy_1$ and $z_2 = x_2 + iy_2$ where $(z_2 \neq 0)$

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operties of Complex Limits

e: Compute the limit

$$\lim_{z \to i} z^2 + z + 1 .$$

$$= \left(\lim_{z \to i} z\right)\left(\lim_{z \to i} z\right) + \lim_{z \to i} z + \lim_{z \to i} 1$$

$$= (i)(i) + i + 1$$

$$= -1 + i + 1 = i$$

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(i) Square root of $\sqrt{(-3)^2}$ = ?

(ii) Limit $\lim_{z \to i} \frac{iz^3 - 1}{z + i}$

(iii) Square of $(-2 - 5i)$

(iv) $f(z) = \frac{1}{z^2 + 4}$ is discontinuous at point.

(v) $f((z)) = z^3 - 3z + 12i + 5$

$f'(z)$ = ?

(vi) if $z = 3 + i4$ and $|z| = 5$

Then $|\bar{z}|$ = ?

(vii) if $w = 1 + 7i$ and

$z = 3 + 2i$

Then find $z \cdot w$ = ?

~~Square root~~

(viii) Simplify. $\left[\cos\frac{\pi}{4} + i\sin\frac{\pi}{4}\right]^{12}$

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$$\frac{z-i}{z^2+1}$$

$z \to i$

$$\lim_{z \to i} \frac{z-i}{z^2-(-1)} \qquad -1 = i^2$$

$$\frac{z-i}{z^2-i^2}$$

~~$z = i$~~

$$\frac{}{(z-i)(z+i)}$$

$$\lim_{z \to i} \frac{1}{z+i}$$

$$\frac{1}{2i}$$

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$$\lim_{z \to i} \frac{z^3}{z^2 + i}$$

Date

$$\frac{i^3}{i^2 + i}$$

$$\frac{i^2(i^2)}{i^2(i+1)}$$

$$\frac{-1}{1+i}$$

$$= \frac{-1}{1+i} \times \frac{1-i}{1-i}$$

$$\frac{-1(1-i)}{1+1}$$

$$\frac{-1}{2}$$

$$\frac{-1(1-i)}{2} = -\frac{1}{2} + \frac{1}{2}i$$

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$f(z) = z^2$ $\quad f'(z) = 2z$

$z = x + iy$ $\quad f(z) = z^2$

$z^2 = (x+iy)^2 = x^2 - y^2 + i2xy$

$u(x,y) = x^2 - y^2$ $\quad v(x,y) = 2xy$

$u_x = 2x$ $\quad v_y = 2x$

$u_y = -2y$ $\quad v_x = 2y$

$u_x = v_y$ $\quad u_y = -v_x$

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$$f(z) = (z - 4i)^8 \quad , z = 3 + 4i$$

$$f'(z) = 8(z - 4i)^7 (z - 4i)' =$$

$$f'(3+4i) = 8(3 + 4i - 4i)^7$$

$$= 8 \cdot 3^7$$

$$= 17496$$

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Product and Powers in Exponential Form

Product in exponential form

For complex number $z = re^{i\theta}$

$z^n = r^n e^{in\theta}$ for $n = 0, \pm 1, \pm 2, \pm 3, \ldots$

Example:

$$z = 7e^{i\frac{\pi}{4}}$$

$$z^{100} = 7^{100} e^{i\frac{\pi}{4}(100)}$$

$$z^{1000} = 7^{1000} e^{i\frac{\pi}{4}(1000)}$$

Product and Powers in Exponential Form

Example: Calculate $(\sqrt{3}+i)^3$ in exponential form

$r = \sqrt{3+1} = 2$

$\theta = \tan^{-1}\frac{1}{\sqrt{3}} = \frac{\pi}{6}$

$$\sqrt{3}+i = 2e^{i\frac{\pi}{6}}$$

$$(\sqrt{3}+i)^3 = 2^3 e^{i\frac{\pi}{6}(3)}$$

$$= 8e^{i\frac{\pi}{2}}$$ Ans

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differenciate.

(iv) $(\sqrt{3} - i)^{1/3}$, exponencial form.

(5 Marks)

1.3

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$$(\sqrt{3}-i)^{\frac{1}{3}}$$

Solution;

$$\sqrt{3}-i = re^{i\theta}$$

$$r = 2$$

$$\theta = -\frac{\pi}{6}$$

$$(\sqrt{3}-i)^{\frac{1}{3}} = \sqrt[3]{2}\ e^{-\frac{i\pi}{18}} \quad ans$$

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$(1-i)^4$ =?

$$(1-i)^2 = (1)^2 + (i)^2 - 2(1)(i)$$
$$= 1\text{-}1\text{-}2i = 2i$$
$$(1-i)^4 = (1-i)^2(1-i)^2$$
$$= (2i)(2i)$$
$$= 4i^2 = 4(-1) = -4$$

how that

$$L.H.S = \frac{1}{1/z} = \frac{1}{z^{-1}} \qquad \because \frac{1}{z} = z^{-1}$$
$$= \frac{1}{z^{-1}} \cdot \frac{z}{z} = \frac{1.z}{z^{-1}.z}$$
$$= \frac{z}{1} = z = R.H.S$$

use the associative and commutative laws for m

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$$-8i = 8\exp\left[i\left(\frac{-\pi}{2} + 2k\pi\right)\right].$$

$k = 2$ find $(8i)^{1/3}$.

$$(-8i)^{1/3} = \left(8\exp\left[i\left(\frac{-\pi}{2} + 2k\pi\right)\right]\right)^{1/3}.$$

$$= 8^{1/3}\exp\left[i\left(\frac{-\pi}{6} + \frac{2k\pi}{3}\right)\right].$$

$k = 2$.

$$= 2^{3\times 1/3}\exp\left[i\left(\frac{-\pi}{6} + \frac{4\pi}{3}\right)\right].$$

$$= 2\exp\left[i\left(\frac{-\pi + 8\pi}{6}\right)\right].$$

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(i) $\dfrac{(\cos 2\theta + i\sin 2\theta)^5 (\cos 3\theta - i\sin 3\theta)^6}{(\cos 4\theta - i\sin 4\theta)^7 (\cos 5\theta + i\sin 5\theta)^8}$

$= \dfrac{(\cos 2\theta + i\sin 2\theta)^5 (\cos(-3\theta) + i\sin(-3\theta))^6}{(\cos(-4\theta) + i\sin(-4\theta))^7 (\cos 5\theta + i\sin 5\theta)^8}$

$= \dfrac{(\cos 10\theta + i\sin 10\theta)(\cos(-18\theta) + i\sin(-18\theta))}{(\cos(-28\theta) + i\sin(-28\theta))(\cos 40\theta + i\sin 40\theta)}$ by De. Moivre's theorem

$= \dfrac{\cos(10\theta - 18\theta) + i\sin(10\theta - 18\theta)}{\cos(-28\theta + 40\theta) + i\sin(-28\theta + 40\theta)}$

$= \dfrac{\cos(-8\theta) + i\sin(-8\theta)}{\cos 12\theta + i\sin 12\theta}$

$= \cos(-8\theta - 12\theta) + i\sin(-8\theta - 12\theta) = \cos(-20)\theta + i\sin(-20)\theta$

$= \cos 20\theta - i\sin 20\theta$

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$$\int_0^1 (x - x^3)\, dx$$

$$\left| \frac{x^2}{2} - \frac{x^4}{4} \right|_0^1$$

$$= \left[\frac{(1)^2}{2} - \frac{(1)^4}{4} \right] - \left[\frac{(0)^2}{2} - \frac{(0)^4}{4} \right]$$

$$= \frac{1}{2} - \frac{1}{4}$$

$$= \frac{2-1}{4} = \frac{1}{4} \text{ Ans.}$$

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Derivative of:

$f(z) = \frac{z^3}{z+i}$ at $z = i$

$$f'(z) = \frac{(z+i)3z^2 - z^3(1)}{(z+i)^2}$$

$$= \frac{3z^2(z+i) - z^3}{(z+i)^2}$$

$$f'(i) = \frac{3(i^2)(i+i) - (i^3)}{(i+i)^2}$$

$$= \frac{3(-1)(2i) - (-i)}{(2i)^2}$$

$$= \frac{-6i + i^2}{-4} \Rightarrow \frac{3}{2}i + \frac{1}{4}$$

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(5) Evaluate

$\frac{1}{2+i} + 2$

$\rightarrow \frac{1+i}{2+i} + 2$

(6) verify

$(3,1)(3,-1)\left\{\frac{1}{5}, \frac{1}{10}\right) = (2,1)$

$(3+i)(3-i)\left\{\frac{1}{5} + \frac{1}{10}i\right)$

$(9-i^2)\left(\frac{1}{10} + \frac{1}{10}i\right)$

$(10)\left(\frac{1}{5} + \frac{1}{10}i\right)$

$\frac{2}{10} \times \frac{1}{5} + \frac{10}{10} \times \frac{1}{10}i$

$(2+i)$ (2,1)

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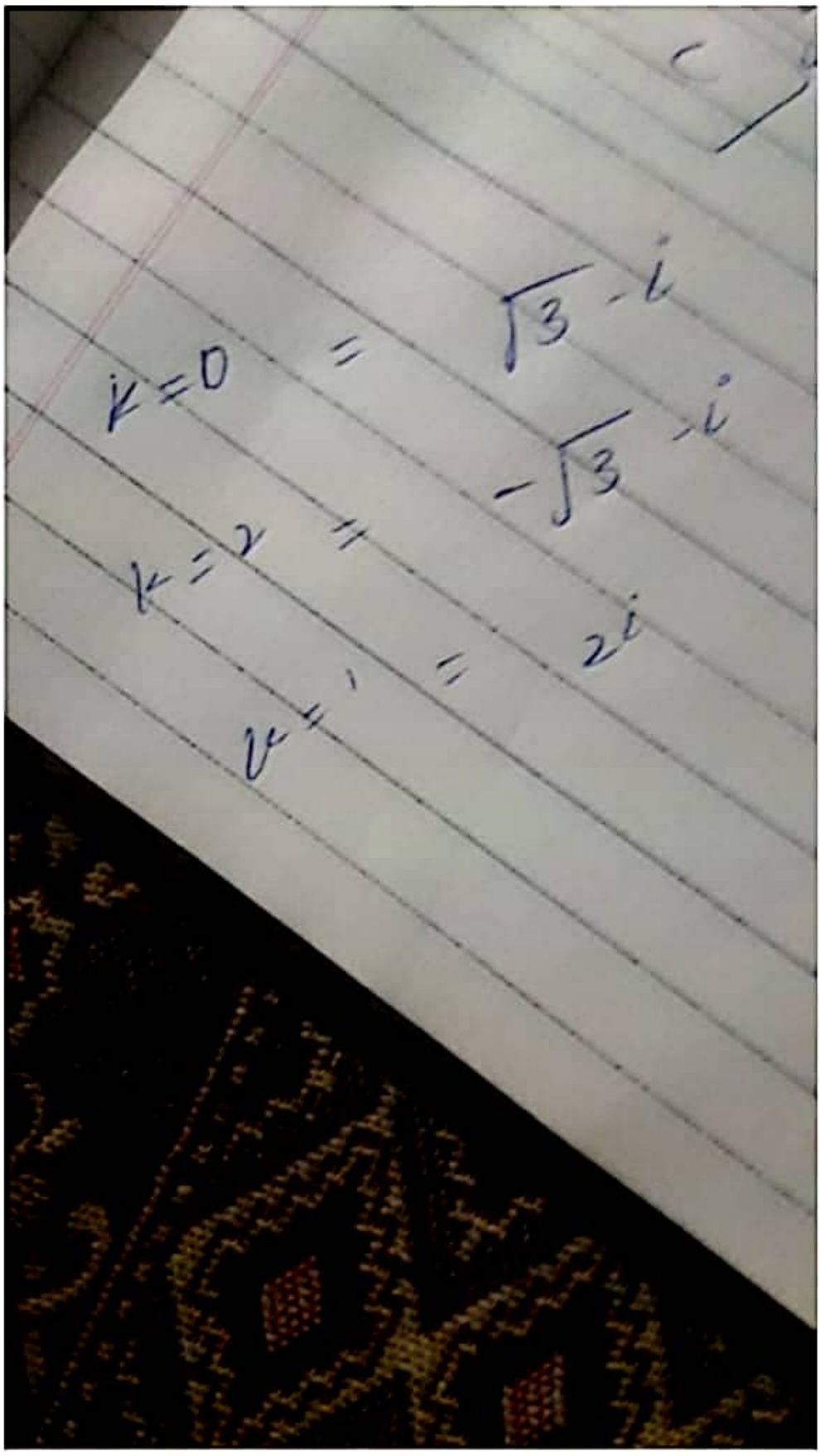

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e form

$L(z) = Az + B$

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$$g(f(z)) = \frac{1}{z^3} = \frac{1}{-i}$$

$$= \frac{1}{-i} \times \frac{i}{i}$$

$$= i$$

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Section 10

Examples:

1. Find the square root of 2i, Topic 12 time 15 mint

Answer:

We can write square root of 2i as $(2i)^{\frac{1}{2}}$

$$2i = 2\left(\cos\frac{\pi}{2} + i\sin\frac{\pi}{2}\right) = 2\exp\left[i\left(\frac{\pi}{2} + 2k\pi\right)\right] \qquad k=0,1$$

$$=2e^{i\left(\frac{\pi}{2}+2k\pi\right)}$$

Taking square root

$$(2i)^{\frac{1}{2}} = \left(2e^{i\left(\frac{\pi}{2}+2k\pi\right)}\right)^{\frac{1}{2}} = \sqrt{2}e^{i\left(\frac{\pi}{4}+k\pi\right)}$$

When k=0

$$\sqrt{2}e^{i\left(\frac{\pi}{4}+0\pi\right)} = \sqrt{2}e^{i\left(\frac{\pi}{4}\right)} = \sqrt{2}\left(\cos\frac{\pi}{4} + i\sin\frac{\pi}{4}\right) = 1+i$$

When k=1

$$\sqrt{2}e^{i\left(\frac{\pi}{4}+1\pi\right)} = \sqrt{2}e^{i\left(\frac{5\pi}{4}\right)} = \sqrt{2}\left(\cos\frac{5\pi}{4} + i\sin\frac{5\pi}{4}\right) = -1-i = -(1+i)$$

So the roots of 2i are $\pm(1+i)$

1. Find the square root of $1-\sqrt{3}i$, Topic 12 time 15 mint

Solution:

$$1-\sqrt{3}i = \cos\left(-\frac{\pi}{6}\right) + i\sin\left(-\frac{\pi}{6}\right) = 2e^{i\left(-\frac{\pi}{6}+2k\pi\right)}$$

Taking square root

$$\left(1-\sqrt{3}i\right)^{\frac{1}{2}} = \left(2e^{i\left(-\frac{\pi}{6}+2k\pi\right)}\right)^{\frac{1}{2}} = \sqrt{2}e^{i\left(-\frac{\pi}{12}+k\pi\right)} \qquad k=0,1$$

When k=0

$$\left(1-\sqrt{3}i\right)^{\frac{1}{2}} = \sqrt{2}e^{i\left(-\frac{\pi}{12}+k\pi\right)}$$

$$= \sqrt{2}e^{i\left(-\frac{\pi}{12}\right)} = \sqrt{2}\left(\cos\left(-\frac{\pi}{12}\right) + i\sin\left(-\frac{\pi}{12}\right)\right)$$

$$= \sqrt{2}\left(\cos\left(\frac{\pi}{12}\right) - i\sin\left(\frac{\pi}{12}\right)\right) = -\sqrt{2}\left(\frac{\sqrt{3}-i}{2}\right) = -\frac{\sqrt{3}-i}{\sqrt{2}}$$

When k=1

$$\left(1-\sqrt{3}i\right)^{\frac{1}{2}} = \sqrt{2}e^{i\left(-\frac{\pi}{12}+1\pi\right)}$$

$$= \sqrt{2}e^{i\left(\frac{11\pi}{12}\right)} = \sqrt{2}\left(\cos\left(\frac{11\pi}{12}\right) + i\sin\left(\frac{11\pi}{12}\right)\right)$$

$$= \sqrt{2}\left(\frac{\sqrt{3}-i}{2}\right) = +\frac{\sqrt{3}-i}{\sqrt{2}}$$

So the roots of $1-\sqrt{3}i$ are $\pm\frac{\sqrt{3}-i}{\sqrt{2}}$

2. $(-16)^{\frac{1}{4}} = (-1*16)^{\frac{1}{4}} = \left(i^2 \times 2^4\right)^{\frac{1}{4}} = 2(i)^{\frac{1}{2}} = 2\sqrt{i}$

We find the square root of i.

Topic 25 examples:

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4. show that $|e^{i\theta}-1|=2$

Solution:

$$
\begin{aligned}
&LHS=|e^{i\theta}-1| \\
&=|\cos\theta+i\sin\theta-1| \qquad 0\le\theta\le 2\pi \\
&when\ \ \theta=\pi \\
&=|\cos\pi+i\sin\pi-1| \\
&=|-1+0-1| \\
&=|-2|=2=RHS
\end{aligned}
$$

5. By writing the individual factors on the left in exponential form, performing the needed operations and changing back to rectangular coordinates, show that

a): $i(1-\sqrt{3}i)(\sqrt{3}+i)=2(1+\sqrt{3}i)$

Solution:

Exponential form are

$$i=e^{i\frac{\pi}{2}}=\cos\frac{\pi}{2}+i\sin\frac{\pi}{2}=i$$

$$(1-\sqrt{3}i)=2\cos\frac{\pi}{3}-2i\sin\frac{\pi}{3}=2e^{-i\frac{\pi}{3}}$$

$$(\sqrt{3}+i)=2\cos\frac{\pi}{6}+2i\sin\frac{\pi}{6}=2e^{i\frac{\pi}{6}}$$

$$
\begin{aligned}
&LHS=i(1-\sqrt{3}i)(\sqrt{3}+i) \\
&=e^{i\frac{\pi}{2}}\left(2e^{-i\frac{\pi}{3}}\right)\left(2e^{i\frac{\pi}{6}}\right) \\
&=4e^{i\frac{\pi}{2}-\frac{\pi}{3}+\frac{\pi}{6}} \\
&=4e^{i\frac{3\pi-2\pi+\pi}{6}} \\
&=4e^{i\frac{\pi}{3}} \\
&=2(2e^{i\frac{\pi}{3}}) \\
&=2(1+\sqrt{3}i) \\
&=RHS
\end{aligned}
$$

b): $LHS=\dfrac{5i}{2+i}=\dfrac{5e^{i\frac{\pi}{2}}}{\sqrt{5}e^{i}}$

8. Prove: $\exp\left(i\frac{\theta_1+\theta_2}{2}\right)\exp\left(i\frac{\theta_1-\theta_2}{2}\right)=\exp(i\theta_1)$

Solution:

$$
\begin{aligned}
&LHS=\exp\left(i\frac{\theta_1+\theta_2}{2}\right)\exp\left(i\frac{\theta_1-\theta_2}{2}\right) \\
&=\left(\cos\left(\frac{\theta_1+\theta_2}{2}\right)+i\sin\left(\frac{\theta_1+\theta_2}{2}\right)\right)\left(\cos\left(\frac{\theta_1-\theta_2}{2}\right)+i\sin\left(\frac{\theta_1-\theta_2}{2}\right)\right) \\
&=\left(\cos\left(\frac{\theta_1+\theta_2}{2}\right)\cos\left(\frac{\theta_1-\theta_2}{2}\right)-\sin\left(\frac{\theta_1+\theta_2}{2}\right)\sin\left(\frac{\theta_1-\theta_2}{2}\right)\right) \\
&\quad+i\left(\sin\left(\frac{\theta_1+\theta_2}{2}\right)\cos\left(\frac{\theta_1-\theta_2}{2}\right)+\cos\left(\frac{\theta_1+\theta_2}{2}\right)\sin\left(\frac{\theta_1-\theta_2}{2}\right)\right) \\
&=\cos\left(\frac{\theta_1+\theta_2}{2}+\frac{\theta_1-\theta_2}{2}\right)+i\sin\left(\frac{\theta_1+\theta_2}{2}+\frac{\theta_1-\theta_2}{2}\right) \\
&=\cos(\theta_1)+i\sin(\theta_1) \\
&=e^{i\theta_1}=\exp(i\theta_1)=RHS
\end{aligned}
$$

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$$\frac{z_1}{z_2}=\frac{8+i}{3+8i}$$

$$=\frac{8+i}{3+8i}\times\frac{3-8i}{3-8i}$$

$$=\frac{24+64i-3i-8i^2}{(3)^2-(8i)^2}$$

$$=\frac{24+64i-3i-8(-1)}{9-64(-1)}$$

$$=\frac{24+64i-3i+8}{9-64(-1)}$$

$$=\frac{32+61i}{73}$$

3/21

Q5: Simplify $(\sqrt{3}+i\,3)-i(5-\sqrt{3}i)$

Solution:

9:07 pm 2/3

$$(\sqrt{3}+i\,3)-i(5-\sqrt{3}i)$$

$$=\sqrt{3}+3i-5i+\sqrt{3}i^2$$

$$=\sqrt{3}-2i+\sqrt{3}(-1)$$

$$=\sqrt{3}-2i-\sqrt{3}$$

$$=\sqrt{3}-\sqrt{3}-2i$$

$$=0-2i$$

$$=-2i$$

Q6: Find *Re(z)* and *Im(z)* complex numbers, $z = -3 + 9i$.

Solution:

$Re(z) = -3$ and $Im(z)=9$

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MTH632 (Solved-Excerc...

SOLVED EXERCISE MTH-632

Complex Analysis & Differential Geometry

Instructor

Dr. Sohail Iqbal

Midterm's Syllabus Solved Exercises

Book Name: Brown-Churchill-Complex Variables and Application 8th edition

Chapter 1 Section 2,3,4,8,10,

Chapter 2 Section 12,18,20,23,25,26

Section 2 ***Basic Algebraic Operations***

Question No.1: **Verify that**

a): $(\sqrt{2}-i)-i(1-\sqrt{2})=-2i$

proof:

$$\begin{aligned} L.H.S &= (\sqrt{2}-i)-i(1-\sqrt{2}) \\ &= \sqrt{2}-i-i+\sqrt{2}i^2 \\ &= \sqrt{2}-2i+\sqrt{2}(-1) \\ &= \sqrt{2}-2i-\sqrt{2} \\ &= -2i = R.HS \end{aligned}$$

b):

$$\begin{aligned} L.H.S &= (2,-3)(-2,1) \\ &= (2-3i)(-2+i) \\ &= 2(-2+i)-3i(-2+i) \\ &= -4+2i+6i-3i^2 \\ &= -4+8i-3(-1) \\ &= -4+8i+3 \\ &= -1+8i = R.HS \end{aligned}$$

c):

$$\begin{aligned} L.H.S &= (3,1)(3,-1)\left(\frac{1}{5},\frac{1}{10}\right) \\ &= (3+i)(3-i)\left(\frac{1}{5}+\frac{1}{10}i\right) \\ &= \left(3(3-i)+i(3-i)\right)\left(\frac{1}{5}+\frac{1}{10}i\right) \\ &= \left(9-3i+3i-i^2\right)\left(\frac{1}{5}+\frac{1}{10}i\right) \\ &= \left(9-(-1)\right)\left(\frac{1}{5}+\frac{1}{10}i\right) \\ &= 10\left(\frac{1}{5}+\frac{1}{10}i\right) \\ &= 2+i \\ &= (2,1) = R.HS \end{aligned}$$

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Example: The Polar representation of z = -1-i is given by:

$$z = \sqrt{2}\left(\cos\left(-\frac{3\pi}{4}\right) + i\sin\left(-\frac{3\pi}{4}\right)\right)$$

or $\frac{\pi}{4}$

$$e^{i\theta} = \cos\theta + i\sin\theta$$

Exponential Form of z is:

$$-i\left(\frac{3\pi}{4}\right) \qquad \sqrt{2}\,-i\left(\frac{3\pi}{4}\right)$$

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$$-\pi < \theta \leq \pi$$

Example: Calculate Polar representation of $z = -1 - i$

$$r = |z| = \sqrt{(-1)2 + (-1)2} = \sqrt{1+1} = \sqrt{2}$$

$$\tan^{-1}\left(\frac{y}{x}\right) = \tan^{-1}\left(\frac{-1}{-1}\right) = \tan^{-1}(1) = \frac{\pi}{4}$$

$$\theta = \arg z = \frac{\pi}{4} + \pi = \frac{5\pi}{4}$$

$$-\pi < \theta \leq \pi \qquad \text{Arg } z = \frac{5\pi}{4}$$

Polar representation of $\mathbf{z = \sqrt{2}(cos\frac{5\pi}{4} + isin\frac{5\pi}{4})}$

If x = 0 then we can not use the following to find arg z

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Q1 Find square roots of $2i$

Let $z = \sqrt{2i}$

by using exponential form $z = re^{i\theta}$

$\Rightarrow r^2 e^{i(2\theta)} = 2i = 2e^{i(\pi/2 + 2n\pi)}$

$r^2 = 2 \Rightarrow r = \sqrt{2}$

$2\theta = \frac{\pi}{2} + 2n\pi \Rightarrow \theta = \frac{\pi}{4} + n\pi$

So, $z = \sqrt{2}\, e^{i(\pi/4 + 2n\pi)}$

$z = \sqrt{2}\left[\cos\left(\frac{\pi}{4} + 2n\pi\right) + i\sin\left(\frac{\pi}{4} + 2n\pi\right)\right]$

$z = \sqrt{2}\left[\pm\frac{1}{\sqrt{2}} \pm \frac{1}{\sqrt{2}}i\right]$

$z = \pm 1 \pm i$

$z = 1+i \quad , \quad z = -1-i$

given by

$$z = \sqrt[n]{r_0}\, exp\left[i\left(\frac{\theta_0}{n} + \frac{2k\pi}{n}\right)\right], k = 0, \pm 1, \pm 2, \ldots$$

Example: Find square roots of $2i$

$2i = 2(\cos \pi/2 + i \sin \pi/2)$

$z_0 = 2e^{i\pi/2}$

$z = \sqrt{2}\, exp\left[i\left(\frac{\pi/2}{2} + \frac{2k\pi}{2}\right)\right] k \in \mathbb{Z}$

$k = 0$

$z_1 = 1 + i$, $z_2 = -1 - i$

$\theta_0 + 2k\pi, k \in \mathbb{Z}$

$\frac{\theta_0}{n} + \frac{2k\pi}{n}, k \in \mathbb{Z}$

$exp\left[i\frac{\theta_0}{n} + \frac{2k\pi}{n}\right], k \in \mathbb{Z}$

Example

$$f(z) = 1 + z + z^2 + z^3 + z^4 + \cdots$$

is NOT a polynomial.

It is not polynomial because polynomial is always finite sum of term. In this case, the terms are not finite and its an ongoing process. So, it is not polynomial. Infact it is a series.

oots of

Relation between roots of a complex number and roots of unity

Consider equation

Then the quadratic form the above equation

ample: Find all values of $(2i)^{1/3}$.

step 1: $z_0 = 2i = 2\cos(\pi/2) + i\sin\pi/2$

$z_0 = 2e^{i(\pi/2)}$

$z_0 = 2e^{i(\pi/2 + 2k\pi)} \quad k \in \mathbb{Z}$

2: $z_0^{1/3} = \sqrt[3]{2}\, e^{i\left(\frac{\pi}{6} + \frac{2k\pi}{3}\right)} \quad k \in \mathbb{Z}$

3: $k = 0, 1, 2$

$c_0 = \sqrt[3]{2}\, e^{i(\pi/6)} = \sqrt[3]{2}\left(\frac{\sqrt{3}}{2} + i/2\right)$

$c_1 = c_0 w_3, \quad c_2 = c_0 w_3^2$

ple: Find roots of the following equation

$$z^2 + (2+i)z + \frac{3}{4} = 0$$

$a=1, b=2+i \quad c=3/4$

$$\frac{-(2+i) + [(2+i)^2 - 4(1)(3/4)]^{1/2}}{2(1)}$$

$$\frac{-(2+i) + [4+i^2+2(2i) - 3]^{1/2}}{2}$$

$$\frac{-(2+i) \pm [4-1+4i-3]^{1/2}}{2}$$

$$\frac{-(2+i) + [4i]^{1/2}}{2}$$

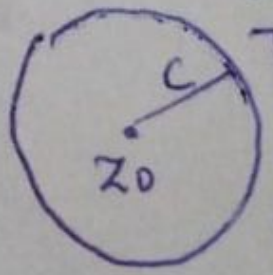
Here $[4i]^{1/2} = \sqrt{2}+i\sqrt{2}, -2\sqrt{2}-i\sqrt{2}$

So, $\frac{-(2+i)+(\sqrt{2}+i\sqrt{2})}{2}, \frac{(-2+i)-(-2\sqrt{2}-\sqrt{2}i)}{2}$

END

- Roots of Unity
- Relation between roots of complex numbers and roots of unity
- Quadratic formula

A circle in plane is a set of points whose distance from a fixed point, say z_0 is constant



The point z_0 is called centre, the constant is called radius

* Equation of a circle of radius ε cent z_0 is $|z-z_0| = \varepsilon$

A closed disc with centre z_0 and ra $|z-z_0| \leq \varepsilon$

$$\lim_{2+i} \; z + 2z - 8 + 5i$$

$$(2+i)^2 + 2(2+i) - 8 + 5i$$

$$4 + i^2 + 4i + 4 + 2i - 8 + 5i$$

$$4 - 1 + 4i + 4 + 2i - 8 + 5i$$

$$4 - 1 + 4 - 8 + 4i + 2i + 5i$$

$$-1 + 11i$$

c) $(1-i)^4 = ?$

$$(1-i)^2 = (1)^2 + (i)^2 - 2(1)(i)$$
$$= 1-1-2i = 2i$$
$$(1-i)^4 = (1-i)^2(1-i)^2$$
$$= (2i)(2i)$$
$$= 4i^2 = 4(-1) = -4$$

2. Show that

$$L.H.S = \frac{1}{1/z} = \frac{1}{z^{-1}} \qquad \because \frac{1}{z} = z^{-1}$$
$$= \frac{1}{z^{-1}} \cdot \frac{z}{z} = \frac{1.z}{z^{-1}.z}$$
$$= \frac{z}{1} = z = R.H.S$$

3. use the associative and commutative laws for multiplication to

$$L.H.S = (z_1 z_2)(z_3 z_4)$$
$$= (z_2 z_1)(z_3 z_4) \qquad \because (z_1 z_2) = (z_2 z_1) \quad \textit{commutative la}$$
$$= (z_2)(z_1 z_3)(z_4) \qquad \because (z_1)(z_3 z_4) = (z_1 z_3)(z_4) \quad \textit{associati}$$
$$= (z_1 z_3)(z_2)(z_4) \qquad \because (z_2)(z_1 z_3) = (z_1 z_3)(z_2) \quad \textit{commut}$$
$$= (z_1 z_3)(z_2 z_4) \qquad \because (z_2)(z_4) = (z_2 z_4)$$
$$= R.H.S$$

5. Drive expression

$$\frac{z_1}{z_2} = \frac{x_1 x_2 - y_1 y_2}{x_2^2 + y_2^2} + i\frac{y_1 x_2 - x_1 y_2}{x_2^2 + y_2^2} \quad (z_2 \neq 0)$$

Let $z_1 = x_1 + iy_1$ and $z_2 = x_2 + iy_2$ where $(z_2 \neq 0)$

). $f(z) = \frac{z^3}{(z+i)^3}$ at $z = i$ (5)

$f(z) = x^2 - y^2 - 2y + i(2x - 2xy)$ where $z = x + iy$

Use $x = \frac{z + \bar{z}}{2}$ & $y = \frac{z - \bar{z}}{2i}$ to write $f(z)$ in terms of z (5 masks

$P(z) = 1 - z - z^2 + z^3$ is continuous at point $z = i$ in complex plan (3)

$f(z) = \frac{(1+z^2)^4}{z^2}$, $(z \neq 0)$ find $f'(z)$ (3masks)

Evaluate $\lim_{z \to 4i} \frac{z^2 + 16}{z - 4i}$ (2 masks)

Show $\lim_{z \to \infty} \frac{4z^2}{(z-1)^2} = 4$ (2masks)

→

Q1 Re z and Im z

$z = x + iy \qquad \bar{z} = x - iy$

$z + \bar{z} = x + iy + x - iy$

$z + \bar{z} = 2x$

$2x = z + \bar{z}$

$x = \frac{z + \bar{z}}{2}$

Img z $\quad z - \bar{z} = x + iy - x$

$z - \bar{z} = +2iy$

$y = \frac{z - \bar{z}}{2i}$

$Re z = \frac{z + \bar{z}}{2} \qquad Img z = \frac{z - \bar{z}}{2i}$

is given by $|z - z_0| = \varepsilon$

defined by $|z - z_0| \leq \varepsilon$ ✓

$z^2 + 3z + 1$

$z = i$

i^2

$i^2 + 3i + 1$

$1 + 3i + 1$

$3i$

$(2, 3i)$ $(3, 2i)$

---

(5) $\lim_{z \to \infty} \frac{3n^2 - 1}{2n^2 + 2} + \frac{n-1}{n+1} i$

Ans $3/2 + i$

(1) $\lim_{z \to 4i} \frac{z^2+16}{z-4i}$

or

$\lim_{z \to 4i} \frac{z^2+16}{z-4i}$

(2) $\lim_{z \to i} \frac{z^4-1}{z^3+i}$

(3) Continuity

$f(z) = z^4 + 9z^2 - 15z + i$

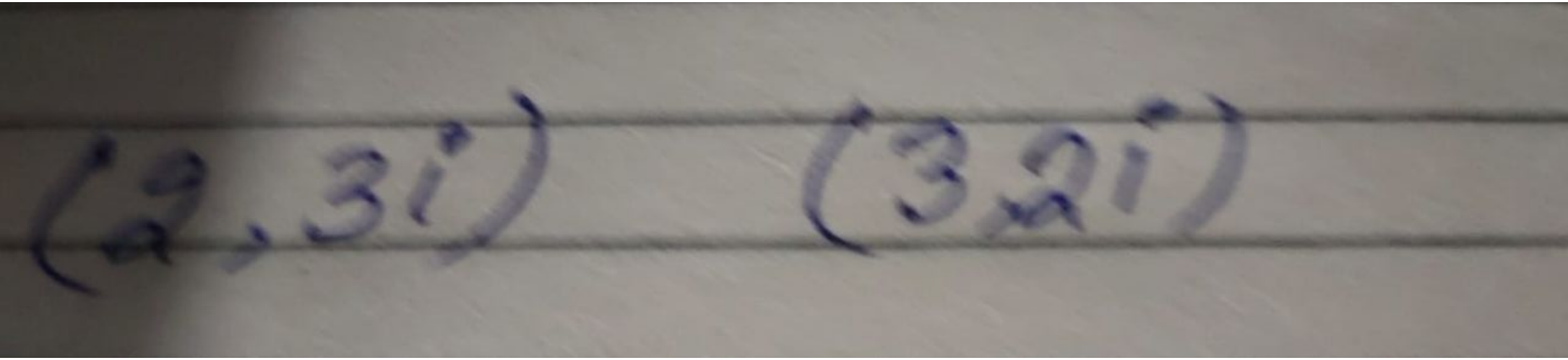
at $z = i$

(4) $(-2-2i)(-1+i)$ exponential form

$$(3+\sqrt{-3})(3-\sqrt{-3})$$

$$(3)^2 - (\sqrt{-3})^2$$

$$9+3 = 12$$


(2, 3i) (3, 2i)

(Mth632)

i) $\lim_{z \to i} \frac{z^4 - 1}{z^2 + 1}$ (3 Marks)

ii) $z_1 = \frac{1}{e^{i\frac{\pi}{3}}}$, $z_2 = \frac{1}{e^{i\frac{\pi}{2}}}$

find $\frac{z_1}{z_2}$. (3 Marks)

(iii) $f(x) = (z + 3i + 5)^7$, $z = -3i + 5$

differenciate.

(iv) $(\sqrt{3} - i)^{11/3}$, exponencial form.

(5 Marks)

(v) Evaluate $(-4 - 4i)^3$

(5 Marks)

(ix) $\lim_{z \to (i+1)} (z^2 - 2z + 1)$

(x) magnitude of $\exp(-2+3i)$

(xi) $\frac{1}{re^{i\theta}}$ The $U_r = ?$

U_r tha

U_y tha

U_x tha kuch clear nai tha.

(xii) Close disk ki mathematical form puchi hui thi.

$$z^{1/3} = (-4-4i)^{1/3}$$

$$z = -4-4i$$

$$r = 4\sqrt{2}$$

$$\theta = \tan^{-1}(1) = \frac{\pi}{4}$$

3rd quadrant

$$-\pi + \frac{\pi}{4}$$

$$-\frac{3\pi}{4}$$

$$z^{1/3} = re^{i\theta} \quad i\left(-\frac{3\pi}{4}\right)$$

$$= 4\sqrt{2}\, e$$

$$= (4\sqrt{2})^{1/3}\left(\cos\left(-\frac{3\pi}{12}\right) + i\sin\left(-\frac{3\pi}{12}\right)\right)$$

$f(z) = x + iy$

$x = \frac{z+\bar{z}}{2}$ $\quad y = \frac{z-\bar{z}}{2i}$

$= \left(\frac{z+\bar{z}}{2}\right)^2 - \left(\frac{z-\bar{z}}{2i}\right)^2 - 2\left(\frac{z-\bar{z}}{2i}\right) + i\left(2\left(\frac{z+\bar{z}}{2}\right)\right.$

$\left. - 2\left(\frac{z+\bar{z}}{2}\right)\left(\frac{z-\bar{z}}{2}\right)\right)$

$= \frac{z^2+\bar{z}^2+2z\bar{z}}{4} - \frac{z^2+\bar{z}^2-2z\bar{z}}{4} - z + \bar{z} + i\left(z+\bar{z} - \left(\frac{z^2-\bar{z}^2}{2}\right)\right)$

$= \frac{z^2+\bar{z}^2+2z\bar{z}-z^2-\bar{z}^2+2z\bar{z}}{4} - z + \bar{z} + i\left(\frac{2z+2\bar{z}-z^2-\bar{z}^2}{2}\right)$

$= \frac{4z\bar{z}}{4} - z + \bar{z} + i\left(\frac{2z+2\bar{z}-z^2-\bar{z}^2}{2}\right)$

(i) Square root of $\sqrt{(-3)^2}$ = ?

(ii) $\lim_{z \to i} \frac{iz^3 - 1}{z + i}$

(iii) Square of $(-2 - 5i)$

(iv) $f(z) = \frac{1}{z^2 + 4}$ is discontinuous at point.

(v) $f((z)) = z^3 - 3z + 12i + 5$

$f'(z) = ?$

(vi) if $z = 3 + i4$ and $|z| = 5$

Then $|\bar{z}| = ?$

(vii) if $w = 1 + 7i$ and

$z = 3 + 2i$

Then find $z \cdot w = ?$

~~Square root~~

(viii) Simplify. $\left[\cos\frac{\pi}{4} + i\sin\frac{\pi}{4}\right]^{12}$

$$= \frac{\dot{z}^2 - 2z}{\dot{c}^0} \quad \text{Ans}$$

Date:

$$\frac{z^2 - 2\bar{z}}{i}$$

$$z^{1/n} = r^{1/n}\left(\cos\left(\frac{\theta+2k\pi}{n}\right)+i\sin\left(\frac{\theta+2k\pi}{n}\right)\right),\ k=0,1,2,3,\ldots,n-1.$$

Date:

$$(x^2 - y^2 - 2y) + i(2x + 2xy)$$

$$\left(\frac{z+\bar{z}}{2}\right)^2 - \left(\frac{z-\bar{z}}{2i}\right)^2 - 2\left(\frac{z-\bar{z}}{2i}\right)$$

$$\frac{z^2+\bar{z}^2+2z\bar{z}}{4} - \frac{(z^2+\bar{z}^2-2z\bar{z})}{-4} - 2\left(\frac{z-\bar{z}}{2i}\right)$$

$$\frac{z^2+\bar{z}^2+2z\bar{z}}{4} + \frac{z^2+\bar{z}^2-2z\bar{z}}{4} - \left(\frac{z-\bar{z}}{i}\right)$$

$$\frac{2z^2+2\bar{z}^2}{4}$$

$$\left(\frac{z^2+\bar{z}^2}{2}\right) - \left(\frac{z-\bar{z}}{i}\right)$$

$$2x + 2xy$$

$$2\left(\frac{z+\bar{z}}{2}\right) + 2\left(\frac{z+\bar{z}}{2}\right)\left(\frac{z-\bar{z}}{2i}\right)$$

$$i\left(\left(z+\bar{z}\right) + \frac{z^2-\bar{z}^2}{2i}\right)$$

$$\frac{z^2+\bar{z}^2}{2} - \frac{(z-\bar{z})}{i} + i(z+\bar{z}) + \frac{z^2-\bar{z}^2}{2}$$

$$\frac{z^2+\bar{z}^2}{2} + \frac{z^2-\bar{z}^2}{2} - \frac{(z-\bar{z})}{i} + i(z+\bar{z})$$

$$z^2 - \frac{(z-\bar{z})}{i} + i(z+\bar{z})$$

region R that is both closed and bounded, there exists a nonnegative real number M such that

$$|f(z)| \leq M \text{ for all points } z \text{ in } R$$

where equality holds for at least one such z.

Proof: Given f is Continuous. We know that $g(z) = |z|$ is Continuous $g \circ f(z) = g(f(z)) = |f(z)|$

Since composition of Continuous functions is Continuous so $|f(z)|$ is Continuous iff $f(z) = u(x,y) + iv(x,y)$ then $\sqrt{(u(x,y))^2 + (v(x,y))^2}$ is Continuous.

tha

(b) $f(z) = x^2 - y^2 - 2y + i(2x - 2xy)$

where $z = x + iy$ use $x = \frac{z + \bar{z}}{2}$ &

$y = \frac{z - \bar{z}}{2i}$ to write $f(z)$ in term of z.

Consequences of Continuity

Theorem: If a function $f(z)$ is continuous and nonzero at a point z_0, then $f(z) \neq 0$ throughout some neighbourhood of that point.

Proof: For $\varepsilon > 0$ $\exists$ $\delta > 0$ such that

$$|f(z) - f(z_0)| < \varepsilon \text{ whenever } |z - z_0| < \delta$$

Taking $\varepsilon = \frac{|f(z_0)|}{2}$, the above statement becomes

$$|f(z) - f(z_0)| < \frac{|f(z_0)|}{2} \text{ whenever } |z - z_0| < \delta$$

Assume there is a pt in the neighbourhood of z_0 where $f(z)$ is zero. This implies

$$|f(z_0)| < \frac{|f(z_0)|}{2}$$ A contradiction.

Composition of Functions

Theorem: If f is continuous at a point z_0 and g is continuous at $f(z_0)$ then $g \circ f$ is continuous at z_0.

Proof: Since g is cont. at $f(z_0)$ so for $\varepsilon > 0$ $\exists$ $\gamma > 0$ such that

$$|g(z) - g(f(z_0))| < \varepsilon \quad \text{whenever} \quad |z - f(z_0)| < \gamma \quad \text{— I}$$

Since f is cont. at point z_0 so for $\gamma > 0$ $\exists$ $\delta > 0$ s.t.

$$|f(z) - f(z_0)| < \gamma \quad \text{whenever} \quad |z - z_0| < \delta \quad \text{— II}$$

To show: For $\varepsilon > 0$ $\exists$ $\delta > 0$ such that

$$|g(f(z)) - g(f(z_0))| < \varepsilon \quad \text{whenever} \quad |z - z_0| < \delta$$

Combining I & II gives the result.

VU

Continuity of $u(x, y)$

mple: Check whether the following function is nuous at (0,0)

$$u(x,y) = \begin{cases} \dfrac{x^3}{x^2+y^2} & (x,y) \neq (0,0) \\ 0 & (x,y) = (0,0) \end{cases}$$

tion:

$$\lim_{(x,y)\to(0,0)} \frac{x^3}{x^2+y^2} = 0$$

$u(0,0) = 0$

$u(x,y)$ is cont. at $(0,0)$

15:02

Show that the function $f(z) = \bar{z}$ is continuous.

$\lim_{z \to z_0} \bar{z} = \bar{z_0}$ (from previous discussion)

$f(z_0) = \bar{z_0}$

$\Rightarrow$ $f(z)$ is Cont. at z_0

$\Rightarrow$ $f(z)$ is Cont. at each complex number.

: Show that the function $f(z) = \bar{z}$ is continuous.

:

$\lim_{z \to z_0} \bar{z} = \bar{z_0}$ (from previous discussion)

$f(z_0) = \bar{z_0}$

$\Rightarrow$ $f(z)$ is Cont. at z_0

$\Rightarrow$ $f(z)$ is Cont. at each Complex number.

operties of Complex Limits

e: Compute the limit

$$\lim_{z \to i} z^2 + z + 1 .$$

$$= \left(\lim_{z \to i} z\right)\left(\lim_{z \to i} z\right) + \lim_{z \to i} z + \lim_{z \to i} 1$$

$$= (i)(i) + i + 1$$

$$= -1 + i + 1 = i$$

Continuity of Real Functions f(

xample: Check whether the following function is
ntinuous or not

$$f(x) = \begin{cases} x^2 + 1 & x \leq 0 \\ x & x > 0 \end{cases}$$

olution: $x = 1$

$\lim_{x \to 1} f(x) = 1$, $f(1) = 1 \Rightarrow f(x)$ is Cont. at $x = 1$

$\lim_{x \to 0} f(x)$: $\lim_{x \to 0^-} f(x) = \lim_{x \to 0^-} x^2 + 1 = 1$

$\lim_{x \to 0^+} f(x) = \lim_{x \to 0^+} x = 0$

Not Cont. at $x = 0$

VU

Properties of Complex Limits

Theorem: Suppose that $\lim_{z \to z_0} f(z) = A$ and $\lim_{z \to z_0} g(z) = B$

then

1. $\lim_{z \to z_0} c\,f(z) = cA$, c a complex constant.

Given $\lim_{z \to z_0} f(z) = A$

To prove: $\lim_{z \to z_0} c\,f(z) = cA$

$f(z) = u(x,y) + iv(x,y)$, $c = c_0 + ic_1$

$\Rightarrow cf(z) = (u+iv)(c_0+ic_1) = uc_0 + iuc_1 + ivc_0 - c_1 v$

$= (uc_0 - c_1 v) + i(uc_1 + vc_0)$

Properties of Complex Limits

Theorem: Suppose that $\lim_{z \to z_0} f(z) = A$ and $\lim_{z \to z_0} g(z) = B$ then

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$= (uc_0 - c_1 v) + i(uc_1 + vc_0)$

Properties of Complex Limits

Theorem: Suppose that $\lim_{z \to z_0} f(z) = A$ and $\lim_{z \to z_0} g(z) = B$

then

1. $\lim_{z \to z_0} c\, f(z) = cA$, c a complex constant.

If $f(z) = u(x,y) + iv(x,y)$, $A = u_0 + iv_0$

$\Rightarrow \lim_{(x,y) \to (x_0,y_0)} u(x,y) = u_0$ & $\lim_{(x,y) \to (x_0,y_0)} v(x,y) = v_0$

$cA = (c_0 + ic_1)(u_0 + iv_0) = \underbrace{(c_0 u_0 - c_1 v_0)} + i\underbrace{(c_1 u_0 + v_0 c_0)}$

$\lim_{(x,y) \to (x_0,y_0)} uc_0 - vc_1 = \underbrace{u_0 c_0 - v_0 c_1}$

$\lim_{(x,y) \to (x_0,y_0)} uc_1 + vc_0 = \underbrace{u_0 c_1 + v_0 c_0}$

$\Rightarrow \lim_{z \to z_0} cf(z) = cA$

Properties of Complex Limits

Theorem: Suppose that $\lim_{z \to z_0} f(z) = A$ and $\lim_{z \to z_0} g(z) = B$ then

1. $\lim_{z \to z_0} c\, f(z) = cA$, c a complex constant.

Given $\lim_{z \to z_0} f(z) = A$

To prove: $\lim_{z \to z_0} c\, f(z) = cA$

$f(z) = u(x,y) + iv(x,y)$, $c = c_0 + ic_1$

$\Rightarrow cf(z) = (u+iv)(c_0+ic_1) = uc_0 + iuc_1 + ivc_0 - c_1 v$

$= (uc_0 - c_1 v) + i(uc_1 + vc_0)$

Properties of Complex Limits

Theorem: Suppose that $\lim_{z \to z_0} f(z) = A$ and $\lim_{z \to z_0} g(z) = B$

Then

1. $\lim_{z \to z_0} c\, f(z) = cA$, c a complex constant.

$f(z) = u(x,y) + i\, v(x,y)$, $A = u_0 + i v_0$

$\Rightarrow \lim_{(x,y) \to (x_0,y_0)} u(x,y) = u_0$ & $\lim_{(x,y) \to (x_0,y_0)} v(x,y) = v_0$

$cA = (c_0 + i c_1)(u_0 + i v_0) = (c_0 u_0 - c_1 v_0) + i(c_1 u_0 + v_0 c_0)$

View Help

Relation between Real & Complex Limits

Proof: To prove: For $\varepsilon > 0$ $\exists\ \delta$ ⇝ find

$$|f(z) - z_0| < \varepsilon \quad \text{whenever} \quad 0 < |z - z_0| < \delta$$

$$|(u - u_0) + i(v - v_0)| < \varepsilon \quad \text{whenever} \quad 0 < |z - z_0| < \delta$$

↑ Find.

Take $\delta = \min\{\delta_1, \delta_2\}$

$$|(u - u_0) + i(v - v_0)| \leq |u - u_0| + |v - v_0||i|$$

$$= |u - u_0| + |v - v_0| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon$$

$0 < |z - z_0| < \delta_1$ $\quad$ $0 < |z - z_0| < \delta_2$

VU

Example: Calculate the limit $\lim_{(x,y)\to(1,2)} 3xy^2 - y.$

Solution:

$$\lim_{(x,y)\to(1,2)} \overbrace{3xy^2}^{F} - \overbrace{y}^{G}$$

$$= \lim_{(x,y)\to(1,2)} 3xy^2 - \lim_{(x,y)\to(1,2)} y$$

$$= \left(\lim_{(x,y)\to(1,2)} 3x\right)\left(\lim_{(x,y)\to(1,2)} y^2\right) - \lim_{(x,y)\to(1,2)} y$$

$$= 3(1)(2)(2) - (2) = 10$$

Relation between Real & Complex Limits

Proof: $|u-u_0| \le |(u-u_0)+i(v-v_0)|$

Similarly $|v-v_0| \le |(u-u_0)+i(v-v_0)|$

$|u-u_0| \le |f(z)-z_0| < \varepsilon$ whenever $0<|z-z_0|<\delta$

$\Rightarrow |u-u_0| < \varepsilon$ whenever, $0<|z-z_0|<\delta$

Similarly $|v-v_0| < \varepsilon$ whenever $0<|z-z_0|<\delta$

VU

10:33

Relation between Real & Complex Limit

Example: Compute the limit

$$\lim_{z \to i} z^2 + z + 1 .$$

Solution:

$$f(z) = z^2 + z + 1$$

$$f(x+iy) = (x+iy)^2 + (x+iy) + 1$$

$$= (x^2 - y^2) - 2ixy + (x+iy) + 1$$

$$= \underbrace{(x^2 - y^2 + x + 1)}_{u(x,y)} + i\underbrace{(-2xy + y)}_{v(x,y)}$$

$$\lim_{(x,y) \to (0,1)} x^2 - y^2 + x + 1 = -1 + 1 = 0$$

$$\lim_{(x,y) \to (0,1)} -2xy + y = 1$$

$$\Rightarrow \lim_{z \to i} f(z) = 1$$

Relation between Real & Complex Limits

Proof: Conversely assume

$$\lim_{(x,y)\to(x_0,y_0)} u(x,y) = u_0 \text{ and } \lim_{(x,y)\to(x_0,y_0)} v(x,y) = v_0$$

To show:

$$\lim_{z\to z_0} f(z) = u_0 + iv_0$$

Given: For $\varepsilon/2 > 0$ $\exists$ $\delta > 0$ such that $\sqrt{(x-x_0)^2+(y-y_0)^2}$

$|u-u_0| < \varepsilon/2$ whenever $0 < |z-z_0| < \delta_1$

$|v-v_0| < \varepsilon/2$ whenever $0 < |z-z_0| < \delta_2$

VU

Relation between Real & Complex Limits

Proof: Assume

$$\lim_{z \to z_0} f(z) = \overbrace{u_0 + iv_0}^{z_0}$$

To show:

$$\lim_{(x,y)\to(x_0,y_0)} u(x,y) = u_0 \text{ and } \lim_{(x,y)\to(x_0,y_0)} v(x,y) = v_0$$

Given: For $\varepsilon > 0$ $\exists\ \delta > 0$ such that

$|u(x,y) + iv(x,y) - (u_0 + iv_0)| < \varepsilon$ whenever $0 < |z - z_0| < \delta$ }

$\Leftrightarrow$ $|(u - u_0) + i(v - v_0)| < \varepsilon$ whenever $0 < |(x - x_0) + i(y - y_0)| < \delta$

To prove: For $\varepsilon > 0$ $\exists\ \delta > 0$ such that

$|u - u_0| < \varepsilon$ whenever $0 < |(x+iy) - (x_0 + iy_0)| < \delta$ }

Real Limits of Multivariable functions

Example: Show that the function $u(x,y) = \frac{xy}{x^2+y^2}$ does not have a limit as (x,y) approaches $(0,0)$.

Solution: For $y = -x$

$$\lim_{(x,y)\to(0,0)} \frac{xy}{x^2+y^2} = \lim_{x\to 0} \frac{-x^2}{x^2+x^2}$$

along $y=-x$

$$= \lim_{x\to 0} \frac{-x^2}{2x^2}$$

$$= \lim_{x\to 0} \frac{-1}{2} = \frac{-1}{2}$$

VU

View Help

Real Limits of Multivariable function

Example: If $u(x,y) = \dfrac{2x^3}{(x^2+y^2)}$ then show that

$$\lim_{(x,y)\to(0,0)} u(x,y) = 0 .$$

Solution:

$x = r\cos\theta, \quad y = r\sin\theta$

$$u(r\cos\theta, r\sin\theta) = \frac{2r^3\cos^3\theta}{r^2\cos^2\theta + r^2\sin^2\theta} = 2r\cos^3\theta$$

$$|u(x,y)-0| \overset{\text{Polar}}{=\!=} |2r\cos^3\theta| = 2r|\cos^3\theta| \le 2r$$

$$2r < \varepsilon \Rightarrow \boxed{r < \frac{\varepsilon}{2}}$$

To find $\delta > 0$ s.t. $|u(x,y)-0| < \varepsilon$

VU

12:22

Tools View Help

Real Limits of Multivariable functions

Example: If $u(x,y) = \frac{2x^3}{(x^2+y^2)}$ then show that

$$\lim_{(x,y)\to(0,0)} u(x,y) = 0 .$$

Solution:

Find $\delta > 0$ such that

$|u(x,y) - 0| \leq 2h < \varepsilon$ whenever $0 < \underbrace{\sqrt{(x-0)^2+(y-0)^2}}_{h} < \delta$

or $|u(x,y) - 0| \leq 2h < \varepsilon$ • $h < \delta$

choose $\boxed{\delta = \frac{\varepsilon}{2}} \Rightarrow h < \frac{\varepsilon}{2} \Rightarrow \boxed{2h < \varepsilon} \Rightarrow |u-0| < \varepsilon$

VU

$f(z) = (z - 4i)^8$, $z = 3 + 4i$

$f'(z) = 8(z - 4i)^7 (z - 4i)' =$

$f'(3+4i) = 8(3 + 4i - 4i)^7$

$= 8 \cdot 3^7$

$= 17496$

$= (z-4i)^8$

$8(z-4i)^7 \cdot \frac{d}{dz}(z-4i)$

$$|z^3 + 3z^2 - 2z + 1|$$

$$\leq |z^3| + |3z^2| + |-2z| + 1$$

$$\because \quad |z^3| = |z|^3 \qquad |z_1 z_2| = |z_1||z_2|$$

$$|z|^3 + 3|z|^2 + 2|z| + 1$$

$$< z^3 + 3z^2 + 2z + 1 \qquad \text{upper bound}$$

$$|z|^2 = z\bar{z}$$

$z = x + iy$

$|z|^2 = \left(\sqrt{x^2+y^2}\right)^2$

$|z|^2 = x^2 + y^2$

$z = x + iy$

$\bar{z} = x - iy$

$z\bar{z} = x^2 - i^2 y^2$

$z\bar{z} = x^2 + y^2$

$z\bar{z} = x^2 + y^2$

Subjective MTH638

Date:-10-Jan-22
Time : 12:00pm

1. Find derivative $f(z) = (1+4z^2)^3$.

2. $f(z) = (z-5i)^7$ at $z = 10+5i$
Find derivative.

3. Find derivative
$f(z) = (z-4i)^8$ at $z = 3+4i$

4. $f(z) = z^4 - 9z^2 + iz - 2$
Check continuity at $z = -i$

5. Simplify by using
De - Moivre's formula.

$$\frac{(\cos 20 + i\sin 20)^5 (\cos 30 - i\sin 30)^6}{(\cos 40 - i\sin 40)^7 (\cos 50 + i\sin 50)^8}$$

6. $\lim_{z \to 2+i} z^2 - 4z + 2 + 5i$

7. write $(-\sqrt{3}+i)^2$ in th form $a+ib$.

8. $(2i)^{1/2}$ if the exponential form of $2i$ is $2\exp(i\pi/2)$

3. Find derivative

$f(z) = (z-4i)^8$ at $z = 3+4i$

4. $f(z) = z^4 - 9z^2 + iz - 2$

Check continuity at $z = -i$

$$\lim_{z \to i} \frac{z - i}{z^2 + 1}$$

$$\lim_{z \to i} \frac{z - i}{z^2 - (-1)} \qquad -1 = i^2$$

$$\frac{z - i}{z^2 - i^2}$$

$$\frac{z = i}{(z - i)(z + i)}$$

$$\lim_{z \to i} \frac{1}{z + i}$$

$$\frac{1}{2i}$$

$$\lim_{z\to\infty} \frac{z^2+2-1}{z^2+z-1}$$

$$\lim_{z\to\infty} \frac{z^2\left(1+\frac{2}{z^2}\right)}{z^2\left(1+\frac{1}{z}-\frac{1}{z^2}\right)}$$

$$\frac{\left(1+\frac{2}{\infty}\right)}{1+\left(\frac{1}{\infty}\right)-\left(\frac{1}{\infty}\right)}$$

$$1$$

$$\lim_{z\to i} \frac{z-1}{z^2+1}$$

$$\frac{z-1}{z^2-(-1)}$$

$$\frac{z-i}{z^2-i}$$

$$\frac{(z-i)\,1}{(z-i)(z+i)}$$

$$\lim_{z\to i} \frac{1}{i+i}$$

$$\frac{1}{2i}$$

$$z^{1/3} = e^{\frac{i(2\pi k)}{3}}$$

$$\cos\frac{2\pi}{3} + i\sin\frac{2\pi}{3}$$

$$= -\frac{1}{2} - i\frac{\sqrt{3}}{2}$$

Find $w = \frac{1}{2}$ when $z = 2-i$

$$w = \frac{1}{2-i} \times \frac{2+i}{2+i}$$

$$\frac{2+i}{4-(-1)} = \frac{2+i}{5} =$$

$$= \frac{1}{5}(2+i)$$

$$\text{or } \frac{2}{5} + \frac{1}{5}i$$

$$z = \pm 1 \pm i$$

$z = 1+i$, $z = -1-i$

Q2: formula of n^{th} root of Complex number

$$z_0^{1/n} z = \sqrt[n]{r_0}\, \exp\left[i\left(\frac{\theta_0}{n} + \frac{2k\pi}{n}\right)\right]$$

$k = 0, \pm 1, \pm 2, \cdots$

(i) Square root of

(ii) $\lim_{z \to i} \frac{iz^3 - 1}{z + i}$

(iii) Square of $(-2 - 5i)$

(iv) $f(z) = \frac{1}{z^2 + 4}$ is discontinuous at point.

(v) $f((z)) = z^3 - 3z + 12i + 5$

$f'(z) = ?$

(vi) if $z = 3 + i4$ and $|z| = 5$

Then $|\bar{z}| = ?$

(vii) if $w = 1 + 7i$ and

$z = 3 + 2i$

Then find $z \cdot w = ?$

~~Square root~~

(viii) Simplify. $\left[\cos\frac{\pi}{4} + i\sin\frac{\pi}{4}\right]^{12}$

MCQs Easy thy Bahot

① Find $f'(z)$ when $F(z) = \frac{z-i}{z+i}$ and $z = i$

② $\lim_{z \to 2+i} (z^2 - 4z + 9 + 5i)$

③ $\lim_{z \to \infty} \frac{5z^2+1}{3z^2-z}$

④ $\lim_{z \to \infty} \frac{4z^2}{(z-1)^2} = 4$

⑤ Ak question C.R equation vala tha

⑥ $f(z) = x^2 - y^2 - 2y + i(2x - 2xy)$ where $z = x+iy$ use $x = \frac{z+\bar{z}}{2}$ & $y = \frac{z-\bar{z}}{2i}$ to write $f(z)$ in term of z.

$$(1+\sqrt{3}i)^4$$

$1+\sqrt{3}i =$

$$r = \sqrt{1+3} = 2$$

$$\theta = \tan^{-1}(\sqrt{3}/1)$$

$$\pi/3$$

$$= 2e^{i(\pi/3)}$$

$$(\sqrt{3}i)^4 = \left(2(\cos\pi/3 + i\sin\pi/3)\right)^4$$

$$2^4(\cos\frac{4\pi}{3} + i\sin\frac{4\pi}{3})$$

$$2^4 e^{i4\pi/3}$$

$$16e^{i4\pi/3}$$

Q1 Re z and Im z

$z = x + iy \qquad \bar{z} = x - iy$

$z + \bar{z} = x + iy + x - iy$

$z + \bar{z} = 2x$

$2x = z + \bar{z}$

$x = \frac{z + \bar{z}}{2}$

Im z $\quad z - \bar{z} = x + iy - x$

$z - \bar{z} = +2iy$

$y = \frac{z - \bar{z}}{2i}$

$Re z = \frac{z + \bar{z}}{2} \qquad Im z =$

Q No (2)

$f(z) = z^2$ $\quad f'(z) = 2z$

$z = x + iy$ $\quad f(z) = z^2$

$z^2 = (x+iy)^2 = x^2 - y^2 + i2xy$

$u(x,y) = x^2 - y^2$ $\quad v(x,y) = 2xy$

$u_x = 2x$ $\quad v_y = 2x$

$u_y = -2y$ $\quad v_x = 2y$

$u_x = v_y$ $\quad u_y = -v_x$

$$\lim_{z \to i} \frac{z^3}{z^2 + i}$$

$$\frac{i^3}{i^2 + i}$$

$$\frac{i \cdot (i^2)}{i \cdot (i + 1)}$$

$$\frac{-1}{1 + i}$$

$$= \frac{-1}{1+i} \times \frac{1-i}{1-i}$$

$$\frac{-1(1-i)}{1+1}$$

$$\frac{-1(1-i)}{2} = -\frac{1}{2} + \frac{1}{2}i$$

$$f(z) = x^2 - y^2 - 2y + i(2x - 2xy$$

$$x = \frac{z+\bar{z}}{2} \qquad y = \frac{z-\bar{z}}{2i}$$

$$\left(\frac{z+\bar{z}}{2}\right)^2 - \left(\frac{z-\bar{z}}{2i}\right)^2 - 2\left(\frac{z-\bar{z}}{2i}\right)$$

$$+ i\left(2\left(\frac{z+\bar{z}}{2}\right) - 2\left(\frac{z+\bar{z}}{2}\right)\right.$$

$$\frac{z^2+\bar{z}^2+2z\bar{z}}{4} - \left(\frac{z^2+\bar{z}^2-2z\bar{z}}{4}\right) - 2($$

$$\frac{4z\bar{z}}{} - 2(z-\bar{z}) + i(z+\bar{z}-($$

Date:

$$\lim_{z \to \infty} \frac{4z^2}{(z-1)^2} = 4$$

$$\frac{4z^2}{z^2 + 2z + 1}$$

$$\frac{z^2(4)}{z^2\left(1 - \frac{2}{z} + \frac{1}{z^2}\right)}$$

$$\frac{z^2(4)}{z^2\left(1 - \frac{2}{z} + \frac{1}{z}\right)}$$

$$\lim_{z \to \infty} \frac{4}{1 - \frac{2}{\infty} + \frac{1}{\infty}}$$

$$\frac{4}{1 - 0 + 0} = 4 \quad \text{R.H.S}$$

). $f(z) = \frac{z^3}{(z+i)^3}$ at $z = i$ (5)

$f(z) = x^2 - y^2 - 2y + i(2x - 2xy)$ where $z = x + iy$

Use $x = \frac{z + \bar{z}}{2}$ & $y = \frac{z - \bar{z}}{2i}$ to write $f(z)$ in terms of z (5 marks)

$P(z) = 1 - z - z^2 + z^3$ is continuous at point $z = i$ in complex plan (3)

$f(z) = \frac{(1+z^2)^4}{z^2}$, $(z \neq 0)$ find $f'(z)$ (3 marks)

Evaluate $\lim_{z \to 4i} \frac{z^2 + 16}{z - 4i}$ (2 marks)

Show $\lim_{z \to \infty} \frac{4z^2}{(z-1)^2} = 4$ (2 marks)

Date:

$$f(z) = \frac{(1+z^2)^4}{z^2}$$

$$= \frac{4(1+z^2)^{4-1} \cdot 2z}{2z}$$

$$= 4(1+z^2)^3$$

OR Chain Rule

$$\lim_{z \to 4i} \frac{z^2 + 16}{z - 4i}$$

$$\frac{z^2 - (-16)}{z^2 - 4i^2}$$

$$\frac{z^2 - 4i}{z - 4i}$$

$$\frac{(z - 4i)(z + 4i)}{(z - 4i)}$$

$$8i$$

$+i)^3$

$f(z) = x^2 - y^2 - 2y + i(2x - 2xy)$

$x = \frac{z+\bar{z}}{2} \qquad y = \frac{z-\bar{z}}{2i}$

$\left(\frac{z+\bar{z}}{2}\right)^2 - \left(\frac{z-\bar{z}}{2i}\right)^2 - 2\left(\frac{z-\bar{z}}{2i}\right)$

$+ i\left(2\left(\frac{z+\bar{z}}{2}\right) - 2\left(\frac{z+\bar{z}}{2}\right)\right)$

$$\lim_{z\to\infty} \frac{z^2}{z^2+z-1}$$

$$\lim_{z\to\infty} \frac{z^2\left(1+\frac{2}{z}\right)}{z^2\left(1+\frac{1}{z}-\frac{1}{z^2}\right)}$$

$$\frac{\left(1+\frac{2}{\infty}\right)}{1+\left(\frac{1}{\infty}\right)-\left(\frac{1}{\infty}\right)}$$

$$1$$

(5) $$\lim_{z\to i} \frac{z-1}{z^2+1}$$

$$\frac{z-1}{z^2-(-1)}$$

$$\frac{z-i}{z^2-i}$$

$$\frac{(z-i)\,1}{(z-i)(z+i)}$$

$$\lim_{z\to i} \frac{1}{i+i}$$

$$\frac{1}{2i}$$

Cube root of 1

let $Z_0 = 1$

olar representation:

$$Z_0 = e^{i(\frac{\theta}{n} + \frac{2k\pi}{n})}$$

Cube root So,

$$Z_0^{1/3} = e^{i(\frac{\theta}{3} + \frac{2k\pi}{3})}$$

$$Z_0^{1/3} = e^{i(0/3 + 2k\pi/3)}$$

$$Z_0^{1/3} = e^{i(\frac{2\pi k}{3})}$$

$$= -\frac{1}{2} - i\frac{\sqrt{3}}{2}$$

(1) Find derivative of

$$f(z) = \frac{z^3}{(z+i)^3} \text{ at } z = i$$

(2) Find the derivative at $z = 2i$ of $f(z) = (iz^3 + 3z^2)^3$

(3) Find $\lim_{z \to i} f(z)$ if

$$f(z) = 3z^2 - iz + 4 - 5i$$

Course Code: _____

Campus Code: _____

FALL/SPRING MID/FINAL TERM 20__ (Y

YU Yildiz University

Find $\lim_{z \to i} f(z)$ if $z = 2i$?

$f(z) = ?$

(3) Find

) Find $\lim_{z \to \infty} \frac{z^2+2}{z^2+z-i}$ $f(z)$ =

⑤ Find $\lim_{z \to i} \frac{z-i}{z^2+1}$

Rough Sheet

ERM 20__ (Year) Date: __/__/____ Sup

(i) Square root of $\sqrt{(-3)^2}$ = ?

(ii) $\lim_{z \to i} \frac{iz^3 - 1}{z + i}$

(iii) Square of $(-2 - 5i)$

(iv) $f(z) = \frac{1}{z^2 + 4}$ is discontinuous at point.

(v) $f((z)) = z^3 - 3z + 12i + 5$

$f'(z) = ?$

(vi) if $z = 3 + i4$ and $|z| = 5$

Then $|\bar{z}| = ?$

(vii) if $w = 1 + 7i$ and

$z = 3 + 2i$

Then find $z \cdot w = ?$

~~Square root~~

(viii) Simplify. $\left[\cos\frac{\pi}{4} + i\sin\frac{\pi}{4}\right]^{12}$

(ix) $\lim_{z \to (i+1)} (z^2 - 2z + 1)$

(x) magnitude of $\exp(-2+3i)$

(xi) $\frac{1}{re^{i\theta}}$ The $U_r = ?$

U_r tha

U_y tha

U_x tha kuch clear nai tha.

(xii) Close disk ki mathematical form puchi hui thi.

$(-3)^2$

$i^2 + 3i + 1$

$z^2 + 3z + 1$

$\Rightarrow z = i$

$i^2 + 3i + 1$

$1 + 3i + 1$

$3i$

MCQs Easy thy Bahot

① Find $f'(z)$ when $F(z) = \frac{z-i}{z+i}$ and $z = i$

② $\lim_{z \to 2+i} (z^2 - 4z + 9 + 5i)$

③ $\lim_{z \to \infty} \frac{5z^2+1}{3z^2-z}$

④ $\lim_{z \to \infty} \frac{4z^2}{(z-1)^2} = 4$

⑤ Ak question C.R equation vala tha

ⓔ $f(z) = x^2 - y^2 - 2y + i(2x - 2xy)$ where $z = x+iy$ use $x = \frac{z+\bar{z}}{2}$ & $y = \frac{z-\bar{z}}{2i}$ to write $f(z)$ in term of z.

MCQs Easy thy Bahot

① Find $f'(z)$ when $F(z) = \frac{z-i}{z+i}$ and $z = i$

② $\lim_{z \to 2+i} (z^2 - 4z + 9 + 5i)$

③ $\lim_{z \to \infty} \frac{5z^2+1}{3z^2-z}$

④ $\lim_{z \to \infty} \frac{4z^2}{(z-1)^2} = 4$

⑤ Ak question C.R equation vala tha

⑥ $f(z) = x^2 - y^2 - 2y + i(2x - 2xy)$ where $z = x + iy$ use $x = \frac{z+\bar{z}}{2}$ & $y = \frac{z-\bar{z}}{2i}$ to write $f(z)$ in term of z.

Subjective MTH638

Date:- 10-Jan-22
Time : 12:00pm

1. Find derivative $f(z) = (1+4z^2)^3$.

2. $f(z) = (z-5i)^7$ at $z = 10+5i$
Find derivative.

3. Find derivative
$f(z) = (z-4i)^8$ at $z = 3+4i$

4. $f(z) = z^4 - 9z^2 + iz - 2$
Check continuity at $z = -i$

5. Simplify by using
De - Moivre's formula.

$$\frac{(\cos 2\theta + i\sin 2\theta)^5 (\cos 3\theta - i\sin 3\theta)^6}{(\cos 4\theta - i\sin 4\theta)^7 (\cos 5\theta + i\sin 5\theta)^8}$$

6. $\lim_{z \to 2+i} z^2 - 4z + 2 + 5i$

7. write $(-\sqrt{3}+i)^2$ in th form $a+ib$.

8. $(2i)^{1/2}$ if the exponential form of $2i$ is $2\exp(i\pi/2)$

MTH632 MID TERM CURENT PAPERS

(VISIT VURANK FOR MORE)

Q1:- Evaluate $\lim_{z \to \infty} \frac{z-1}{z^2+2+1}$

Q2 $\lim_{z \to i} \frac{z-1}{z^2+1}$

Q3 $\lim_{z \to 2+i} 2z^2 + 4 - 3z + 5i$

Q4:- find $f'(y) \Rightarrow f(y) = (y + 6y^2)^3$

Q5:- Find value of $(1+\sqrt{3}i)^4$

Q6:- Find $(1+\sqrt{3})^{1/3}$ exponential for
was given.

MCQS:- (i) Z-W (2) Conjugate ble
(3) reciprocal of i (4) ~~polar form of~~
(5) Almost paper was easy. MCQ
were too much easy.

Subjective MTH632

1. Show that polynomial $P(z) = 1 - 2 - z^2 + z^3 - z^4 + z^5$ is continuous at point $z = \frac{1}{2}$ in complex plane.

2. $\lim_{x \to 4} \frac{2 - \sqrt{z}}{4 - z}$

3. $\lim_{z \to 2+i} (z^2 + 4z - 4 + 5i)$

4. $\lim_{z \to 3} (z^2 + 4z + 3)$

5. $f(z) = \frac{(1+z)^4}{z^9}$, find derivative, $f'(z)$.

6. Evaluate $\lim_{z \to 3i} (3z^2 - 4z + 6 - 6i)$

MTh 632

(1) $(\sqrt{2} - i) - i(1 - \sqrt{2}i) = -2i$ (2 Marks) Prove it

(2) $(1+z)^2 = 1 + 2z + z^2$ prove it (2 Mark)

(3) Check continuity of function at $z = i$ (3 Mark)

$$f(z) = \frac{z^3 + 3z + 2}{z^2 + 1}$$

(4) Evaluate $\lim_{z \to 2+i} (z^2 + 2 + 5i - 4\bar{z})$ (3 Marks)

(5) Find $f'(z)$ at $z = i$ (5 Mark)

$$f(z) = \frac{z^3}{(z+i)^3}$$

(6) Evaluate (5 Mark)

$$\frac{1+2i}{3-4i} + \frac{2-i}{5i}$$

MTh 632 2019 (Mids)

Q1:- $(\sqrt{3}\,i)^2$ express it as $a+ib$ (2)

Q:- evaluate $f(z)=\frac{z^2+3z+2}{z^2+1}$ at $z=i$ (2)

Q:- $\lim_{z\to\infty} \frac{5z^2+1}{3z^2-z}$ (3)

Q:- $\lim_{z\to 2+i} z^2-4z+2+5i$ (3)

Q:- $f(z)=z^4-9z^2+iz-2$ check continuity at $z=-i$ (5)

Q:- find. $\lim_{n\to\infty} z_n$ & $z_n=\frac{1}{n}\cos n\left(\frac{\pi}{3}\right)+\sin n(\pi/3)i$ (5)

$\lim_{z \to 2+i} (z^2 - 4z + 2 + 5i)$

$\lim_{z \to 4} \frac{2 - \sqrt{z}}{4 - z}$ (7) (8)

of $f(z) = \frac{z-1}{z+1}$ at $z = i$

① Verify that $(2, -3)(-2, 1) = (-1, 8)$ (2)

② Test continuity of $f(z)$ at $z = i$ for

$$f(z) = \frac{z^2 + 3z + 2}{z^2 + 1}$$ (2)

③ Find $f'(z)$ if $f(z) = \frac{z-1}{2z+1}$ (3)

④ Evaluate $\lim_{z \to 2+i} (z^2 - 4z + 2 + 5i)$ (3)

⑤ Evaluate $\lim_{z \to 4} \frac{2 - \sqrt{z}}{4 - z}$ (5)

⑥ Find derivative of $f(z) = \frac{z-1}{z+1}$ at $z = i$ (5)

Rough Sheet

FALL/SPRING MID/FINAL TERM 2018 (Year)

Campus Code: VEHARI

Course Code: MTH 632

Date: 23/12/2018

Superintendent Signature: _____

Student Signature: _____

Student ID: _____

① verify $(2,-3)(-2,1) = (-1,8)$

② $\lim_{z \to i} \frac{z-i}{z^2+1}$

③ $\lim_{z \to 4i} \frac{z^2+16}{z-4i}$

④ Find derivative of $f(z) = (iz^3+3z^2)^3$ at $z = 2i$

⑤ $\lim_{n \to \infty} z_n$ where $z = \frac{1}{n}\left(\cos\left(\frac{n\pi}{3}\right) + \sin\left(\frac{n\pi}{3}\right)i\right)$

⑥ Cauchy Riemann eqns in Polar form

Show that $f'(z) = \frac{1}{2f(z)}$

Moreover $f(z) = \sqrt{\theta}\, e^{i\theta/2}$

Note: Submit this sheet to Superintendent, before leaving the Examination Center.

VU

Rough Sheet

FALL/SPRING MID/FINAL TERM 20[ ] (Year) Date: [ ] / [ ] / 20[ ] Superintendent Signature: _____

Campus Code: _____ Student Signature: _____

Course Code: [illegible] Student ID: [illegible]

$2 + i \neq -1 - 3i$

$4e(1-2i)$

$= 1$

$(1+7i)(3+2i)$

$3 + 2i + 21i + 14i^2$

$3 + 23i - 14$

$-11 + 23i$

$\tan^{-1}\left(\frac{-1}{1}\right)$

$\tan^{-1}(-1)$

$= \pi$

$\frac{6i-1}{3}$

$\frac{19i-i}{3}$

$= \frac{17i}{3}$

$\left|e^{(2+i)}\right|$

e^{2}

$e^{2} \cdot e^{i}$

e^{2}

(24) $1 + x^3 + 5x - 10x^2$

$3x^2 + 5$

$= (2,-5)(2,1)$

$= -4+5, -6+2$

$= -1, 4$

(25) $\lim_{x \to 4} \frac{2-\sqrt{x}}{4-x}$

$= \lim_{x \to 4} \frac{2-\sqrt{x}}{(2-\sqrt{x})(2+\sqrt{x})}$

$= \frac{1}{2+\sqrt{4}}$

$= \frac{1}{2+2}$

$= \frac{1}{4}$

(26) $z_1 - z_2$

$= (5+i) - (-8+3i)$

$= (5+i+8-3i)$

$= 13 - 2i$

$= 13 + 2i$

$\lim_{z \to 1-i} \left(z^2 + 2\right)$

$\lim_{z \to (1-i)} \left(\frac{1}{z^2} + 2\right)$

$= \frac{1+2z^2}{z^2}$

$= \frac{1+2(1-i)^2}{(1-i)^2}$

$= \frac{1+2(1+i^2-2i)}{1+i^2-2i}$

$= \frac{1+2-4+4i}{1-1-2i} = 2$

$5z + 5 - 10$

$5i - 5$

(27) $(1+z)^2 = 1 + 2z + z^2$

$= (1+z)(1+z)$

$= (1+z^2, 2+2z)$

$= 1 - z^2 + 2z$

$(1+2i)^2$

$= 1 + 4i + 4i^2$

$= 1 - 4 + 4i$

$= -3 + 4i$

$(1+x+iy)^2$

$= (1+x)^2 + 2(1+x)(iy)^2$

$= 1 + x^2 + 2x + 2iy + 2iy$

$= 1 + 2x + x^2 + 2iy + 2iy$

$= 1 + 2(x+iy) + x^2$

(28) $f(z) = (iz^3 + 3z^2)^3$

$f'(z) = 3\left[iz^3 + 3z^2\right]^2 \frac{d}{dz}(iz^3 + 3z^2)$

$= 3(iz^3 + 3z^2)^2 \cdot (3iz^2 + 6z)$

$= 3\left[1(i)^3 + 3(i)^2\right]^2 (3i(i)$

$= 3\left[8i^3 + 12i^2\right]^2 \left[8i^3 + 1\right.$

$= 3(0)$

$= 0$

Note: Submit this sheet to Superintendent, before leaving the Examination Center.

Mth 632

Reduce $\frac{1+2i}{3-4i} + \frac{2-i}{5i}$ (4)

in Real form

Express $-1+i$ in exponential form (3) marks

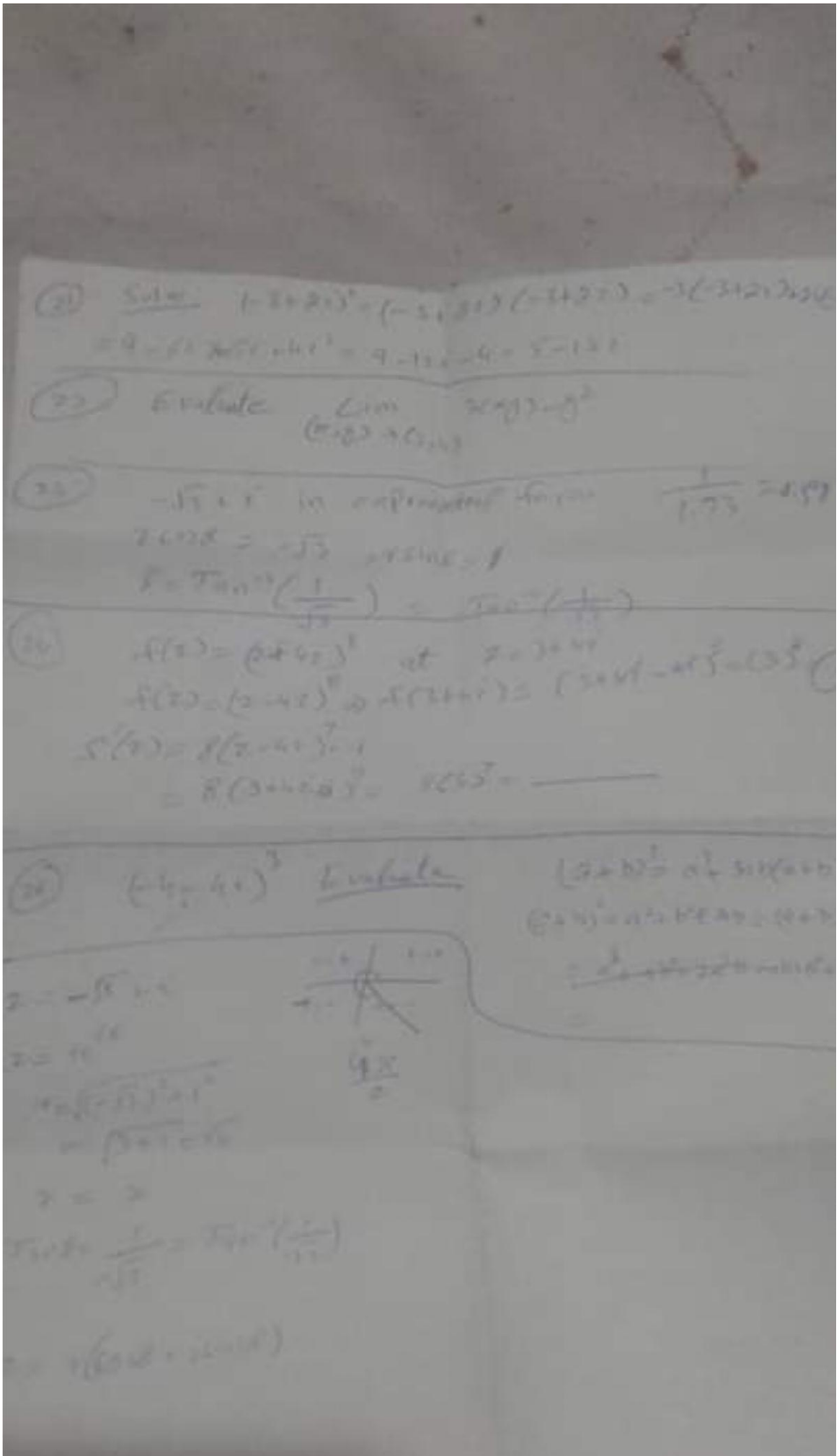
Express $\sqrt{3}-i$ in exponential form 2 marks

Mcqz

Mcqz are very very easy and we mostly were 1-20 module

Cauchy-Raimann rule

$f'(z) = z - \bar{z}$ 5 marks



Math 632

Q1) Evalute $(-4-4i)^3$

Q2) Express $(-\sqrt{3}+i)$ in the form $a+ib$

Q3) $z = x+iy$

$$x = \frac{z+\bar{z}}{2} \qquad y = \frac{z-\bar{z}}{2i}$$

Q4) $(1+z)^2 = 1+z^2+2z$ prove

Student ID:

① $\dfrac{3+i}{-1+2i} \div \dfrac{2+3i}{}$

② $\sqrt{3^2+2^2} = \sqrt{9+4} = \sqrt{13}$

③ $z = 1e^{i\theta}$

$\dfrac{1}{z} = \dfrac{1}{1e^{i\theta}}$

④ $f(z) = \dfrac{3iz}{5}$

$= \dfrac{3i \cdot 1^2}{5} = 6i$

⑤ $f(z) = \dfrac{3iz - i}{3}$

$= \dfrac{3i \cdot 2 - i}{3}$

$= \dfrac{6i - i}{3}$

$= 5i$

⑥ $\sqrt{6^2+7^2}$

$= \sqrt{36+49}$

$= \sqrt{85}$

⑦ $(4-i)^2$

$= 16 + i^2 - 2 \cdot 4 \cdot i$

$= 16 - 1 - 8i$

$= 15 - 8i$

$2z + i = 0$

$-i = 2z$

$\dfrac{-i}{2} = z$

⑧ $f(z) = z^2 + z + 1$

$= (i-1)^2 + i - 1 + 1$

$= i^2 - 1 - 2i + i - 1 + 1$

$= -2i + i = i(-2+1)$

$= -i$

⑨ $f(z) = z^2 + 3z - 1$

$= i^2 + 3i - 1$

$= -1 + 3i - 1$

$= -2 + 3i$

⑩ $f(z) = 2z^3 + 2z - 7$

$f'(z) = 6z^2 + 2$

⑪ $\dfrac{1+2i}{3-4i} + \dfrac{2-i}{5i}$

$= \dfrac{1+2i}{3-4i} \times \dfrac{3+4i}{3+4i} + \dfrac{2-i}{5i} \times \dfrac{i}{i}$

$= \dfrac{(1+2i)(3+4i)}{(3-4i)(3+4i)} + \dfrac{2i - i^2}{5i^2}$

$= \dfrac{3 + 4i + 6i + 8i^2}{(3)^2 - (4i)^2} + \dfrac{2i - (-1)}{5(-1)}$

$= \dfrac{3 + 10i + 8(-1)}{9 - 16(-1)} + \dfrac{2i+1}{-5}$

$= \dfrac{3 + 10i - 8}{9 + 16} + \dfrac{2i+1}{-5}$

$= \dfrac{-5 + 10i}{25} - \dfrac{2i+1}{5}$

$= \dfrac{-5 + 10i - 5(2i+1)}{25}$

$= \dfrac{-5 + 10i - 10i - 5}{25}$

$= \dfrac{-10}{25} = -\dfrac{2}{5}$

⑫ Limit $(x,y) \to (3,4)$ $3xy - y^2$

$3 \cdot 3 \cdot 4 - 4^2$

$= 36 - 16$

$= 20$

⑬ Limit $z \to 2+i$ $\left(z^2 - 4z + 2 + 5i\right)$

$= (2+i)^2 - 4(2+i) + 2 + 5i$

$= 4 + i^2 + 4i - 8 - 4i + 2 + 5i$

$= 4 - 1 + 4i - 8 - 4i + 2 + 5i$

$= -3 + 5i$

Sheet to Superintendent, before leaving the Examination Center.

Q2. The number $z = x+iy$ can be written in polar form as ANS: $r\cos\theta + ir\sin\theta$

Q3.IF $z = e^{i\theta}$, then $z' =$ ANS: $e^{-i\theta}$

Q4. The absolute value of complex number Z=1-i is $\sqrt{2}$

Q5. The absolute value of complex number Z=a+ib is 1) a^2+b^2 ,2) a+b , 3) $z\bar{z}$, 4) $\sqrt{z\bar{z}}$

Q6. The function f(z)= $\frac{1}{z^2+1}$ is defined everywhere in the finite plane except at the point 1) $z = \pm 1$, 2) $z = \pm 2$,3) $z = \pm i$,

4) $z = \pm 2i$

Q7. let a and b denote complex constants, then $\lim_{z \to z_0}$ az+b= 1)az+b, 2) az_0+b 3) a+b 4) b

Q8. The magnitude of exp(2+3i) is 1) exp(2) 2) exp(3) 3) exp(2*3) 4) exp(2/3)

Q9. square of 1+2i is -3+4i

Q10. let f(z)= $\frac{1}{z}$ is discontinuous at z= 1) 0 2) 1 3) -1 4) i

Q11. let $f(z) = \frac{1}{z^2+1}$ is discontinuous at z = 1) $\pm i$ 2) ± 1 3) ± 2 4) $\pm 2i$

Q12. Let $f(z) = \frac{1}{z-3i}$ is discontinuous at z = 1) 3i 2) -3i 3) i 4) -i

Q13. The polar Cauchy – Riemann equation on a pair of real – valued function of two real variables u and v are

1) $ur = v\theta$ and $u\theta = -vr$ 2) $ur = v\theta$ and $u\theta = vr$ 3) $rur = v\theta$ and $u\theta = -rvr$ 4) $ru\theta = vr$ and $ur = -rv\theta$

Q14. The limit of f(z)=z^2-2z+1 at z=i+1 is 1) 1 2) -1 ,3) i 4) -i

Q15. The limit of f(z) = z^2+3z-1 at z= i is 1) 3i-2 2) 3i 3) 2i-3

Q16. let f(z)=12z^2+15z-30 then f'(z) is 1) z+15 2) 12z^2+15 3) 36z^2+15 4) 36z^2+15

Q17. Multiply w=(3+2i) with z =(1+7i) we get z.w= 1) 3+14i 2) -11+23i 3) 21+14i 4) 21+2i

Q18. conjugate of complex numbers z = -3-3i is 1) -3+3i 2) 3+3i 3) -3+6i 4) 3-3i

Q19. The sum of two complex numbers z=3+i , w=-1+2i is z+w= 1) 2+3i 2) 3+2i 3) -1+i

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Q20. Solve $(-3+2i)^2$ (2)

Q21. Compute the exponential form of $\sqrt{3}-i$ (2)

Q22. Verify that ,(3,1)(3,-1)$(\frac{1}{5},\frac{1}{10})$= (2,1) (3)

Q23. By using Cauchy – Riemann equations (in polar form) , find $f'(z)$ of the $f(z) = \frac{1}{z^2} = \frac{1}{r^2}(\cos 4\theta - i\sin 4\theta)$ (3)

Q24.Using the definition $\frac{dw}{dz} = \lim_{\Delta z \to 0} \frac{\Delta w}{\Delta z}$, prove that $\frac{dw}{dz} = -\frac{1}{z^2}$ when $w = \frac{1}{z}(z \neq 0)$. (5)

Q25. Evaluate , $\lim_{z \to i} \frac{z^2+z-2+i}{z^2-2z+1}$ (5)

Subjective

Mth632

➡️Limit put krna 2 questions

➡️Continuity check krna

➡️Differential find krna

➡️Cauchy Reimn theorem ka ak question tha 5 marks

★ 7:58

Mth632

Mcqs koi b file sy nahi tha dub new the and solve karny waly thy

Q1. Find Rez and Imz

Q2. Show Rez=z+zbar/2

Q3. Find limit z--2+i $(z^2+2z+4-8+5i)$

Q4. Find limit z--i (z^3/z^2+i)

Q5 |z|=1 show 1<equal |z-2|<equal 3

Q6 find derivative of

$F(x)= z^3/(z+i)^2$

Mth 632 8:30am
Mcq bhot asan thy basic calculations krni the, file main sy 2 mcq ay huay thy. Aur short aur long b bhot asan thy. Limit aur derivative find krny waly 5 short aur long ay huay thy. Aur last wala quetion ni dekh saka time out ho gya tha.